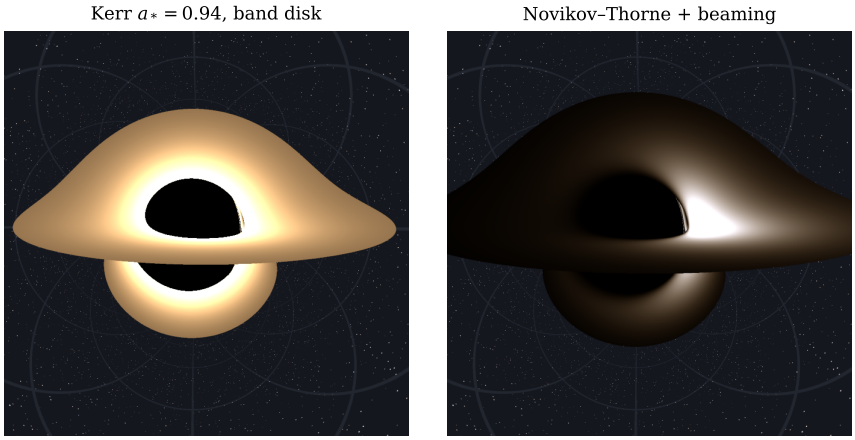
SGR A*

A First-Principles Relativistic Simulator of the Galactic-Centre Black Hole

Shadow metrology, S-star orbit fitting, and a free-quadrupole test of the no-hair theorem — with a complete account of the methods, tests, and development of the instrument



C. Excelsis

Independent research report

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Abstract

We present a ground-up, physically rigorous simulator of the Galactic-centre black hole Sgr A*, built on exact Kerr geodesics and validated end-to-end against published strong-field observations. Null and timelike geodesics are integrated in a second-order Mino-time formulation of the Carter equations with per-step constraint projection, conserving (E, L_z, Q) to better than 10^{-6} relative over complete orbits. A CPU reference ray tracer reproduces the analytic Schwarzschild shadow to a relative accuracy of 5×10^{-16} and the Bardeen critical curve for spinning holes; a double-precision GPU compute port agrees with the CPU ground truth to a mean pixel residual of 0.00289. Fitting 13 orbital and potential parameters to the 145 astrometric and 44 radial-velocity measurements of the star S2 (Gillessen et al., 2017) with full general-relativistic timelike geodesics recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8272 \pm 182$ pc, within 0.14σ and 0.03σ of the GRAVITY interferometric values, at a reduced χ^2 of 0.959. The model reproduces the gravitational-redshift excess at the 2018 pericentre passage ($205\,\mathrm{km\,s^{-1}}$ peak) detected by GRAVITY Collaboration (2018), and the Schwarzschild precession of 11.98 arcmin per orbit confirmed by GRAVITY Collaboration (2020). At the GRAVITY mass and distance the simulated photon ring diameter is $49.68\,\mu$as, inside the EHT measurement of $51.8 \pm 2.3\,\mu$as (Event Horizon Telescope Collaboration, 2022). Freeing the quadrupole moment in a Hartle–Thorne extension yields $\varepsilon_q = 0.00 \pm 0.93 \times 10^5$ (68 per cent), consistent with the no-hair theorem. An affine-invariant ensemble MCMC run through the identical forward model (32 walkers $\times$ 6000 steps) confirms the least-squares covariances globally — the posterior is single-moded with $M = 4.502^{+0.628}_{-0.603} \times 10^6\,\mathrm{M}_\odot$ — and a joint fit of five S-stars (S2, S1, S8, S12, S13) sharing one potential tightens the mass to $4.461 \pm 0.119 \times 10^6\,\mathrm{M}_\odot$, a 39 per cent improvement over S2 alone, while demonstrating concretely why single short-arc stellar fits cannot weigh the black hole.

This report is the long-form account of that programme. Part I is a plain-language primer for readers without a physics background. Parts II–IV derive,

from first principles, every formula the simulator implements; document the numerical methods and their error analysis; present the full results with their statistical treatment; and record the complete development history, test suite, and validation gates through which the instrument was built.

Notation and Conventions

Geometric units $G = c = 1$ are used throughout unless stated; masses, lengths and times are quoted in units of the gravitational radius M (for Sgr A*: $GM/c^2 = 5.1243^{-1}$ scaling on the sky, $GM/c^3 = 21.17$ s). Metric signature $(-,+,+,+)$; Boyer–Lindquist coordinates (t, r, θ, ϕ).

The complete symbol table, with the equation or section where each symbol is defined, is Appendix E; it is placed at the back so it can cross-reference the definitions, and duplicated there by design so a reader can reach it from either end of the report.

Reproducibility Statement

Every quantitative statement in this report that is not a cited literature value is typeset from a macro in `numbers.tex`, generated by `scripts/make_numbers.py` from machine-readable artefacts produced by the simulation; every figure is generated by `make figures` (matplotlib) from the same artefacts. The single command `make reproduce` re-runs the full chain — unit tests, reference renders, sub-ring metrology, orbit fits, calibration, sweeps, and figure generation — from source. A number that cannot survive this cycle cannot appear in this document.

Part I

Foundations for the Non-Specialist

Chapter 1

Gravity, Mass, and the Idea of a Black Hole

This first part of the report assumes no physics background at all. It explains, in ordinary language, every idea the technical chapters build on. If you are comfortable with general relativity, skip directly to Part II.

1.1 Throwing things upward

Throw a ball into the air and it comes back. Throw it harder and it climbs higher before it comes back. This everyday fact hides a precise question: *is there a throwing speed so great that the ball never comes back at all?*

There is. For every planet or star there is a special speed, the *escape velocity*, at which a thrown object just barely coasts away forever, slowing all the while but never quite stopping. From the surface of the Earth it is about 11 kilometres per second — thirty times the speed of sound. From the surface of the Sun it is about 620 kilometres per second. The rule of thumb is simple: the more mass squeezed below you, and the closer you stand to it, the faster you must go to leave.

Worked example: How fast must you throw?

The escape velocity has a one-line recipe. A thrown object escapes if its energy of motion, $\frac{1}{2}v^2$ for every kilogram of it, at least matches the energy needed to climb out of the gravity well, Gm/r per kilogram — where m is the planet's mass, r its radius, and G is Newton's gravitational constant, 6.67×10^{-11} in SI units. Setting the two equal and solving for v:

$$v_{\text{esc}} = \sqrt{\frac{2Gm}{r}}.$$

For the Earth, $m = 5.97 \times 10^{24}$ kg and $r = 6\,371$ km, so $2Gm = 7.97 \times 10^{14}$ m^3/s^2; dividing by $r = 6.371 \times 10^6$ m gives 1.25×10^8 m^2/s^2, and the square root is 11 200 m/s — the 11 km/s quoted above. Notice that nothing in the formula cares what the ball is made of: only total mass and radius enter. That indifference is the loophole a black hole exploits.

That second part — *the closer you stand* — is the key to everything in this report. Squeeze the same mass into a smaller ball and its surface moves closer to the centre, so the escape velocity from that surface goes up, even though the total mass never changed. Squeeze the Sun to a radius of three kilometres and the escape velocity at its surface would reach the fastest speed there is: the speed of light, about 300 000 kilometres per second.

1.2 The point of no return

Light's speed is not just fast; physics as we understand it (and as Chapter 2 sketches) says nothing can move faster. So an object compressed within that critical size has a startling property: at some radius, not even light travels fast enough to escape. Whatever falls inside can send no signal out. The sphere where escape would require exactly the speed of light is called the *event horizon*: an "event" happening inside it can never be seen from outside, even in principle.

An object like this is a *black hole* — black because no light leaves it, and a hole because things can fall in but not climb out. The critical radius is named the *Schwarzschild radius* after Karl Schwarzschild, who solved Einstein's equations for a single round mass within weeks of their publication in 1915. The Schwarzschild radius grows in direct proportion to the mass: double the mass and the horizon doubles in size. For the Earth it is about nine millimetres — the Earth would have to be crushed to the size of a marble. For the Sun, about three kilometres.

For the black hole at the centre of our Galaxy, which weighs about 4.3 million Suns, it is around 13 million kilometres — some eighteen times the radius of the Sun, yet still tiny on galactic scales.

Two cautions about the picture this conjures. First, the horizon is not a surface you could touch: nothing is *there*. It is a boundary drawn in empty space, the line on the map beyond which no path leads back out. A falling astronaut would notice nothing special while crossing it. Second, the mass is not spread throughout the dark sphere. As far as the theory can say, everything that fell in is concentrated at the centre in a region where the equations lose their meaning — the *singularity*. The singularity is a statement of ignorance rather than a physical prediction: it marks where the current theory of gravity stops being trustworthy. Fortunately, for everything in this report only the *outside* of the horizon matters, and there the theory is precise and testable.

Worked example: The size of Sgr A*'s horizon

Ask the escape-velocity recipe a pointed question: at what radius does v_{esc} reach the speed of light? Set $v_{\text{esc}} = c$ and solve for the radius:

$$R_S = \frac{2Gm}{c^2}.$$

For Sgr A*, $m = 4.297$ million solar masses $= 4.297 \times 10^6 \times 1.989 \times 10^{30}$ kg $= 8.55 \times 10^{36}$ kg. Then $2Gm = 1.14 \times 10^{27}$ in SI units, and dividing by $c^2 = 9.0 \times 10^{16}$ m^2/s^2 gives

$$R_S = 1.27 \times 10^{10} \text{ m} \approx 12.7 \text{ million km},$$

about eighteen Sun-radii, as quoted in the text; light itself would need 42 seconds to cross that distance. One honest caution: we used Newton's energy formula in a regime where Newton's theory fails. By a happy algebraic coincidence the fully relativistic calculation (Chapter 8) returns *exactly* the same R_S — which is why this little derivation, essentially the one John Michell wrote down in 1784, is worth showing at all.

1.3 Falling in: the gentle giant surprise

One more everyday idea unlocks a real result. Stand on the Earth and your feet are pulled slightly harder than your head — they are closer to the centre. Near a compact mass this difference, the *tidal* force, can become violent: fall feet-first toward a small black hole and the pull on your feet so exceeds the pull on your head that you would be stretched like taffy (astrophysicists genuinely say

spaghettification).

Here is the surprise: the bigger the black hole, the *gentler* its horizon. The tidal stretch across a falling body depends on the mass divided by the *cube* of the distance, and since the horizon radius grows in proportion to the mass, the stretch *at the horizon* falls off as one over the mass squared. Crossing the horizon of a 10-solar-mass black hole, the difference in pull across a two-metre astronaut is about 2×10^8 m/s^2 — twenty million times Earth's gravity, lethal long before arrival. Crossing the horizon of Sgr A*, four hundred thousand times heavier, the same difference is about one *millimetre* per second squared — less than the tug your feet feel from standing on a kerb. An astronaut would drift across Sgr A*'s point of no return without a twinge, noticing nothing at all; the drama is reserved for much deeper in. This is why nothing dramatic marks the horizon in the renders of this report: general relativity says, correctly, that locally there is nothing there (the caution of the previous section made precise).

1.4 Sagittarius A*, our own giant

Point a telescope at the constellation Sagittarius and you look toward the centre of the Milky Way, 8 277 parsecs away (a *parsec* is about 3.26 light-years; the light arriving tonight left before the last ice age peaked). At the exact dynamical centre sits a compact radio source named Sgr A* ("Sagittarius A star"). Around it, stars swarm at extraordinary speeds — a few pass their closest approach at several thousand kilometres *per second*, about a hundred times faster than the Earth orbits the Sun. Something enormously heavy and enormously compact must be holding them: the measured mass is 4.297 ± 0.012 million solar masses, packed into a region smaller than the orbit of Mercury. No cluster of dim stars, no exotic gas cloud survives scrutiny as an alternative; a supermassive black hole is the only explanation standing (Chapter 7 gives the observational history).

Sgr A* is the star of this report for a practical reason: it is the *nearest* supermassive black hole, and the only one where two entirely different kinds of measurement meet. We can follow individual stars looping around it year by year (Chapters 5, 14, 15), and we can photograph the glowing ring of light bent around its horizon (Chapters 3, 11). This report describes a simulator built from first principles that reproduces both kinds of measurement from the same underlying equations.

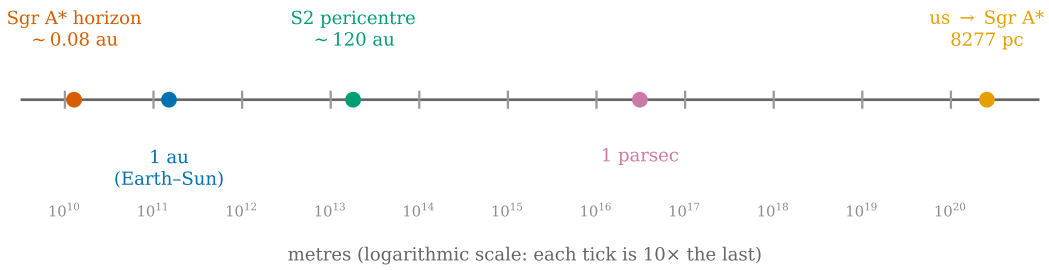


Figure 1.1: The scales of the problem on a single logarithmic line, from the Sgr A* horizon to its distance from us. Each tick is ten times the previous one: the distance to Sgr A* is about twenty billion times the size of its horizon — comparable to viewing a coin at the distance of the Moon.

1.5 A sense of scale

The numbers in this subject span so many powers of ten that it is worth pausing to place them on one line (Fig. 1.1).

Astronomers measure distances within a planetary system in *astronomical units* (au): one au is the Earth–Sun distance, about 150 million kilometres. The horizon of Sgr A* spans about a sixth of an au. The star S2, the most important character of Part III, dives to about 120 au from the black hole — about three times the distance of Pluto from the Sun — which sounds close until you recall that this is still about 1 500 horizon radii out.

For things on the sky, astronomers use angles. A degree is what your little finger covers at arm's length; an *arcsecond* is 1/3600 of a degree, roughly a coin seen from four kilometres away; and the measurements in this report are made in *milliarcseconds* (a thousandth of an arcsecond) and *microarcseconds* (μas, a millionth). The photon ring of Sgr A* spans about 52 μas — the apparent size of a doughnut on the Moon. That such an angle is measurable at all, by linking radio telescopes across the whole Earth into one instrument, is among the technical wonders of modern astronomy.

One more unit will keep appearing: the *gravitational radius*, written M in this report. It is the natural yardstick of a black hole — half the Schwarzschild radius — and it lets us describe any black hole, whatever its mass, with the same numbers. "The disk's inner edge sits at $6\,M$" is true of a stellar black hole and of

Sgr A* alike; only the metre-stick changes. Chapter 6 says more about this trick of working in natural units.

Terms you just met.

Escape velocity The launch speed at which an object barely coasts away from a mass forever.

Event horizon The boundary around a black hole from inside which nothing, not even light, can escape. Not a material surface.

Schwarzschild radius The size of the event horizon of a non-spinning mass; proportional to the mass.

Singularity The central region where the known equations of gravity break down; hidden inside the horizon.

Sgr A* The 4.3-million-solar-mass black hole at the centre of the Milky Way, 8 277 parsecs away.

au, parsec, milli-/microarcsecond Units for distances and sky angles; see Fig. 1.1.

Gravitational radius (M) The natural length unit of a black hole; the horizon of a non-spinning hole sits at $2\,M$.

→ *Full treatment: the observational case for Sgr A* in Chapter 7; the exact horizon formulas in Chapter 8.*

Chapter 2

Spacetime and Einstein's Gravity, Gently

2.1 Gravity is not a pull

Newton described gravity as a force: the Earth pulls the apple, the Sun pulls the Earth. His description is astonishingly accurate for slow objects far from big masses, and Chapter 5 will use it freely. But it leaves a mystery — how does the Sun reach across empty space to tug on the Earth? — and it fails, measurably, exactly in the situations this report cares about.

Einstein's 1915 answer, *general relativity*, removes the mystery by removing the force. In his picture, mass does not pull on distant objects. Instead, mass changes the shape of *spacetime* — the four-dimensional fabric of the three space directions plus time — in its neighbourhood, and objects then simply travel through that reshaped fabric in the straightest possible lines. The famous slogan, due to John Wheeler, is: *matter tells spacetime how to curve, and spacetime tells matter how to move.*

A useful (if imperfect) picture is a stretched rubber sheet with a bowling ball resting in the middle. Marbles rolled across the sheet curve toward the ball, not because the ball grips them, but because the surface they travel on slopes. Each marble is doing the most natural thing possible — going straight ahead on a surface that is not flat. The Earth orbits the Sun for the same reason: it moves straight ahead through a region of spacetime that the Sun has curved, and "straight ahead" there is a loop.

The straightest possible path through curved spacetime has a name that appears on nearly every page of Part II: a *geodesic*. On the Earth's curved surface,

geodesics are the great circles that airliners follow; in curved spacetime, geodesics are the paths of everything that falls freely — thrown stones, orbiting planets, and light itself. The entire simulator described in this report is, at its heart, a machine for computing geodesics very accurately.

2.2 Light bends

Because light also travels through spacetime, its path also bends where spacetime curves. The effect is tiny near the Sun — starlight grazing the solar surface deflects by less than two arcseconds, first measured during the 1919 eclipse — but near a black hole the bending becomes wild.

Fig. 2.1 shows real computed light paths around a non-spinning black hole, calculated with the same equations the simulator uses. A ray passing far away bends gently, like the starlight of 1919. A ray passing closer bends more. And there is a critical aiming distance at which the bending wins completely: the ray wraps around the hole and can circle it — light in orbit. Aim any closer and the ray spirals in and is gone. The dividing line between "escapes, however bent" and "captured" is what carves out the black-hole *shadow*, the subject of Chapter 3.

This bending also acts as a distorting lens. A black hole with a glowing disk around it shows you the disk twice over: light from the part of the disk hidden *behind* the hole is bent up and over the top (and under the bottom), so the far side appears above and below the dark centre. That is why the famous images — the Event Horizon Telescope's photographs and the film *Interstellar*'s Gargantua — have their characteristic crossed-halo look. Every one of those features appears, for the right physical reasons, in the renders of Part III.

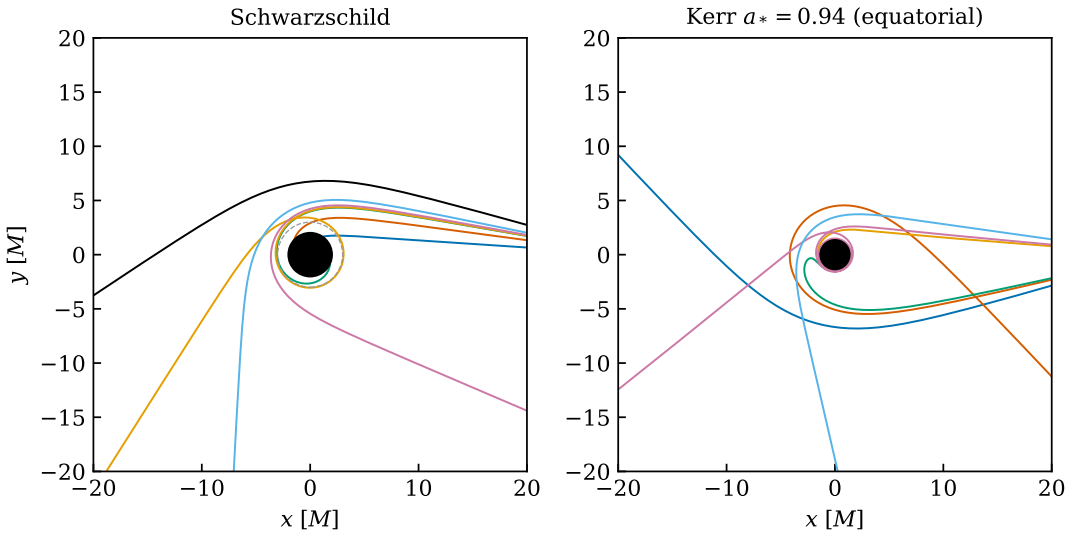


Figure 2.1: Computed paths of light rays in the equatorial plane of a black hole, from the simulator's own geodesic integrator. Left: a non-spinning (Schwarzschild) hole — rays aimed farther out escape with a bend; a critical ray circles on the dashed photon sphere; closer rays are captured. Right: a rapidly spinning (Kerr) hole — the spacetime itself circulates, so rays travelling with the spin behave differently from rays travelling against it (Chapter 3).

Worked example: The 1919 deflection, in two lines

General relativity gives a compact formula for the bending of a light ray that passes a mass m at closest distance r (it follows from the same equations the simulator integrates, Chapter 9):

$$\delta = \frac{4Gm}{c^2 r}.$$

The combination Gm/c^2 — a mass expressed as a length — will follow us through the whole report; for the Sun it is 1.48 km. A ray grazing the solar surface has $r = 696\,000$ km, so $\delta = 4 \times 1.48/696\,000 = 8.5 \times 10^{-6}$ radians. Multiplying by 206 265 (arcseconds per radian) gives $1.75''$ — precisely the deflection Eddington's expeditions went to measure in 1919, and double what a naïve Newtonian calculation predicts. The factor of two is pure spacetime curvature: half the bend comes from time running slow, half from space itself being curved.

2.3 Time runs slow near mass

Curved spacetime does something with no Newtonian counterpart at all: it changes the rate of time. Clocks deeper in a gravitational well genuinely run slower as seen from outside — not because of any mechanical fault, but because "one second" is itself a local quantity, like "north", that depends on where you are.

The effect is real and routine: the satellites of the GPS system carry clocks about 20 000 km above the Earth, where time runs faster than at the ground by about 45 microseconds per day. Uncorrected, satellite navigation would drift by kilometres within a day. Near a black hole the same effect grows enormous; at the horizon, an outside observer would see a falling clock slow to a stop.

Worked example: The GPS correction, by the numbers

The fractional rate difference between a clock at radius r_1 and one at radius r_2 is, to an excellent approximation, $(Gm/c^2)\,(1/r_1 - 1/r_2)$. For the Earth, $Gm/c^2 = 4.4$ *millimetres* — the Earth's mass written as a length. With the ground at $r_1 = 6\,371$ km and the GPS orbit at $r_2 = 26\,560$ km:

$$\frac{1}{r_1} - \frac{1}{r_2} = 1.19 \times 10^{-7}\ \text{m}^{-1}, \qquad 0.0044 \times 1.19 \times 10^{-7} = 5.3 \times 10^{-10}.$$

Multiply by the 86 400 seconds in a day: 46 microseconds — the satellite clocks run *fast* by that much, because they sit higher in the well. (Special relativity's time dilation from orbital speed works the other way and claws back about 7 μs, for a net 39 μs a day.) Light travels twelve kilometres in 39 μs: ignore relativity today and satellite navigation is off by kilometres by tomorrow.

For this report the significance is very concrete: when the star S2 dives to its closest approach to Sgr A*, its light is emitted from slightly deeper in the black hole's gravitational well, where time runs slower. Every wave of its light is stretched on the climb out — the wavelength shifts toward the red. The GRAVITY instrument measured exactly this *gravitational redshift* in 2018, and reproducing that measurement from first principles is one of this report's central results (Chapters 4, 15).

Worked example: S2's slowed clock, on an envelope

How large is the effect for S2 at closest approach? The fractional redshift of light climbing out from radius r is roughly $Gm/(c^2 r)$. For Sgr A*, $Gm/c^2 = 6.35 \times 10^9$ m — half the horizon radius computed in Chapter 1 — and S2's closest approach is $r \approx 120$ au $= 1.80 \times 10^{13}$ m:

$$z_{\text{grav}} \approx \frac{6.35 \times 10^9}{1.80 \times 10^{13}} = 3.5 \times 10^{-4}.$$

A wavelength stretch of 3.5×10^{-4} masquerades, in a spectrograph, as an extra recession speed of $3.5 \times 10^{-4} \times c \approx 106$ km/s. That — about a hundred km/s of "impossible" extra redshift, switching on only around closest approach — is the size of the signal GRAVITY hunted in 2018, and Chapter 15 shows our full model reproducing it with no adjustable amplitude.

2.4 How we know: a century of passed tests

It is fair to ask why anyone trusts this theory enough to build a thesis on it. The answer is a hundred years of passed examinations, each sharper than the last. The 1919 eclipse measurement of bent starlight (worked above) made the theory famous; the anomalous 43-arcseconds-per-century creep of Mercury's orbit — known and unexplained for six decades before Einstein — was retro-dicted exactly, with no adjustable constant. The twentieth century added laboratory redshift measurements (light climbing a 22-metre tower in 1959), the slowing of radar echoes grazing the Sun, and the patient 20 000-fold-verified spin-down of the Hulse–Taylor binary pulsar, which loses orbital energy at precisely the rate gravitational-wave emission predicts. The twenty-first century has heard those waves directly (LIGO, 2015) and photographed photon rings (EHT, 2019 and 2022). Every test in this list interrogates a different limb of the theory — weak field, strong field, static, dynamical — and none has ever returned a confirmed violation. The measurements this report reproduces (the S2 redshift and precession, the ring diameter) are the newest entries in that ledger, and the no-hair test of Chapter 17 is an attempt to *add* one: the point of precision reproduction is to build apparatus wherever the theory could still fail.

2.5 What a metric is (without tensors)

One piece of vocabulary unlocks all of Part II. To compute anything, we need coordinates — labels like latitude and longitude that name the points of spacetime. But coordinates are just names; they do not tell you how *far apart* two nearby points really are. On the Earth, one degree of longitude is 111 km at the equator and zero at the pole: the conversion from coordinate differences to true distances varies from place to place.

The rulebook for that conversion is called the *metric*. Give the metric the small coordinate steps between two nearby events (a bit of north–south, a bit of east–west, a bit of time) and it returns the true separation — including the true amount of elapsed clock time. All of the curvature of spacetime is encoded in how this rulebook changes from place to place; slow-running clocks and bending light are both just entries in the rulebook. Einstein's field equations determine the metric produced by a given arrangement of mass; for a lone spinning black hole the exact answer is known and is the famous *Kerr metric* (Chapters 3, 8),

on which this entire simulator is founded.

When Part II writes symbols such as g_{tt} or g_{rr}, nothing deeper is meant than "the entry of the rulebook that converts time-steps (or radius-steps) into true separations, at this location".

Terms you just met.

Spacetime Space and time treated as one four-dimensional fabric, whose shape is what we experience as gravity.

General relativity (GR) Einstein's theory: mass curves spacetime; freely falling objects follow the straightest paths in it.

Geodesic The straightest possible path through curved spacetime; the track of anything falling freely, including light.

Gravitational lensing The bending of light by curved spacetime; a black hole shows the far side of its own disk, bent over the top.

Time dilation Clocks deeper in a gravitational well run slower; measurable with GPS satellites and with the star S2.

Gravitational redshift Light climbing out of a gravitational well is stretched to longer, redder wavelengths.

Metric The location-dependent rulebook that converts coordinate differences into true distances and times; in GR it carries all of gravity.

→ Full treatment: the Kerr metric in Chapter 8; geodesics and their conserved quantities in Chapter 9; the traced light-bending figures in Chapters 9 and 11.

Chapter 3

Spinning Black Holes and the Shadow

3.1 Black holes spin

Almost everything in the universe rotates — planets, stars, whole galaxies — and when a rotating object collapses, its spin is not lost. It concentrates, the way a figure skater speeds up when she pulls her arms in. Black holes born from collapsing, swirling matter are therefore expected to spin, often ferociously.

A non-spinning black hole is described exactly by Schwarzschild's 1915 solution of Chapter 1. The spinning case resisted for another half-century, until Roy Kerr found the exact solution in 1963. The *Kerr metric* — the rulebook of Chapter 2 for a spinning hole — is the mathematical foundation of this entire report, and it says that a black hole is completely described by just two numbers: its mass and its spin. The spin is usually quoted as a dimensionless number a_* between 0 (not spinning) and 1 (the theoretical maximum, spinning so fast the horizon would tear open). For Sgr A* the working estimate used throughout this report is $a_* = 0.94$ — a rapid spinner.

3.2 Frame dragging: spacetime as a whirlpool

A spinning mass does something Newton never imagined: it drags spacetime around with it. Close to a spinning black hole, "standing still" is simply not on the menu — space itself circulates like water around a drain, and everything in it, light included, is carried along. This *frame dragging* has been measured in Earth

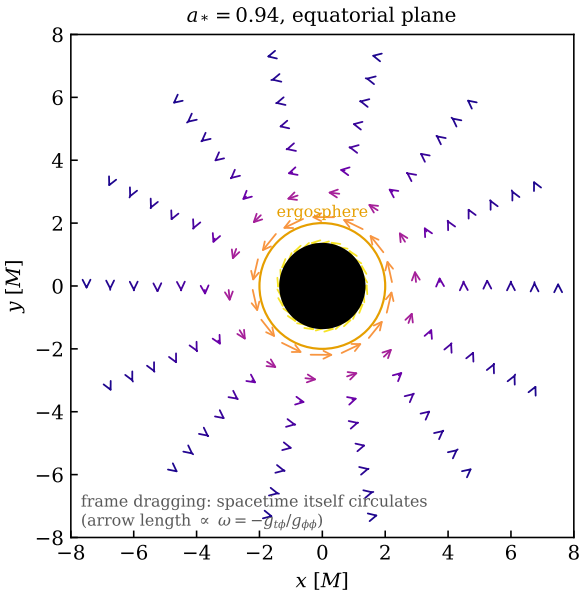


Figure 3.1: Frame dragging around a black hole with $a_* = 0.94$, computed from the Kerr rulebook. Each arrow shows how far a "stationary" marker at that radius is swept around per unit time; inside the orange boundary (the ergosphere) refusing to rotate is impossible.

orbit (by the Gravity Probe B gyroscopes and the LAGEOS satellites) at the level of fractions of a degree per century; near Sgr A* it becomes a dominant feature of the landscape (Fig. 3.1).

There is a region, outside the horizon but close to it, where the dragging is so strong that *nothing* can hold still — to hover without circulating you would need to move faster than light. This region is the *ergosphere*, a flattened shell thickest over the equator (Fig. 3.2). Objects inside it can still escape — it is not a point of no return — but they cannot avoid being swept around.

3.3 The anatomy of a spinning hole

Fig. 3.2 labels the landmarks that Part II computes exactly, for a non-spinning and a rapidly spinning hole side by side.

- The *event horizon* (black): the point of no return. Spin shrinks it, from $2\,M$ at $a_* = 0$ toward $1\,M$ at the extreme.
- The *ergosphere* (orange): the no-standing-still zone, touching the horizon at

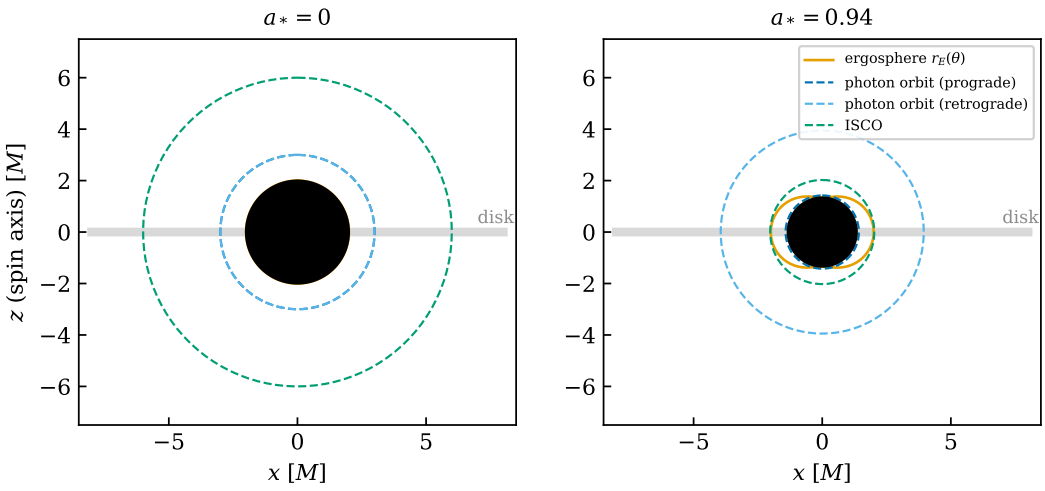


Figure 3.2: Cross-sections through a non-spinning ($a_* = 0$) and a rapidly spinning ($a_* = 0.94$) black hole, drawn to scale from the exact formulas of Chapter 8. Spin shrinks the horizon, opens up an ergosphere, splits the photon orbit into prograde/retrograde pairs, and pulls the innermost stable orbit (ISCO) dramatically inward.

the poles and bulging at the equator.

- The *photon orbits* (blue): radii where light can circle the hole. For a spinning hole there are two equatorial ones — a closer orbit for light moving *with* the rotation (*prograde*) and a farther one for light moving *against* it (*retrograde*). Spacetime's whirlpool helps the co-rotating light hold a tighter circle.

- The *ISCO* (green), the innermost stable circular orbit: the closest circle on which matter can orbit steadily. Any closer and the slightest nudge sends it spiralling in. The ISCO is the inner edge of the glowing disk of Chapter 4, and it moves inward spectacularly with spin — from $6\,M$ for $a_* = 0$ to about $2\,M$ at $a_* = 0.94$. Much of what changes visually with spin in our renders traces back to this single fact.

Worked example: Spin landmarks with a pocket calculator

Part II derives every radius in Fig. 3.2; here is a taste of how cheaply the first one comes. With spin, the horizon radius becomes (in units of the gravitational radius M of Chapter 1, in which the no-spin horizon sits at $2\,M$):

$$r_+ = M\left(1 + \sqrt{1 - a_*^2}\right).$$

For Sgr A*'s $a_* = 0.94$: $a_*^2 = 0.884$, so $\sqrt{1 - 0.884} = \sqrt{0.116} = 0.341$, and $r_+ = 1.34\,M$ — the horizon has shrunk by a third from its no-spin size. The other landmarks take a few more lines of the same arithmetic (Chapter 8 shows all of it): the equatorial light orbits land at $1.43\,M$ (prograde) and $3.95\,M$ (retrograde) — split wide apart by the whirlpool — and the ISCO at $2.02\,M$, barely outside the horizon. Those numbers are exactly what Fig. 3.2's right panel draws.

3.4 Where spin comes from, and how anyone measures it

Two natural questions before the shadow. *Why* should Sgr A* spin fast? Because it eats: gas spiralling in through a disk (Chapter 4) arrives carrying rotation, and every meal torques the hole toward faster spin. A classic calculation shows that steady disk-feeding drives the spin toward — but never past — about $a_* = 0.998$, which is why that particular number appears as the "near-extremal" column of the tables in Part II. Black holes born from mergers or fed chaotically can end up anywhere on the dial, so the spin of any individual hole is an empirical question.

And *how* is a spin ever measured? Never directly — no surface, no visible rotation. Every method reads the spin through the ISCO of the previous section: the disk's inner edge sits at the ISCO, the ISCO's radius depends on spin, and so anything sensitive to how deep the disk reaches carries a spin signature. For stellar-mass black holes in X-ray binaries, astronomers fit the temperature of the disk's glow or the shape of an iron spectral line bent by the deep orbits. For Sgr A*, none of these tools yet works cleanly: the ring *size* barely depends on spin (this chapter, third fact below), and the star S2 orbits far too high to feel it (Chapter 17). The $a_* = 0.94$ used throughout this report is therefore an adopted *working value* — a high spin consistent with what current modelling prefers, chosen so that spin-dependent effects are prominent in the renders — and

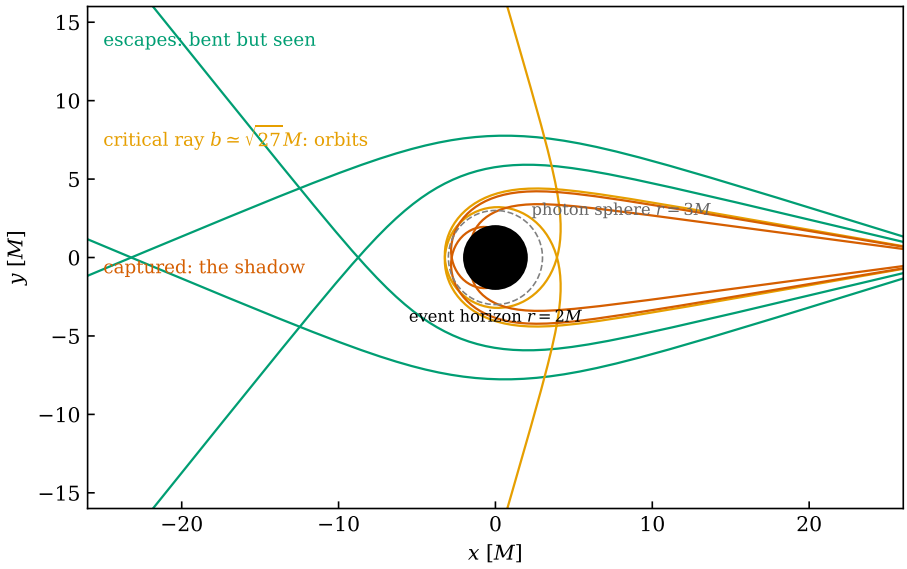


Figure 3.3: How the shadow forms, computed from the exact bending equation. Green rays escape (bent); the orange critical ray circles on the photon sphere; red rays are captured. The shadow is the set of camera directions whose rays end on the horizon — its edge is light that grazed the photon sphere.

Chapter 19 is careful to show which conclusions would and would not change at other spins.

3.5 The shadow and the photon ring

Now surround the hole with light — a glowing disk, or simply the stars behind it — and ask what a distant camera sees. Chapter 2 showed that rays aimed too close are captured. Point the camera at the hole, and there is a patch of sky, for every pixel of which the answer is "that line of sight ends on the horizon". That dark patch is the *shadow* (Fig. 3.3).

Three facts about the shadow do a great deal of work in this report.

First, *the shadow is bigger than the horizon.* Light does not have to hit the horizon to be lost — anything aimed within the critical distance spirals in. For a non-spinning hole the shadow's radius is $\sqrt{27} \approx 5.2\,M$, about 2.6 times the horizon radius. Our simulator reproduces this number to sixteen digits (Chapter 11), which is the sort of check that separates a scientific instrument from a pretty picture.

Second, *the shadow's edge is made of light that almost stayed.* Rays that graze the critical boundary can circle the hole once, twice, three times before escaping to the camera. Each extra half-turn stacks another, exponentially thinner and fainter image of the whole sky (and of the disk) right at the shadow's rim. This onion of nested images is the *photon ring*, and its sub-rings are analysed quantitatively in Chapter 13. The Event Horizon Telescope's measurement of this ring for Sgr A* — $51.8 \pm 2.3\,\mu$as across — is one of the two observational anchors of this whole project.

Third, *spin distorts the shadow.* Frame dragging lets prograde light survive closer in, so the shadow's edge on the side rotating toward you pulls inward: the silhouette flattens on one side and shifts sideways, by an amount our simulator measures precisely (Chapters 11, 19). Viewed nearly pole-on — our actual view of Sgr A* — the distortion is frustratingly small, a few per cent, which is why the ring's *size* alone cannot pin down the spin (the "spin–inclination degeneracy" that Chapter 19 maps out).

Worked example: From shadow size to sky size

Two numbers from this chapter already predict what the Event Horizon Telescope should see. The no-spin shadow radius is $\sqrt{27}\,M = 5.20\,M$, a diameter of $10.4\,M$. Chapter 19 works out the sky conversion for Sgr A*: one M subtends 5.1243 μas from Earth. A first estimate of the shadow on the sky is therefore

$$10.4 \times 5.12 \approx 53\ \mu\text{as}.$$

Feeding in the spin and our nearly pole-on viewing angle reshapes and slightly shrinks the outline: the full Kerr calculation of Chapter 19 gives 49.68 μas. Both sit beside EHT's measured ring of $51.8 \pm 2.3\ \mu$as — a pocket-calculator estimate from two formulas lands within a few per cent of a planet-spanning radio measurement. The remaining per-cent-level work is what the rest of this report is for.

Terms you just met.

Kerr metric The exact spacetime of a spinning black hole (Kerr 1963); fully specified by mass and spin.

Spin parameter a_* Dimensionless spin from 0 to 1; Sgr A* is modelled at $a_* = 0.94$.

Frame dragging The whirlpool-like circulation of spacetime around a spinning mass.

Ergosphere The region where standing still is impossible; escape is still possible from inside it.

Prograde / retrograde Moving with / against the black hole's rotation; prograde orbits survive closer in.

Photon sphere / photon orbits Radii at which light circles the hole; unstable, so grazing light eventually escapes or falls.

ISCO Innermost stable circular orbit — the inner edge of the disk; moves from $6\,M$ (no spin) toward the horizon as spin grows.

Shadow The dark patch of captured lines of sight; radius $\sqrt{27}\,M$ for no spin, subtly reshaped by spin.

Photon ring The stack of exponentially fainter images of the scene hugging the shadow's edge.

→ *Full treatment: Kerr geometry and all characteristic radii in Chapter 8; the traced shadow and critical curve in Chapter 11; sub-ring measurements in Chapter 13.*

Chapter 4

Light, Heat, and Colour: Reading the Disk

4.1 Hot things glow, and colour is a thermometer

Heat a poker in a fire and it glows dull red; hotter, and it turns orange, then yellow, then blue-white. This everyday sequence is one of the most useful facts in astrophysics: a dense hot body glows with a spectrum — a recipe of colours — that depends on its temperature and on almost nothing else. Physicists call this ideal glow *blackbody radiation*, and the temperature-to-colour dictionary is exact: around 3 000 degrees looks ember-orange, 6 000 degrees looks white like the Sun's surface, 20 000 degrees looks blue-white. The renders in this report use precisely this dictionary — every pixel's colour comes from a computed temperature pushed through the same colour-matching tables that describe human vision (Chapter 12), not from an artist's palette.

Worked example: Wien's thermometer

The dictionary's headline entry is Wien's displacement law: a hot body at temperature T glows most brightly at the wavelength

$$\lambda_{\text{peak}} = \frac{2.90 \times 10^{6}\ \text{nm K}}{T}.$$

The Sun's surface, at $T \approx 5\,800$ K, peaks at $2.90 \times 10^{6}/5\,800 = 500$ nm — blue-green light; the Sun looks white only because the glow is a broad mixture centred there. A 3 000 K ember peaks at 970 nm, in the infrared — our eyes catch only its red tail, hence "red-hot". A 20 000 K surface peaks at 145 nm, deep in the ultraviolet, and the visible scraps of that spectrum read as blue-white. Every rendered pixel in Part III runs this logic in exact form: temperature in, full spectrum out, folded through the eye's three colour sensitivities (Chapter 12).

4.2 The accretion disk

Gas falling toward a black hole almost never falls straight in. Like water circling a drain, it has some rotation, so it settles into a flattened, swirling *accretion disk*. Within the disk, gas at each radius orbits at its own rate — inner gas circles faster than outer gas, just as Mercury outpaces Neptune — and the resulting internal friction does two things: it lets the gas drift gradually inward, and it heats the disk white-hot. The energy released is prodigious. Matter spiralling through a disk onto a rapidly spinning black hole can radiate away tens of per cent of its entire rest-mass energy — compared with less than one per cent for nuclear fusion. Our simulator computes this efficiency exactly (it is the η of Chapter 8): about 5.72 per cent for a non-spinning hole, rising to 17.96 per cent at Sgr A*'s spin, because the disk's inner edge (the ISCO of Chapter 3) reaches so much deeper into the well.

The model disk used throughout Part III is the classic *Novikov–Thorne* thin disk: geometrically thin like a vinyl record, hottest just outside the ISCO, cooling outward. It is an idealisation — the real flow around Sgr A* is a hot, puffed-up, far dimmer affair — and Chapter 22 is careful about which conclusions survive that difference. The disk model gives our images their brightness and colour; crucially, the *geometry* of shadow and photon ring comes from the spacetime itself, not from the disk details.

4.3 Why one side of the disk blazes

Every published image of a black-hole disk — the EHT rings, *Interstellar*'s Gargantua, the renders in this report — is conspicuously brighter on one side. That asymmetry is a speed effect, not a lighting effect.

The inner disk orbits at an appreciable fraction of the speed of light. When a source of light moves toward you that fast, three effects conspire: its light is squeezed forward like spray from a moving hose (*beaming*), arrives blueshifted, and arrives at a faster clip. The approaching side of the disk therefore appears far brighter — in our reference render at $a_* = 0.94$, viewed nearly edge-on, the approaching limb outshines the receding one by a factor of 9.60 in total light (Chapter 12). The receding side is correspondingly dimmed and reddened. The whole effect is packaged in a single number per light ray, the *redshift factor* g: the ratio of the light's frequency as received to its frequency as emitted. Brightness scales as g^4 — the fourth power — which is why the asymmetry is so dramatic. If a simulated disk ever looks symmetric, its g^4 is wrong; this was one of our hard validation gates (Chapter 21).

Worked example: Why the fourth power is dramatic

Let one limb of the disk approach so that its light arrives with all frequencies raised twenty per cent ($g = 1.2$), and the other limb recede with them lowered twenty per cent ($g = 0.8$). Twenty per cent sounds modest, but brightness follows g^4:

$$\frac{(1.2)^4}{(0.8)^4} = \left(\frac{1.2}{0.8}\right)^4 = 1.5^4 \approx 5.1$$

— a five-fold contrast from a twenty-per-cent frequency shift. In our reference render of Sgr A*'s disk the factor g actually spans 0.139 (deep, receding, gravitationally dimmed gas) to 1.440 (the fast approaching limb), so the most extreme pixel pair differs in g^4 by a factor of order ten thousand; integrated limb against limb, the measured contrast is 9.60× (Chapter 12). This is why no honest render of an orbiting disk can ever look symmetric.

4.4 Reading a single line of light

Everything Part III learns from S2's speed comes through one narrow feature in its spectrum. A star's atmosphere absorbs light at razor-sharp, fixed wavelengths set by atomic physics; the workhorse here is a hydrogen line called Brγ ("Brackett

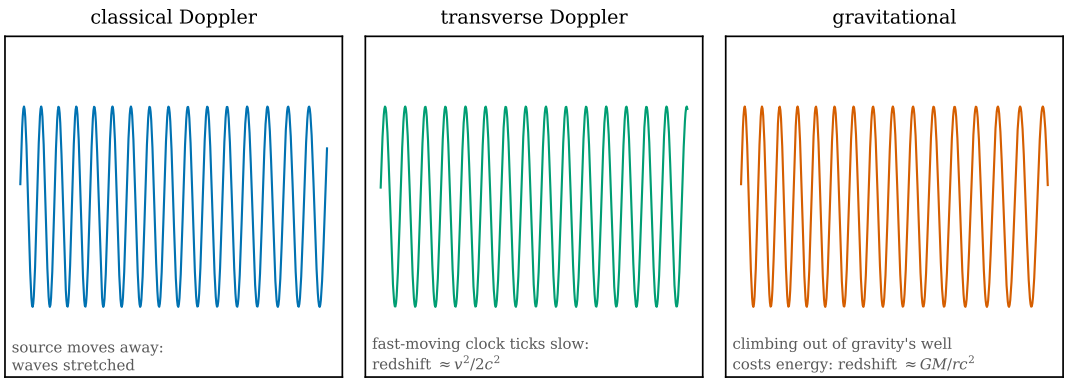


Figure 4.1: The three ways a wavelength gets shifted. Classical Doppler: motion along the line of sight stretches or compresses the wave train. Transverse Doppler: a fast clock ticks slowly, so even sideways motion reddens the light — a purely relativistic effect. Gravitational: light climbing out of a gravity well arrives stretched.

gamma"), at 2.16612 micrometres in the infrared. Two facts make it the right tool. First, its rest wavelength is known to sub-km/s equivalent precision from laboratory physics, so any measured offset is pure motion (or gravity). Second, it sits in the infrared "K band", where the thirty magnitudes of dust between us and the Galactic centre — which extinguish visible light by a factor of 10^{12} — thin to a mere factor of ten: toward Sgr A*, the infrared is the only window there is.

The measurement itself is conceptually a ruler read: disperse the star's light, find the line's centre, and convert the fractional wavelength offset into a speed (1 part in 10^5 of the wavelength $\approx$ 3 km/s). When Part III quotes "radial velocity $\pm$30 km/s", that error bar is the practical limit of finding a slightly smeared line-centre on a faint star — and when the relativistic terms below add their $\sim$200 km/s at pericentre, they move the line by seven such error bars: comfortably measurable, which is exactly why the 2018 pericentre passage was such an occasion.

4.5 The three redshifts

That factor g mixes three physically distinct effects, and pulling them apart is exactly what the S2 measurements of Part III require (Fig. 4.1).

1. *Classical Doppler*: the familiar ambulance-siren shift. Motion away from you stretches the waves (redshift); motion toward you compresses them (blueshift). This is by far the largest term for an orbiting star, and Newton would have predicted it too.

2. *Transverse Doppler*: special relativity's addition. A fast mover's clock runs slow, so its emitted waves come out at a reduced rate — a redshift that appears even when the motion is purely sideways. It scales with speed squared, so it only matters when things get truly fast.

3. *Gravitational redshift*: general relativity's addition, from Chapter 2 — light climbing out of a gravitational well is stretched. It scales with depth in the well.

For the star S2 at its 2018 closest approach, the first term is several thousand km/s, while the second and third are each worth only about a hundred km/s — and, by a neat algebraic coincidence worked out in Chapter 15, almost exactly equal to each other. The GRAVITY collaboration's 2018 detection of that little two-hundred-km/s surplus over the Newtonian prediction was the first direct test of general relativity beside a supermassive black hole. Reproducing it, with no tunable knob, is one of this report's headline results: our model peaks at 205 km/s at epoch 2018.379 (Chapter 15).

Worked example: The two small redshifts at S2's pericentre

Both relativistic terms can be sized on the back of an envelope. At closest approach S2 moves at $v \approx 7\,700$ km/s (Chapter 5 computes this from the orbit), so $v/c = 0.0256$. The transverse-Doppler redshift is half the square of that: $z_{\rm trans} = \frac{1}{2}(0.0256)^2 = 3.3 \times 10^{-4}$, worth ≈ 98 km/s. The gravitational redshift was estimated in Chapter 2: 3.5×10^{-4}, or ≈ 106 km/s. Their sum, ≈ 204 km/s, is the extra redshift no Newtonian analysis can produce — and the full model of Chapter 15 indeed peaks at 205 km/s. The near-equality of the two terms is no accident of arithmetic: at the pericentre of a very eccentric orbit the speed is close to the free-fall speed, $v^2 \approx 2Gm/r$, which forces $\frac{1}{2}v^2/c^2$ to sit close to $Gm/(c^2 r)$ automatically.

Terms you just met.

Blackbody radiation The temperature-determined glow of a dense hot body; the exact temperature-to-colour dictionary used in our renders.

Accretion disk The flattened swirl of hot gas spiralling into a black hole; modelled here as the Novikov–Thorne thin disk.

Accretion efficiency (η) The fraction of rest-mass energy radiated by matter spiralling in: 5.72% (no spin) to 17.96% (Sgr A*'s spin).

Doppler beaming The forward-throwing of light by a fast-moving source; makes the approaching disk limb $\sim$9.60$\times$ brighter in our reference render.

Redshift factor g Received frequency over emitted frequency for one light ray; observed brightness scales as g^4.

Classical / transverse / gravitational redshift The three distinct wavelength shifts; separating them at S2's pericentre is the GRAVITY 2018 test.

→ Full treatment: disk physics and the g^4 law in Chapter 12; the redshift decomposition and the S2 result in Chapters 14 and 15.

Chapter 5

Orbits, S-Stars, and Weighing a Black Hole

5.1 Kepler's ellipse

Four hundred years before anyone imagined black holes, Johannes Kepler discovered that planets travel on *ellipses* — squashed circles — with the Sun at one *focus*, a point offset from the centre (Fig. 5.1). Three of his terms are used on nearly every page of Part III:

- the *semi-major axis* a — half the long diameter, the orbit's overall size;
- the *eccentricity* e — how squashed and off-centre the ellipse is, from 0 (circle) toward 1 (a stretched sliver);
- the *pericentre* and *apocentre* — the points of closest and farthest approach. (For orbits around stars you may know these as perihelion and aphelion.)

Kepler's second discovery is visible in the dots of Fig. 5.1: bodies move *fast near pericentre and slow near apocentre.* His third links the orbit's size and period to the central mass — bigger orbits take longer, and, decisively for us, *at fixed size, a heavier central mass means a faster orbit.* Measure the size and the period of any orbit and the central mass falls out. This is how we weigh everything in astronomy, from planets to galaxies — and it is exactly how Sgr A* is weighed.

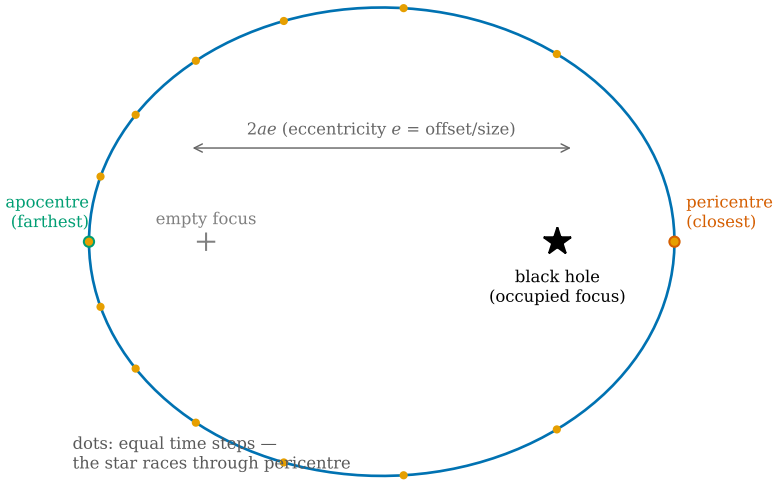


Figure 5.1: The anatomy of a Kepler orbit. The attracting mass sits at one focus; eccentricity measures how off-centre it is. The dots mark equal time steps: the orbiting body loiters near apocentre and races through pericentre.

5.2 The S-stars, and S2 above all

Buried in the central light-month of the Galaxy is a swarm of ordinary young stars on extraordinary orbits — the *S-stars*. Since 1992, two teams (in Garching and in Los Angeles) have photographed the region every year, watching individual stars trace out their ellipses around an invisible focus. The star *S2* is the jewel of the collection: bright enough to measure precisely, with a period of just 16 years — short enough that we have now watched it complete more than one and a half full loops. Its orbit is extreme by any planetary standard: eccentricity 0.88, pericentre speed near 7 700 km/s (2.6 % of the speed of light), diving to about 1 500 horizon radii from the black hole.

This report fits the complete published S2 data set — 145 positions and 44 velocities spanning 1992–2016 (Chapter 15) — plus four more S-stars in a joint fit (Chapter 16).

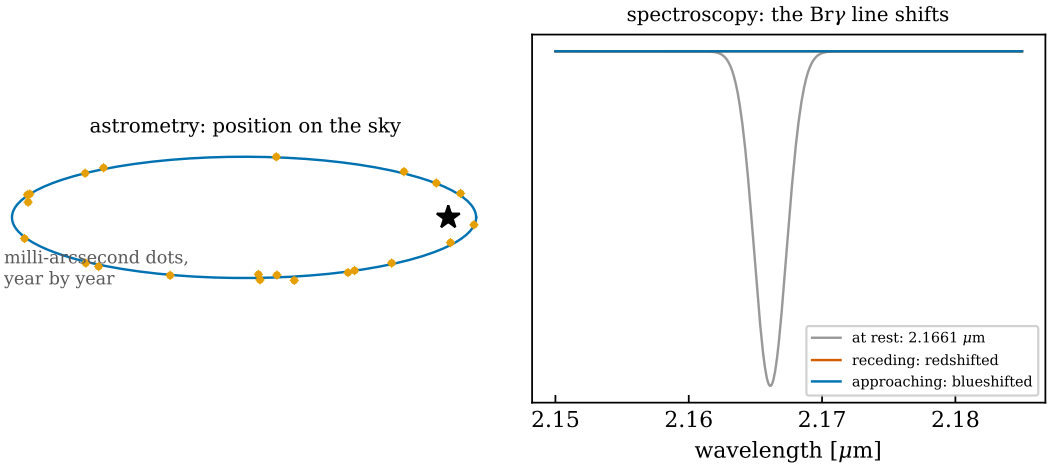


Figure 5.2: The two observation channels. Left — astrometry: the star's position on the sky, measured year after year in milliarcseconds, traces the projected ellipse. Right — spectroscopy: the wavelength of a known line in the star's light shifts with its line-of-sight speed (the Doppler effect of Chapter 4).

Worked example: How fast is S2 at pericentre?

Newton's toolkit — accurate to about a per cent even this close to a black hole — gives the speed anywhere on a Kepler orbit through the *vis-viva* relation, $v^2 = Gm\,(2/r - 1/a)$. There is a convenient set of units: with r and a in au and v in km/s, Gm for one solar mass is 887 (because the Earth, at $r = a = 1$ au, orbits at 29.8 km/s). For Sgr A*, multiply by 4.27 million. S2's orbit has $a \approx 1\,033$ au and eccentricity $e = 0.883$, putting pericentre at $r_p = a(1-e) = 121$ au:

$$v_p^2 = 4.27 \times 10^6 \times 887 \times \left(\tfrac{2}{121} - \tfrac{1}{1033}\right) = 5.9 \times 10^7 \ (\mathrm{km/s})^2,$$

so $v_p \approx 7\,700$ km/s — the 2.6% of light speed quoted above. The same formula at apocentre ($r_a = a(1+e) = 1\,944$ au) gives just 480 km/s: sixteen times slower. That factor is Kepler's second law made vivid, and it is why the dots of Fig. 5.1 bunch up at the far end of the ellipse.

5.3 Two channels: positions and speeds

Everything we know about the S-stars comes through two complementary kinds of measurement (Fig. 5.2).

Astrometry is simply measuring where the star is on the sky, epoch by epoch —

in milliarcseconds, using adaptive-optics cameras on the world's largest telescopes and, since 2017, the GRAVITY interferometer, which links four 8-metre telescopes into one virtual 130-metre one. Astrometry sees the orbit *projected* on the sky, like the shadow of a hoop tilted against a wall.

Spectroscopy measures how fast the star moves *along* the line of sight, via the Doppler shift of a hydrogen absorption line in its spectrum (the Brγ line, in the infrared where the Galactic dust is transparent). This "radial velocity" channel sees exactly the component of motion that astrometry cannot.

Neither channel alone determines everything; together they overdetermine the orbit. And one beautiful consequence deserves its own sentence: because astrometry gives the orbit's *angular* size while the velocities give its *physical* scale, combining them yields the *distance to the Galactic centre* — a geometric distance measurement across 26 000 light-years with no cosmic distance ladder involved. Our fit recovers both the mass and this distance (Chapter 15).

Worked example: Weighing Sgr A* in three lines

The fitted S2 orbit of Chapter 15 has angular semi-major axis 124.82 milliarcseconds, period 16.0561 years, and the distance comes out at 8272 parsecs. Kepler's third law then weighs the black hole with startling economy. First convert the angle to a physical length — this is what the parsec is *for*: one au subtends exactly one arcsecond at one parsec, so

$$a = 0.12482'' \times 8\,272 \text{ pc} = 1\,033 \text{ au}.$$

Second, apply Kepler's third law in solar units (mass $= a^3/P^2$, with a in au, P in years, mass in Suns):

$$m = \frac{(1\,033)^3}{(16.06)^2} = \frac{1.10 \times 10^9}{258} \approx 4.27 \text{ million solar masses}.$$

Three lines of arithmetic land within a fraction of a per cent of the full relativistic fit, 4.270 ± 0.195 million (Chapter 15) — because S2 spends most of its 16 years in territory where Newton is an excellent approximation. Everything beyond these three lines — precession, redshift excess, the no-hair test — lives in the small, precious relativistic corrections.

5.4 How anyone measures a milliarcsecond

The precision behind Fig. 5.2 deserves a moment of awe, and a sketch of how it is possible at all. Starlight reaching a ground-based telescope has been rumpled

by the atmosphere: without correction, any star blurs into a dancing patch about one arcsecond across — a thousand times coarser than the measurements Part III fits. Three inventions close the gap. *Adaptive optics* measures the atmosphere's rumpling several hundred times a second (using a bright reference star, or a laser-created artificial one) and bends a deformable mirror to cancel it, restoring the telescope to its physics-limited sharpness — for an 8-metre telescope in the infrared, about 57 milliarcseconds. *Precision centroiding* then exploits the fact that locating the *centre* of a sharp round blob is far easier than resolving detail inside it: with enough photons, the centre is pinned to a hundredth of the blob's width or better — sub-milliarcsecond. Finally, *interferometry* links telescopes together: the GRAVITY instrument combines the four 8-metre VLT telescopes, 130 metres apart, into one virtual aperture whose sharpness scales with the 130 metres rather than the 8 — reaching tens of *micro*arcseconds on S2. The data fitted in Part III spans this whole instrumental history, which is why the published error bars range from several milliarcseconds (1990s imaging) down to fractions of one (recent epochs), and why an honest fit must weight each epoch by its own uncertainty — the χ^2 machinery of Chapter 6.

5.5 Einstein's fingerprints on the orbit

If gravity were exactly Newtonian, S2's ellipse would repeat itself forever, fixed in space. General relativity disagrees in two measured ways.

First, the *redshift excess* of Chapter 4: at the 2018 pericentre passage, S2's spectral lines arrived about 200 km/s redder than any Newtonian orbit could explain — time dilation plus gravitational redshift, detected by GRAVITY in 2018.

Second, *orbital precession*: the ellipse itself slowly pivots. Each orbit, the pericentre direction advances by a small angle, so the path traces a rosette rather than a closed ring (Fig. 5.3). For Mercury this famous advance is 43 arcseconds per century and was the first triumph of general relativity; for S2 it is about 12 *arcminutes* per 16-year orbit — some fifty thousand times faster — and GRAVITY confirmed it in 2020. Our simulator reproduces both effects with no adjustable amplitudes: the fitted mass and geometry force them (Chapter 15).

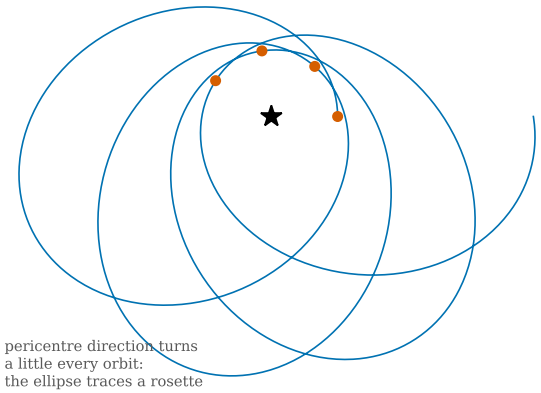


Figure 5.3: Relativistic precession: the orbital ellipse pivots a little each revolution, tracing a rosette. The dots mark successive pericentre directions. (Exaggerated for visibility: S2's advance is 12′ per orbit; this drawing uses thousands of times more.)

Worked example: The rosette angle

The pericentre advance per orbit has a closed form, derived in Chapter 14:

$$\Delta\varpi = \frac{6\pi\, Gm}{c^2\, a\,(1-e^2)}.$$

Every ingredient is already on the table. The mass-as-length: $Gm/c^2 = 4.27 \times 10^6 \times 1.48\ \text{km} = 0.0421$ au. The denominator: $a(1-e^2) = 1\,033 \times (1 - 0.883^2) = 228$ au. So

$$\Delta\varpi = \frac{6\pi \times 0.0421}{228} = 3.5 \times 10^{-3}\ \text{radians} = 0.20^\circ \approx 12\ \text{arcminutes per orbit},$$

the very advance GRAVITY confirmed in 2020 (the full model gives 11.98′; Chapter 15). Mercury's celebrated 43 arcseconds per *century* is this same formula fed with one solar mass and Mercury's orbit — S2's advance per unit time is some fifty thousand times larger, which is what makes a sixteen-year experiment possible in a human career.

5.6 Thirty-four years in one diagram

Fig. 5.4 places the milestones — from the first infrared images of the S-stars to the EHT picture of the ring — on one line, ending with the simulator this

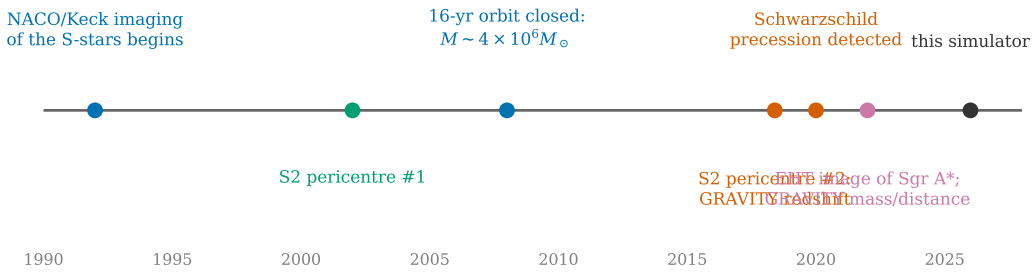


Figure 5.4: Milestones of the Galactic-centre programme covered in this report.

report describes: a single instrument that reproduces, from first principles, the measurements of that whole timeline.

Terms you just met.

Ellipse, focus, semi-major axis, eccentricity The shape language of orbits (Fig. 5.1).

Pericentre / apocentre Closest and farthest points of an orbit; S2's pericentre is ∼120 au at 2.6% of light speed.

S-stars, S2 The stars orbiting Sgr A*; S2, with its 16-year period, is the most precisely measured.

Astrometry Measuring sky positions over time (milliarcseconds); gives the projected orbit.

Radial velocity (spectroscopy) Measuring line-of-sight speed from Doppler-shifted spectral lines.

Geometric distance Combining both channels yields the distance to the Galactic centre directly.

Precession / rosette The slow pivot of the orbital ellipse predicted by GR; 12′ per orbit for S2, confirmed 2020.

→ *Full treatment: the relativistic orbit model in Chapter 14; the fit and both GR effects in Chapter 15; the multi-star fit in Chapter 16.*

Chapter 6

The Toolkit in Plain Words

This last primer chapter collects the working tools — units, statistics, and one big theorem — that the technical chapters use constantly.

6.1 Natural units: setting $G = c = 1$

Physicists working with black holes almost always set the gravitational constant G and the speed of light c equal to 1. This is not a trick or an approximation; it is a choice of units, exactly like measuring road distances in hours of driving. Once you do it, mass, length, and time all become interchangeable, and a single symbol M can serve as the yardstick for everything: lengths are quoted in units of the gravitational radius ("the ISCO is at $6\,M$"), times in units of the light-crossing time of that radius. For Sgr A*, one M of length is about 0.042 au and one M of time is about 21 seconds. The huge practical benefit is universality: every formula and every rendered image in Parts II–III applies to *any* black hole; only the final conversion to kilometres, seconds, or microarcseconds depends on the actual mass (that conversion is Chapter 19).

Worked example: One M, three ways

For Sgr A*, the yardstick M — the gravitational radius Gm/c^2 — evaluates, in everyday units, to

$$M = 4.297 \times 10^6 \times 1.48 \text{ km} \approx 6.35 \times 10^9 \text{ m} = 0.0424 \text{ au},$$

using the handy fact that one solar mass equals 1.48 km of length. As a *time*, divide by c once more: $6.35{\times}10^9/3.00{\times}10^8 = 21.2$ seconds — the natural tick of Sgr A* (one orbit at the innermost stable circle takes about two dozen ticks, some eight and a half minutes). As an *angle* seen from Earth, divide instead by the distance, 8 277 pc $= 2.554 \times 10^{20}$ m, giving 2.49×10^{-11} radians; radians convert to microarcseconds via $\times\, 2.063 \times 10^{11}$, so one M subtends 5.12 μas — the 5.1243 μas per M used throughout Chapter 19. Mass as kilometres, seconds, and microarcseconds: one number, three costumes.

A related habit: Part II speaks of *vectors* (arrows with direction and length — a velocity is one) and occasionally *tensors*. For reading this report it is enough to know that a tensor is a machine that eats vectors and returns numbers, in a way that all observers agree on. The metric of Chapter 2 is the only tensor we ever really need: it eats two little coordinate steps and returns their true separation.

6.2 Error bars, and the meaning of χ^2

Every real measurement comes with an uncertainty — an *error bar* — and the whole craft of fitting models to data is bookkeeping those error bars honestly. The universal bookkeeping device is called *chi-squared* (χ^2), and it answers one question: *by how many error bars, in total, does the model miss the data?* (Fig. 6.1). For each data point, take the miss, divide by that point's error bar, square it (so misses in either direction count alike, and big misses count a lot), and add up over all points.

Two rules of thumb make χ^2 readable. A model that is right, fitted to honest error bars, should miss each point by about one error bar on average — so χ^2 divided by the number of data points (the *reduced* χ^2) should be near 1. Much bigger means the model (or the error bars) are wrong; much smaller means the error bars were pessimistic. Our headline S2 fit lands at $\chi^2_\nu = 0.959$ (Chapter 15) with the published error bars used exactly as given.

And once the best knob-settings are found, *how well known* is each knob? Wiggle a knob away from its best value and χ^2 climbs; the range over which

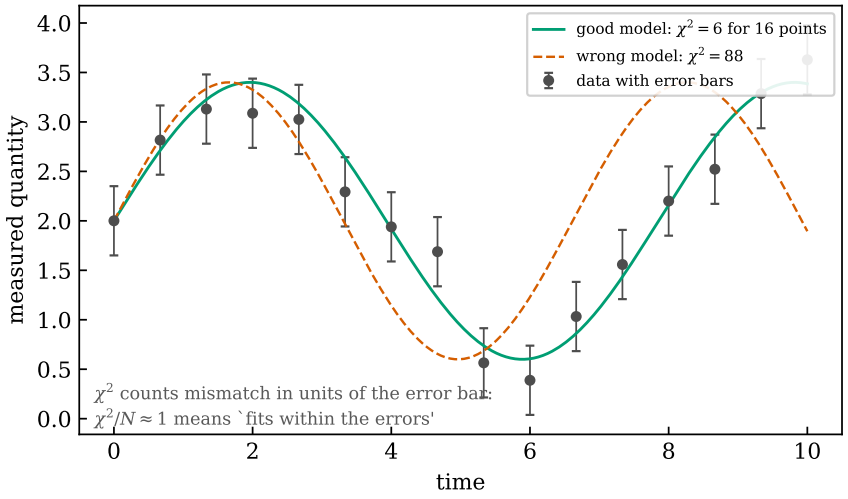


Figure 6.1: χ^2 made visible. Two candidate models through the same data: the good one misses each point by about one error bar ($\chi^2 \approx$ number of points); the wrong one racks up an enormous total. Fitting means turning model knobs until χ^2 is as small as it gets.

it climbs by less than a set amount is the knob's *confidence interval* — the "±" quoted after every fitted number in this report. When we write $M = 4.270 \pm 0.195$ million suns, the ± is precisely such an interval.

Worked example: A five-point χ^2, by hand

Toy data: five measurements with error bars, and a candidate model curve. Suppose the model misses the five points by 0.8, −1.2, 0.3, 1.5, −0.6 error bars respectively. Then

$$\chi^2 = 0.64 + 1.44 + 0.09 + 2.25 + 0.36 = 4.8,$$

about one per point ($\chi^2_\nu \approx 1$): a healthy fit. Now slide the whole curve up by two error bars. The misses become 2.8, 0.8, 2.3, 3.5, 1.4, and

$$\chi^2 = 7.84 + 0.64 + 5.29 + 12.25 + 1.96 = 28$$

— nearly six per point. No committee is needed to reject the second model; the "ruling" is nothing more than this arithmetic done honestly. The real fits of Chapter 15 do exactly the same sum, with hundreds of measurements and thirteen knobs.

6.3 When two knobs move together: degeneracy

One statistical idea deserves its own section, because it shapes the headline results of Part III. Sometimes two model knobs affect the data in nearly the same way, so turning one up and the other down leaves the fit almost unchanged: the knobs are *degenerate*, and the data constrain some *combination* of them tightly while leaving the individual values loose.

The mass of Sgr A* and our distance to it are the canonical example. The star's *angular* orbit (astrometry) fixes only how mass and distance-cubed combine — a heavier hole farther away draws the same picture on the sky. The star's *speeds* (spectroscopy) fix a different combination — a heavier hole farther away must move its stars faster. Each channel alone leaves a whole family of acceptable $(M, \text{distance})$ pairs; only together do they pick the single crossing point. Even then, the crossing is at a shallow angle: the allowed region is not a neat circle but a long, tilted ellipse, and quoting "$M = 4.270 \pm 0.195$" without also stating the distance assumption would quietly misrepresent what is known. The corner-plot figures of Chapter 15 exist precisely to display such ellipses; the spin–inclination degeneracy of the ring size (Chapter 3) is the same phenomenon wearing imaging clothes. When a later chapter says a parameter direction is "weakly constrained", this picture — a long valley in the χ^2 landscape — is always what is meant.

6.4 MCMC: mapping uncertainty by wandering

For a model with thirteen knobs, the surface of χ^2 over all knob combinations is a thirteen-dimensional landscape, and simple wiggling can miss how knobs trade off against one another. The heavy machinery for this is *Markov chain Monte Carlo* (MCMC): release a crowd of random walkers into the landscape, with rules that make them drift toward — and then wander within — the low-χ^2 valleys (Fig. 6.2). After enough steps, *where the walkers spend their time is itself the answer*: dense regions are probable parameter values, sparse regions improbable. Histograms of the walkers' positions give every parameter's uncertainty and every pairwise trade-off at once. Chapter 15 runs exactly this (32 walkers, thousands of steps, the full relativistic model evaluated at every step) and checks that the walker map agrees with the simpler wiggle estimate.

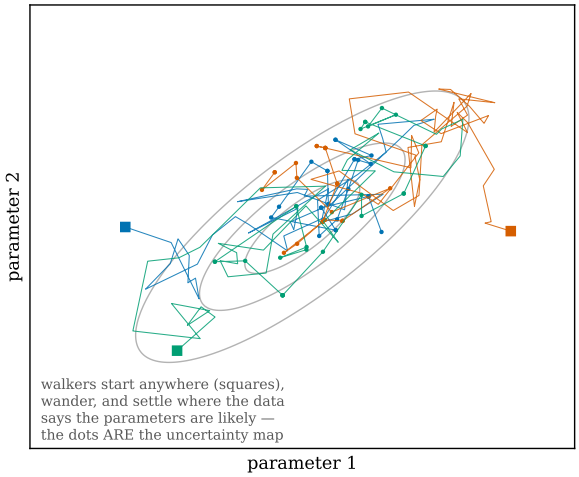


Figure 6.2: Three MCMC walkers released onto a two-parameter landscape (grey contours: the low-χ^2 valley). After an initial wander-in, the walkers trace out the valley; their footprints map the uncertainty, including the diagonal trade-off between the two parameters.

6.5 The no-hair theorem in one picture

One theorem gives this project its sharpest question. General relativity predicts that a black hole, once settled, is the simplest macroscopic object in nature: its exterior gravity is *completely* fixed by two numbers, mass and spin. All other detail of whatever fell in — its shape, its chemistry, its history — is radiated away or swallowed. The theorem's nickname says it: *black holes have no hair*.

This is falsifiable. Any rotating body has a *quadrupole moment* — a measure of how squashed its gravity field is (Fig. 6.3). For planets and stars the squashing is free to vary; for a Kerr black hole it is locked rigidly to the mass and spin. So the test is: measure mass, spin, and quadrupole *independently*, and check the lock. We implement this by adding one knob to the spacetime — a quadrupole surplus called ε_q, zero for a true Kerr hole — and asking the S2 data how far from zero it could be (Chapter 17). The honest answer today: the data are fully consistent with zero, with room to spare (a single star at 1 500 horizon radii barely feels the quadrupole), and the same machinery is ready for the closer stars that future instruments will find.

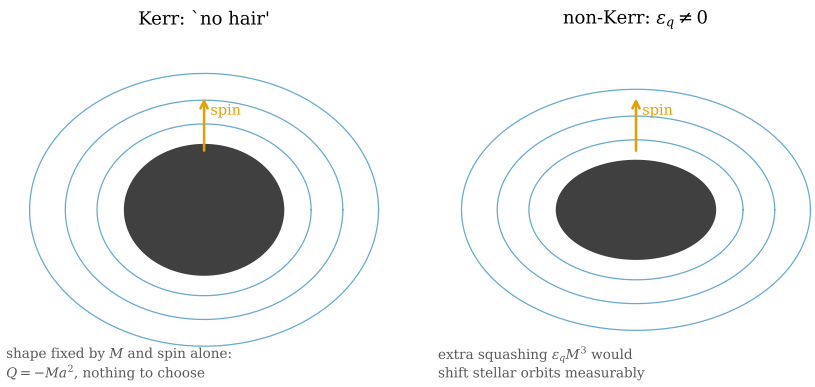


Figure 6.3: The no-hair test in one picture. Left: for a Kerr black hole the field's squashing (quadrupole) is fixed by mass and spin — nothing to choose. Right: if the measured squashing deviated ($\varepsilon_q \neq 0$), stellar orbits would precess differently, and Kerr would be falsified.

Terms you just met.

Geometric units ($G = c = 1$) Measuring everything in units of the black hole's own scale M; for Sgr A*, $1\,M \approx 0.042$ au ≈ 21 s.

Tensor A machine that eats vectors and returns observer-independent numbers; the metric is the one that matters here.

χ^2 / reduced χ^2 Total squared misses in units of the error bars; near 1 per data point means a healthy fit.

Confidence interval The range over which a parameter can move before χ^2 climbs unacceptably; the "±" on every number.

MCMC Mapping parameter uncertainty by letting random walkers populate the low-χ^2 valleys.

No-hair theorem A black hole's exterior gravity is fixed by mass and spin alone; tested here through the quadrupole knob ε_q.

→ *Full treatment: units and calibration in Chapter 19; fitting statistics and MCMC in Chapter 15; the no-hair constraint in Chapter 17.*

Part II

Theory and Methods

Chapter 7

Introduction and Strong-Field Context

7.1 Sagittarius A*: a dual strong-field laboratory

The compact radio source Sgr A* at the dynamical centre of the Milky Way is the nearest supermassive black hole and the only one for which two independent strong-field probes are available at once: the resolved orbits of the S-star cluster within a few thousand gravitational radii (Gillessen et al., 2017; GRAVITY Collaboration, 2018, 2020, 2022), and the horizon-scale image of its photon ring obtained by the Event Horizon Telescope (Event Horizon Telescope Collaboration, 2022). Together they anchor the mass ($4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$), the distance ($8277 \pm 9$ pc), and the angular size of the emission ring ($51.8 \pm 2.3\,\mu$as), making Sgr A* the cleanest laboratory for testing general relativity (GR) in the strong-field, low-velocity regime.

The case that the central object is a black hole was assembled over five decades. The compact radio source itself was found in 1974 (Balick & Brown, who named it); infrared proper-motion monitoring of the innermost stars began in the early 1990s with the Garching and UCLA groups, and by the early 2000s individual stars — S2 above all — had been watched through enough of an orbit to make the enclosed mass and its compactness inescapable: several million solar masses inside a pericentre of $\sim$120 au excludes every cluster of dark stellar remnants on stability grounds, and the 2018–2020 relativistic measurements closed the remaining loopholes at the orbital scale. That programme was recognised with the

2020 Nobel Prize in Physics (Genzel and Ghez, with Penrose for the theoretical side); the 2022 EHT image then carried the evidence from $\sim 10^3\, R_S$ down to the photon shell itself. This report takes the black-hole interpretation as its working hypothesis and asks the quantitative next question: does one Kerr metric, with one (M, R_0, a_*), account for *all* of it at once?

A geometric accident makes Sgr A* uniquely suitable: it subtends the largest angular gravitational radius of any black hole on the sky. Stellar-mass holes are millions of times closer per unit mass but billions of times smaller; the giant of M87 is fifteen hundred times more massive but two thousand times farther. At $\theta_g = 5.1243\,\mu$as, Sgr A* narrowly beats M87* and dwarfs everything else — which is why the only two horizon-scale images in existence are of precisely these two objects, and why the S-star laboratory has no analogue anywhere else in the sky.

7.2 The no-hair theorem as a falsifiable target

The no-hair theorem states that the exterior spacetime of an astrophysical black hole is fully specified by its mass M and spin a; in particular the mass quadrupole moment is locked to $Q = -Ma^2$. Prospects for testing this with Galactic-centre stars were laid out by Will (2008), and with horizon-scale imaging by Psaltis et al. (2016). Both programmes require a forward model that is exact in the strong field: post-Newtonian expansions degrade at the S2 pericentre ($r_{\rm peri} \approx 1470\, R_S$), and the photon ring is intrinsically a property of exact null geodesics (Bardeen, 1973; Johnson et al., 2020).

It helps to place the two probes on the map of gravity experiments, whose axes are the potential $\epsilon = GM/rc^2$ and the speed v/c. Solar-system tests live at $\epsilon \lesssim 10^{-6}$; binary-pulsar timing reaches $\epsilon \sim 10^{-6}$ orbitally but involves strongly self-gravitating sources; gravitational-wave mergers probe $\epsilon \sim 0.1$ dynamically for seconds. The Galactic-centre programme occupies a distinct corner: S2 at pericentre samples $\epsilon \approx 3.5 \times 10^{-4}$, $v/c \approx 0.026$ — a *stationary* strong-ish field traversed slowly, ideal for precision spectroscopy of the metric itself — while the photon ring samples $\epsilon \sim 0.1$–0.5 with photons, the static strong-field limit. No other object offers both rungs simultaneously, around the same mass, with both anchored to the same dynamical calibration; that dual anchoring is what this report's single-code approach is built to exploit.

7.3 This programme

This report describes such a forward model, built from first principles and validated at every layer. The simulator integrates the Carter-separated geodesic equations (Carter, 1968) in Kerr spacetime, renders the accretion flow with the standard relativistic thin-disk model (Novikov & Thorne, 1973; Page & Thorne, 1974), fits real astrometric and spectroscopic data for the star S2, and embeds a free-quadrupole Hartle–Thorne metric (Hartle & Thorne, 1968; Glampedakis & Babak, 2006) to constrain departures from Kerr. The same geodesic core runs on the GPU for interactive exploration, and every number quoted here is regenerated from the code by a single reproduction target.

7.4 Lineage of the imaging problem

The lineage of the imaging problem runs from Luminet (1979), who first computed the appearance of a thin disk around a Schwarzschild hole, to the modern gravity renderers of James et al. (2015); our CPU reference tracer follows the same backward ray-tracing logic but is held to numerical standards set by the analytic theory (Chapter 10).

The theoretical apparatus this simulator rests on was assembled over half a century. Kerr (1963) found the spinning vacuum solution; Boyer & Lindquist (1967) gave it the coordinates in which it is most usable, and in which the horizon structure is transparent. Carter (1968) discovered the fourth constant of motion that renders the geodesics integrable — without it, no machine-precision ray tracer would be possible — and Bardeen et al. (1972) worked out the circular-orbit structure (ISCO, photon orbits, binding energy) in the closed forms this code implements verbatim. Bardeen (1973) computed the analytic shadow boundary, the "critical curve" against which our Kerr renders are validated to machine precision.

The imaging problem then developed in parallel with the means to solve it. Cunningham & Bardeen (1973) traced the appearance of a star orbiting an extreme-Kerr hole; Luminet (1979) produced the first image of a thin accretion disk around a Schwarzschild hole, complete with the lensed secondary image that gives every black-hole picture its characteristic halo; and four decades later James et al. (2015) brought the same backward ray-tracing to cinema audiences through *Interstellar*'s Gargantua. Our CPU reference tracer follows this lineage but is held to the numerical standards of the analytic theory (Chapter 11) rather than

to visual plausibility.

The observational payoff arrived in the last decade. The GRAVITY Collaboration detected the gravitational redshift of S2 at its 2018 pericentre (GRAVITY Collaboration, 2018) — confirmed independently by Do et al. (2019) — measured the Schwarzschild precession of its orbit (GRAVITY Collaboration, 2020), and combined multiple stellar orbits into the most precise mass and distance to date (GRAVITY Collaboration, 2022). In the same window the Event Horizon Telescope resolved the horizon-scale emission ring of Sgr A* (Event Horizon Telescope Collaboration, 2022), whose diameter is set by the photon shell of Chapter 11. This report's forward model reproduces findings from both of these observational programmes from one geodesic core.

7.5 Contributions

The specific contributions of this programme are:

1. a turning-point-free second-order Mino-time integration of the Carter equations with per-step constraint projection, which makes traced shadow boundaries independent of integrator tolerance (Chapter 10);

2. machine-precision validation of the traced Schwarzschild shadow against Synge's formula and of the Kerr shadow against the analytic critical curve (Chapter 11);

3. a 13-parameter full-geodesic fit to the public S2 data that recovers the GRAVITY mass and distance and reproduces both published relativistic effects with no tunable amplitudes (Chapter 15);

4. a quantified no-hair constraint from S2 alone, with a covariance forecast for continued monitoring (Chapter 17);

5. a validated double-precision GPU implementation of the same geodesic core (Chapter 18); and

6. a single-command reproduction pipeline in which every number in this report is regenerated from source (Appendix D).

7.6 Reader's guide

The report is organised in four parts so that readers with different backgrounds can enter at different depths.

Part I — Foundations for the non-specialist assumes no physics background. It explains, in plain language and with almost no equations, every concept the later chapters rely on: black holes and horizons, curved spacetime, spin and the shadow, the accretion disk and its colours, stellar orbits, and the statistical toolkit. Each primer chapter ends with a short list of the terms just introduced and a pointer to the chapter where the full treatment lives. Readers comfortable with general relativity can skip Part I entirely.

Part II — Theory and Methods derives, from first principles, every formula the simulator implements: the Kerr geometry and its characteristic radii (Chapter 8), the Carter-separated geodesics and their second-order Mino-time form (Chapter 9), and the numerical integration machinery with its error analysis (Chapter 10). Each derivation cites the source function that implements it.

Part III — Results presents the science: shadow and critical-curve metrology (Chapter 11), the relativistic thin disk (Chapter 12), photon-ring substructure (Chapter 13), the S-star forward model (Chapter 14), the S2 fit and its statistics (Chapter 15), the wider S-star cluster (Chapter 16), and the no-hair constraint (Chapter 17).

Part IV — Engineering, validation, and closing documents the GPU implementation (Chapter 18), the Sgr A* calibration (Chapter 19), the interactive instrument (Chapter 20), the development history and complete test suite (Chapter 21), and closes with discussion and conclusions. The appendices carry extended algebra, algorithm pseudocode, number provenance, the reproduction pipeline, and a glossary.

A reader who wants only the headline science should read Chapters 11, 15 and 17, plus the calibration of Chapter 19. A reader evaluating the numerics should read Chapters 10 and 21. Every quantitative statement that is not a cited literature value is typeset from a generated macro and is reproducible by `make reproduce` (Appendix D).

Chapter 8

Kerr Geometry from First Principles

This chapter derives every closed-form Kerr quantity the simulator implements: the metric and its algebraic structure (`kerr_components` in `physics/src/metric.cpp`), the characteristic radii (`r_horizon`, `r_ergosphere`, `r_photon`, `r_isco`), and the circular-orbit energetics (`omega_circular`, `energy_circular`, `angmom_circular`) that feed both the disk model of Chapter 12 and the accretion-efficiency discussion. Everything here is exact; nothing is expanded in powers of the spin.

8.1 The Kerr solution in Boyer–Lindquist coordinates

The unique stationary, axisymmetric, asymptotically flat vacuum black-hole solution of Einstein's equations is the two-parameter family found by Kerr (1963), characterised by mass M and specific angular momentum $a = J/M$ with $|a| \le M$. In the coordinates of Boyer & Lindquist (1967) (t, r, θ, ϕ), with $G = c = 1$, the line element is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 \\ &+ \Sigma\,\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\,\mathrm{d}\phi^2, \end{aligned} \tag{8.1}$$

built from the three auxiliary functions used throughout this report,

$$\Sigma = r^2 + a^2\cos^2\theta, \qquad \Delta = r^2 - 2Mr + a^2, \qquad A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta. \tag{8.2}$$

This is precisely the form implemented by `kerr_components`. Two limits anchor it: $a \to 0$ recovers the Schwarzschild solution in its standard coordinates, and $M \to 0$ recovers flat spacetime in oblate spheroidal coordinates — the residual a then only labels a coordinate choice, which is why all curvature effects below always enter through products Ma or M/r.

That this two-parameter family is not merely *a* solution but *the* solution is the content of the black-hole uniqueness theorems: Israel's theorem forces the static vacuum black hole to be Schwarzschild, and the Carter–Robinson theorem extends the rigidity to the stationary axisymmetric case, singling out Kerr (Wald, 1984; Chandrasekhar, 1983). Whatever messy configuration collapses — a lopsided star, a merging binary — the exterior is expected to ring down to (8.1), radiating away every memory except M and a. (A third parameter, electric charge, is allowed by the theorems but is neutralised in astrophysical environments within moments; we set it to zero throughout.) This rigidity is what Chapter 17 turns into a measurement: if the exterior of Sgr A* is *not* Kerr, no choice of M and a can hide it.

The two parameters can be read off at infinity without ever approaching the hole. Expanding (8.1) for $r \gg M$,

$$g_{tt} \simeq -\left(1 - \frac{2M}{r}\right), \qquad g_{t\phi} \simeq -\frac{2Ma\sin^2\theta}{r}, \tag{8.3}$$

the first entry is the Newtonian potential $\Phi = -M/r$ in disguise — so M is the mass any distant Keplerian orbit weighs — while the cross-term is the gravitomagnetic potential of a body with angular momentum $J = Ma$: spin is stored in the metric's off-diagonal entry. In the multipole language of Chapter 17, these are the statements $M_0 = M$ and $S_1 = Ma$; the first *non-trivial* no-hair lock appears one order deeper, $M_2 = -Ma^2$. Throughout we use the dimensionless spin $a_* = a/M \in [0, 1]$, so that the working Sgr A* model, $a_* = 0.94$, carries angular momentum $J = 0.94\,GM^2/c$ in physical units.

8.1.1 Metric determinant and inverse

The t–ϕ sector of (8.1) is a 2×2 block; its determinant collapses through the identity

$$g_{tt}\,g_{\phi\phi} - g_{t\phi}^2 = -\Delta\sin^2\theta, \tag{8.4}$$

which the reader can verify by direct substitution of (8.2) (the A-terms cancel against the cross term). Multiplying by $g_{rr}g_{\theta\theta} = \Sigma^2/\Delta$ gives the full determinant

$$\det g = -\Sigma^2 \sin^2\theta. \tag{8.5}$$

Equation (8.5) is asserted for random (r, θ, a) samples by a dedicated unit test (Chapter 21); it is a sensitive canary because any sign or factor slip in $g_{t\phi}$ breaks it immediately. The same block structure yields the inverse metric in closed form — the t–ϕ block inverts as a 2×2 matrix and the diagonal r, θ entries invert trivially (`invert_bl_block` in `metric.hpp`):

$$g^{tt} = \frac{g_{\phi\phi}}{-\Delta \sin^2\theta}, \quad g^{t\phi} = \frac{-g_{t\phi}}{-\Delta \sin^2\theta}, \quad g^{\phi\phi} = \frac{g_{tt}}{-\Delta \sin^2\theta}, \quad g^{rr} = \frac{\Delta}{\Sigma}, \quad g^{\theta\theta} = \frac{1}{\Sigma}. \tag{8.6}$$

Two consequences of this block structure are used constantly downstream. First, the invariant volume element is simply $\sqrt{-g} = \Sigma \sin\theta$ — no square roots of Δ survive — which keeps the ZAMO tetrad normalisation of Chapter 11 free of horizon-adjacent cancellations. Second, because the only off-diagonal coupling is t–ϕ, the four-velocity normalisation condition $g_{\mu\nu}u^\mu u^\nu = -1$ is always a *quadratic* in u^t with coefficients read directly from (8.1); that is the equation `solve_ut` solves in closed form when initialising timelike geodesics (Chapter 9), and its discriminant going negative is the code's first line of defence against unphysical initial conditions.

8.1.2 Kerr–Schild form

The Boyer–Lindquist chart is singular at $\Delta = 0$; that surface is a coordinate artefact, not a curvature singularity. The cleanest way to see this is the Kerr–Schild form: there exist coordinates $(\tilde{t}, x, y, z)$ in which

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2Mr^3}{r^4 + a^2 z^2}\, l_\mu l_\nu, \qquad l_\mu = \left(1,\ \frac{rx + ay}{r^2 + a^2},\ \frac{ry - ax}{r^2 + a^2},\ \frac{z}{r}\right), \tag{8.7}$$

where η is the flat metric, l_μ is null with respect to both η and g, and $r(x, y, z)$ is defined implicitly by $\frac{x^2+y^2}{r^2+a^2} + \frac{z^2}{r^2} = 1$. Every metric component in (8.7) is manifestly regular at the horizon.

The route from Boyer–Lindquist to this form is worth recording, because it shows exactly which pathology the BL chart has. Define ingoing Kerr coordinates $(v, r, \theta, \tilde{\phi})$ by absorbing the divergent pieces of radial null motion,

$$\mathrm{d}v = \mathrm{d}t + \frac{r^2 + a^2}{\Delta}\,\mathrm{d}r, \qquad \mathrm{d}\tilde{\phi} = \mathrm{d}\phi + \frac{a}{\Delta}\,\mathrm{d}r. \tag{8.8}$$

Both corrections blow up at $\Delta = 0$ precisely because BL time t and azimuth ϕ are tied to distant static observers, for whom an infalling ray takes infinite t (and winds infinitely in ϕ, the signature twist of rotation) to reach the horizon. Substituting (8.8) into (8.1) cancels every $1/\Delta$:

$$\begin{aligned} \mathrm{d}s^2 = & -\left(1-\frac{2Mr}{\Sigma}\right)\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}v\,\mathrm{d}\tilde{\phi} \\ & - 2a\sin^2\theta\,\mathrm{d}r\,\mathrm{d}\tilde{\phi} + \Sigma\,\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\,\mathrm{d}\tilde{\phi}^2, \end{aligned} \tag{8.9}$$

which is regular for all $r > 0$; the Cartesian relabelling $x+\mathrm{i}y = (r+\mathrm{i}a)\sin\theta\,\mathrm{e}^{\mathrm{i}\tilde{\phi}}$, $z = r\cos\theta$ then produces (8.7). The simulator works in Boyer–Lindquist coordinates throughout — the Carter separation of Chapter 9 lives there — and simply stops rays at $r = 1.001\,r_+$; the Kerr–Schild form is recorded here as the proof that nothing physical happens at the chart boundary. The true curvature singularity sits at $\Sigma = 0$, i.e. $r = 0$, $\theta = \pi/2$: a ring of coordinate radius a in the equatorial plane, safely behind the horizon for all $|a| \le M$.

8.2 Killing structure

The metric (8.1) depends on neither t nor ϕ, so ∂_t and ∂_ϕ are Killing vectors: along any geodesic with four-momentum p^μ,

$$E \equiv -p_t \quad \text{and} \quad L_z \equiv p_\phi \tag{8.10}$$

are conserved. Remarkably, Kerr also possesses a *hidden* symmetry with no coordinate expression: an antisymmetric Killing–Yano tensor $f_{\mu\nu}$ whose square $K_{\mu\nu} = f_{\mu\lambda}f^{\lambda}{}_{\nu}$ is a rank-2 Killing tensor, $\nabla_{(\lambda}K_{\mu\nu)} = 0$. The associated quadratic invariant $K_{\mu\nu}p^\mu p^\nu$ furnishes the fourth constant of motion — Carter's Q (Carter, 1968) — which is derived from the Hamilton–Jacobi side in Chapter 9. Four constants $(E, L_z, Q, p_\mu p^\mu)$ for four momentum components is what makes Kerr geodesics integrable, and hence what makes a machine-precision ray tracer possible at all.

The hidden symmetry is concrete enough to write down. The Killing–Yano two-form of Kerr is (Penrose, 1969; Chandrasekhar, 1983)

$$f = a\cos\theta\,\mathrm{d}r \wedge \left(\mathrm{d}t - a\sin^2\theta\,\mathrm{d}\phi\right) + r\sin\theta\,\mathrm{d}\theta \wedge \left[(r^2+a^2)\,\mathrm{d}\phi - a\,\mathrm{d}t\right], \tag{8.11}$$

antisymmetric and satisfying $\nabla_{(\mu}f_{\nu)\lambda} = 0$; its square $K_{\mu\nu} = f_{\mu\lambda}f^{\lambda}{}_{\nu}$ is the Killing tensor. Two structural facts follow by inspection of (8.11). First, f mixes the (t, ϕ)

and (r, θ) sectors with coefficients depending on r and θ *separately* — the geometric seed of the additive separation $S = S_r(r) + S_\theta(\theta)$ that Chapter 9 exploits. Second, at $a = 0$ the first term dies and $K_{\mu\nu}p^\mu p^\nu$ degenerates to $r^4 p_\theta^2$ plus angular momentum terms — the square of the *total* angular momentum, which is why Carter's Q reduces in the Schwarzschild limit to $L^2 - L_z^2$, the familiar statement that only two of the three angular momentum components are independent once L_z is fixed. The combination the code actually conserves, $Q = K_{\mu\nu}p^\mu p^\nu - (L_z - aE)^2$ (`carter_q`), is chosen so that $Q = 0$ picks out exactly the equatorial geodesics — the convention under which every potential in Chapter 9 is written.

8.3 Frame dragging and the ZAMO

The off-diagonal $g_{t\phi}$ has a direct physical reading. Consider an observer with zero angular momentum, $u_\phi = 0$ (a "ZAMO"). Raising the index,

$$\frac{u^\phi}{u^t} = \frac{g^{\phi t}u_t}{g^{tt}u_t} = -\frac{g_{t\phi}}{g_{\phi\phi}} \equiv \omega(r,\theta) = \frac{2Mar}{A}, \tag{8.12}$$

so an observer who *has* no angular momentum nevertheless *moves*: local inertial frames are dragged around the hole at the rate ω, falling off as $2Ma/r^3$ far away. The ZAMO is the natural camera carrier for ray tracing — it exists everywhere outside the horizon, including inside the ergosphere, and reduces to the static observer at $a = 0$; its orthonormal tetrad is constructed explicitly in Chapter 11 (`zamo_tetrad`).

Three properties of ω deserve record. Far from the hole, $\omega \to 2J/r^3$ with $J = Ma$: the Lense–Thirring precession rate, the same gravitomagnetic circulation measured (at the level of tens of milliarcseconds per year) by the Gravity Probe B gyroscopes and the LAGEOS laser-ranging satellites in Earth orbit — the identical term of the identical metric expansion, sixteen orders of magnitude apart in field strength. At the horizon, using $r_+^2 + a^2 = 2Mr_+$ (from $\Delta = 0$),

$$\omega(r_+) = \frac{2Mar_+}{(r_+^2 + a^2)^2} = \frac{a}{2Mr_+} \equiv \Omega_H, \tag{8.13}$$

independent of θ: the horizon rotates rigidly, like a solid body, with angular velocity Ω_H — for the Sgr A* model spin, $2\pi/\Omega_H \approx 18\,M$, one horizon revolution every six and a half minutes of coordinate time. And between those limits ω interpolates monotonically, which is what the primer's Fig. 3.1 displays. The ZAMO's proper-time normalisation defines the lapse $\alpha = \sqrt{\Delta\Sigma/A}$; $\alpha \to 0$ at the

horizon is the familiar infinite-redshift statement, and the pair (α, ω) is exactly what the camera model of Chapter 11 needs to convert locally measured photon directions into conserved (E, L_z).

8.4 Horizons, ergosphere, and the static limit

Three characteristic surfaces follow from the metric functions alone.

Horizons. The BL chart degenerates where $\Delta = 0$:

$$r_\pm = M \pm \sqrt{M^2 - a^2}. \tag{8.14}$$

The outer root r_+ is the event horizon (`r_horizon`); it shrinks from $2M$ at $a = 0$ to M at extremality, where the square root vanishes. For $|a| > M$ there is no horizon at all — a naked singularity — and the `KerrMetric` constructor refuses such spins by design.

The horizon is not merely a radius; it has intrinsic geometry. Setting $r = r_+$, $\mathrm{d}t = \mathrm{d}r = 0$ in (8.1) and integrating $\sqrt{g_{\theta\theta} g_{\phi\phi}}$ over the sphere gives the horizon area

$$A_H = 4\pi \left(r_+^2 + a^2\right) = 8\pi M r_+, \tag{8.15}$$

where the second equality again uses $r_+^2 + a^2 = 2Mr_+$. Two classical consequences hang on (8.15). First, the *irreducible mass* $M_{\mathrm{irr}} = \sqrt{A_H/16\pi}$: Hawking's area theorem forbids A_H from decreasing in any classical process, so M_{irr} is the part of the mass that can never be extracted. For a Kerr hole $M^2 = M_{\mathrm{irr}}^2 + J^2/(4M_{\mathrm{irr}}^2)$ (Christodoulou's mass formula), and the difference between M and M_{irr} — up to 29% of M at extremality — is rotational energy stored in the dragged spacetime itself, in principle minable via the Penrose process (Penrose, 1969) or its electromagnetic cousin, the Blandford & Znajek (1977) mechanism, which powers the relativistic jets this simulator deliberately does *not* model (Chapter 22 returns to that honest boundary). Second, the area never appears in any optics below: the *shadow* of Chapter 11 is set by the photon orbits at $\sim 3M$, not by r_+ — a distinction the popular identification of "the EHT ring" with "the horizon" blurs.

Static limit and ergosphere. A static observer (fixed r, θ, ϕ) has $u^\mu \propto \partial_t$, which requires $g_{tt} < 0$. From (8.1), $g_{tt} = 0$ where $\Sigma = 2Mr$, i.e.

$$r_E(\theta) = M + \sqrt{M^2 - a^2 \cos^2\theta} \tag{8.16}$$

(`r_ergosphere`). Between r_+ and r_E — the ergosphere — ∂_t is spacelike: no observer can stand still, though escape remains possible. The surface touches the

horizon at the poles and stands off farthest at the equator ($r_E = 2M$ for every spin), as drawn in the primer's Fig. 3.2. Penrose's observation that orbits of negative energy exist inside this region — allowing rotational energy extraction — is not exploited by the simulator, but the ZAMO tetrad's good behaviour there is essential for close-in cameras.

8.5 Equatorial photon orbits: the cubic and its trigonometric solution

In the equatorial plane ($\theta = \pi/2$, $Q = 0$) the radial potential of Chapter 9 for a photon of impact parameter $b = L_z/E$ reduces to

$$\frac{R(r)}{E^2} = \left(r^2 + a^2 - ab\right)^2 - \Delta\,(b-a)^2. \tag{8.17}$$

A circular photon orbit requires a double root: $R(r) = R'(r) = 0$. Dividing the two conditions eliminates b; after simplification (worked in full in Appendix A) the radius of the circular orbit satisfies

$$r^{3/2} - 3Mr^{1/2} \pm 2a\sqrt{M} = 0, \tag{8.18}$$

a cubic in $\sqrt{r}$ (upper sign prograde). Depressed cubics of the form $x^3 - 3px + q = 0$ with three real roots are solved by the trigonometric substitution $x = 2\sqrt{p}\cos\vartheta$, which turns the equation into $\cos 3\vartheta = -q/(2p^{3/2})$. Applying it with $x = \sqrt{r/M}$, $p = 1$, $q = \pm 2a/M$ gives the closed form implemented in `r_photon`:

$$r_{\rm ph}^{\pm} = 2M\left\{1 + \cos\left[\tfrac{2}{3}\cos^{-1}\left(\mp\frac{a}{M}\right)\right]\right\}. \tag{8.19}$$

At $a = 0$ both signs give $\cos^{-1}(0) = \pi/2$ and $r_{\rm ph} = 2M(1 + \cos 60^\circ) = 3M$, the Schwarzschild photon sphere. At extremality the prograde orbit reaches $r_{\rm ph}^{+} = M$ (coincident in coordinate radius with the horizon) while the retrograde orbit retreats to $4M$: frame dragging assists co-rotating light and punishes counter-rotating light, the asymmetry visible in the traced trajectories of Fig. 2.1 and, ultimately, in the displaced shadow of Chapter 11. The corresponding critical impact parameters follow from back-substituting (8.19) into the double-root conditions; they reappear as the Bardeen critical curve in Chapter 11.

Two consistency checks on (8.19) are cheap and instructive. For small spin, $\cos^{-1}(\mp a_*) = \pi/2 \pm a_* + O(a_*^3)$, and expanding the cosine about $\pi/3$ gives

$$r_{\rm ph}^{\pm} = 3M \mp \frac{2a}{\sqrt{3}} + O(a^2/M)\,: \tag{8.20}$$

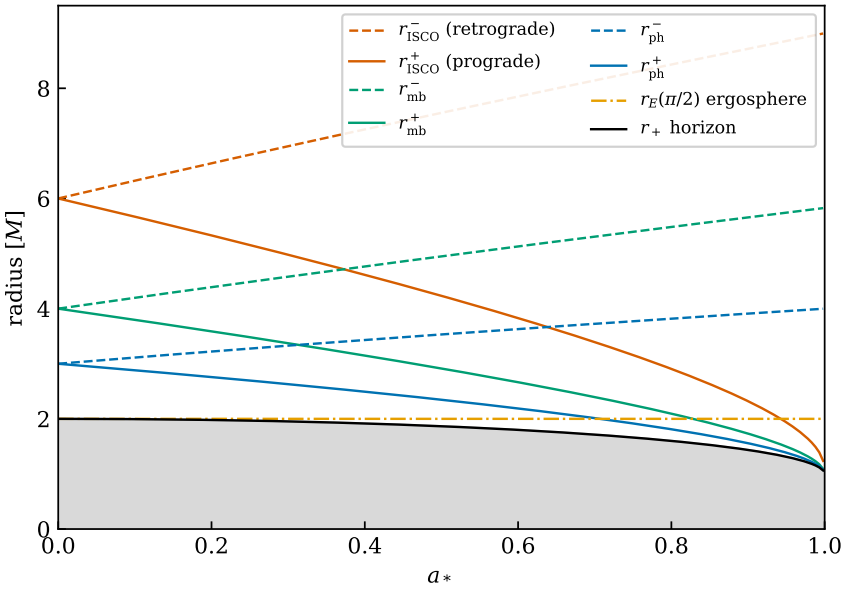


Figure 8.1: Every characteristic radius of the Kerr geometry as a function of spin, from the closed forms of this chapter (`physics_tables` dump; grey region is inside the horizon). Solid: prograde; dashed: retrograde. The prograde ISCO's plunge toward the horizon is the single most consequential feature for disk appearance.

the photon orbits split *linearly* in spin, prograde inward, retrograde outward, at slope $2/\sqrt{3} \approx 1.15$ — this is the leading-order lever by which spin displaces the shadow silhouette. At the working spin $a_* = 0.94$ the exact values from (8.19) are $r^+_{\rm ph} = 1.43\,M$ and $r^-_{\rm ph} = 3.95\,M$ (the `physics_tables` grid of Fig. 8.1): the prograde orbit sits barely $0.08\,M$ outside the horizon at $1.34\,M$, while the retrograde orbit has retreated past the Schwarzschild value — an asymmetry of nearly a factor of three in radius for light, all of it frame dragging.

8.6 Circular timelike orbits: Ω, E, L

For massive particles on circular equatorial orbits, the geodesic equation reduces to one algebraic condition. Writing the orbital angular velocity $\Omega = u^\phi/u^t$ and demanding $\mathrm{d}^2r/\mathrm{d}\tau^2 = 0$ yields (equivalently: extremising the effective potential of Fig. 8.3)

$$\Omega_\pm = \frac{\pm\sqrt{M}}{r^{3/2} \pm a\sqrt{M}}, \tag{8.21}$$

the relativistic generalisation of Kepler's third law (`omega_circular`); the a-term in the denominator is pure frame dragging. The four-velocity normalisation then

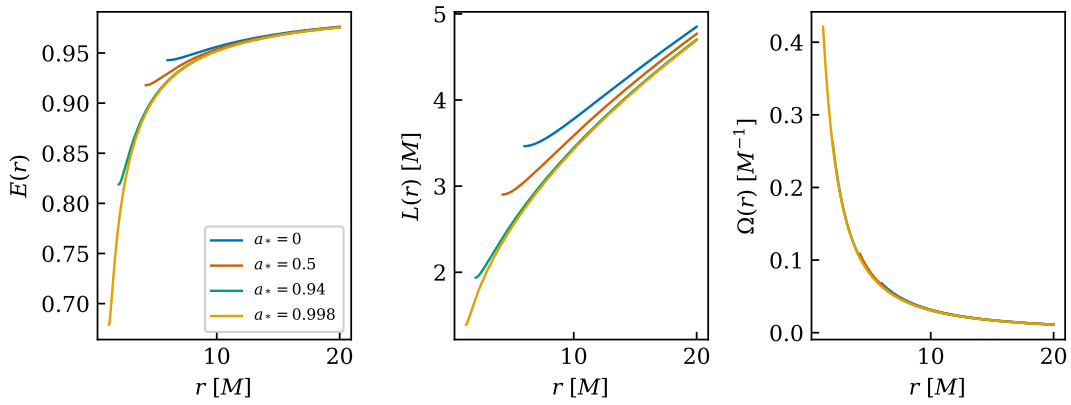


Figure 8.2: Specific energy, angular momentum, and angular velocity of prograde circular orbits, from (8.21)–(8.23) evaluated by `physics_tables`. Curves start at each spin's ISCO.

fixes u^t, and lowering indices gives the specific energy and angular momentum (Bardeen et al., 1972):

$$E(r) = \frac{1 - 2M/r \pm a\sqrt{M/r^3}}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}}, \tag{8.22}$$

$$L(r) = \frac{\pm\sqrt{Mr}\left(1 + a^2/r^2 \mp 2a\sqrt{M/r^3}\right)}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}} \tag{8.23}$$

(`energy_circular`, `angmom_circular`; upper signs prograde). Fig. 8.2 plots all three across the spin range. Two structural features matter downstream. First, the common denominator vanishes at the *photon orbit* — circular orbits become null there, and between the photon orbit and the ISCO they exist but are unstable. Second, $E(r)$ has a minimum exactly at the ISCO, which is the cleanest way to locate it.

8.7 The ISCO and the marginally bound orbit

Stability of a circular orbit requires the effective potential's extremum to be a minimum; the innermost stable circular orbit (ISCO) is the inflection point where $\partial^2 V/\partial r^2 = 0$ joins the circular-orbit conditions. Carrying out the algebra (the full reduction is in Appendix A) collapses the triple condition to a quartic in $\sqrt{r}$ whose relevant root is expressed through the celebrated auxiliary quantities of

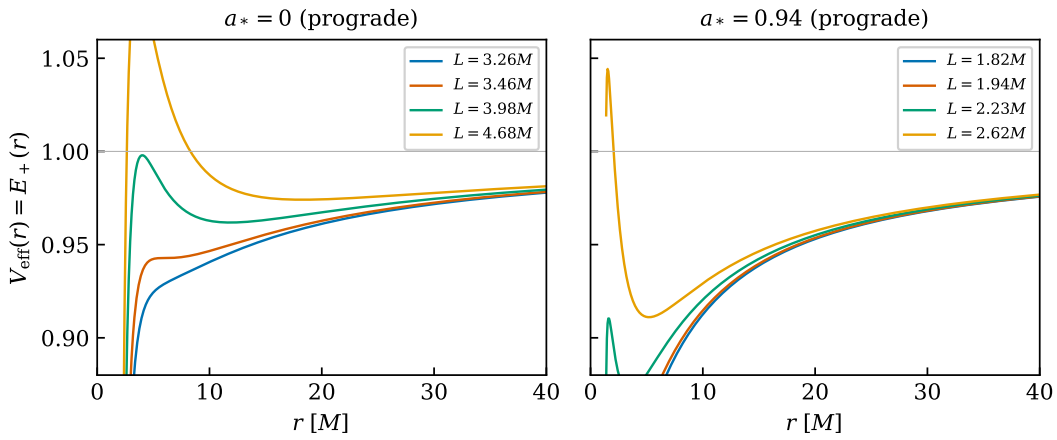


Figure 8.3: The effective potential for equatorial timelike motion, $V_{\rm eff}(r) = E_+(r)$ (the energy at which r is a turning point), at four angular momenta around $L_{\rm ISCO}$. Circular orbits live at the extrema; the inflection where minimum and maximum merge is the ISCO. Left: Schwarzschild; right: $a_* = 0.94$ prograde — the whole structure moves inward and downward (deeper binding).

Bardeen et al. (1972):

$$\begin{aligned} Z_1 &= 1 + (1 - a_*^2)^{1/3}\left[(1 + a_*)^{1/3} + (1 - a_*)^{1/3}\right], \\ Z_2 &= \sqrt{3a_*^2 + Z_1^2}, \\ r_{\rm ISCO} &= M\left[3 + Z_2 \mp \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}\right], \end{aligned} \tag{8.24}$$

with the upper sign prograde (`r_isco`). Before the Z_1, Z_2 machinery, the stability condition itself is compact enough to quote: a circular orbit at radius r is stable iff

$$r^2 - 6Mr \pm 8a\sqrt{Mr} - 3a^2 \;\geq\; 0, \tag{8.25}$$

and the ISCO is its marginal root (Bardeen et al., 1972). Equation (8.25) passes its limits transparently: at $a = 0$ it factors to $r(r - 6M) \geq 0$, hence $6M$; at $a = M$ the prograde form $r^2 - 6Mr + 8M^{3/2}r^{1/2} - 3M^2$ vanishes at $r = M$ and the retrograde form at $r = 9M$. The checks the test suite pins (Chapter 21): at $a_* = 0$, $Z_1 = 3$, $Z_2 = 3$ and $r_{\rm ISCO} = 6M$ exactly; as $a_* \to 1$ the prograde ISCO slides to M while the retrograde one recedes to $9M$; and the prograde radius decreases monotonically with spin (Fig. 8.1).

One more closed form deserves promotion from the appendix, because it compresses the entire energetics of accretion into a single line. At the ISCO — and

only there — the specific energy (8.22) collapses to

$$E_{\mathrm{ISCO}} = \sqrt{1 - \frac{2M}{3\,r_{\mathrm{ISCO}}}}, \tag{8.26}$$

valid for every spin and both senses of rotation (the derivation is three lines from the marginal-stability condition (8.25); Appendix A). It reproduces $\sqrt{8/9}$ at $r_{\mathrm{ISCO}} = 6M$ and $1/\sqrt{3}$ at the extremal prograde $r_{\mathrm{ISCO}} = M$, and it means the whole efficiency curve of Fig. 8.4 is really just the ISCO-radius curve of Fig. 8.1 replotted. The identity doubles as an external check on the code: `physics_tables` evaluates E_{ISCO} through the general formula (8.22), knowing nothing of (8.26), yet its tabulated values reproduce $\sqrt{1 - 2M/3r_{\mathrm{ISCO}}}$ to all six printed decimals across the spin grid (e.g. $E = 0.820404$ against the identity's 0.820406 at $a_* = 0.938$).

A second marginal radius appears in accretion theory: the *marginally bound* orbit, where $E = 1$ — a particle dropped from rest at infinity can just circularise. Setting $E = 1$ in (8.22) gives

$$r_{\mathrm{mb}}^{\pm} = 2M \mp a + 2\sqrt{M(M \mp a)}, \tag{8.27}$$

lying between the photon orbit and the ISCO for every spin (Fig. 8.1).

8.8 Binding energy and accretion efficiency

Equation (8.22) carries a direct astrophysical payload. Gas spiralling quasi-circularly through a disk arrives at the ISCO with specific energy E_{ISCO}, having radiated the binding energy

$$\eta = 1 - E_{\mathrm{ISCO}} \tag{8.28}$$

per unit rest mass. For Schwarzschild, $E_{\mathrm{ISCO}} = \sqrt{8/9}$ and $\eta = 1 - 2\sqrt{2}/3 = 5.72\,\%$. At the Sgr A* working spin $a_* = 0.94$ the efficiency is 17.96 %, and at $a_* = 0.998$ it reaches 32.10 % (Fig. 8.4; values generated by `physics_tables` from (8.22) and (8.24)) — an order of magnitude beyond nuclear fusion's 0.7%, which is why accreting black holes are the most efficient steady engines known. The same E and L profiles drive the Page–Thorne dissipation integral of Chapter 12: the disk's radial brightness profile is nothing but the gradient of this binding-energy release, corrected for the torque bookkeeping.

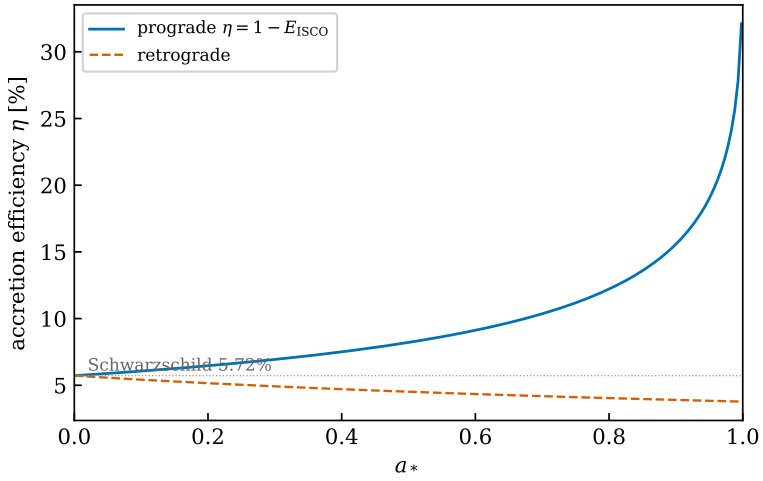


Figure 8.4: Accretion efficiency $\eta = 1 - E_{\mathrm{ISCO}}$ against spin, prograde and retrograde, from `physics_tables`. The Schwarzschild value $1 - \sqrt{8/9} = 5.72\%$ is marked.

8.9 What the rest of the report inherits from this chapter

Because every later chapter leans on these closed forms, it is worth closing with the explicit ledger of Table 8.1: each formula, the function that implements it, and the place its correctness is pinned. The pattern of the table is the pattern of the whole project: no formula ships without a limit it must hit exactly, and no implementation ships without a test that hits it.

Table 8.1: The Chapter 8 ledger: closed form → implementation → anchor. All functions live in `physics/src/metric.cpp`; tests are named as in the Catch2 suite of Chapter 21.

Quantity	Implementation	Anchored by
Metric (8.1)	`kerr_components`	determinant identity (8.5) test
Inverse (8.6)	`invert_bl_block`	$g^{\mu\lambda}g_{\lambda\nu} = \delta^{\mu}_{\nu}$ spot checks
Horizon (8.14)	`r_horizon`	$2M$ at $a_* = 0$; M at $a_* = 1$
Ergosphere (8.16)	`r_ergosphere`	$2M$ equatorial for all spins
Photon orbits (8.19)	`r_photon`	$3M$ at $a_* = 0$; split for $a_* > 0$
ISCO (8.24)	`r_isco`	$6M$ at $a_* = 0$; monotone in spin
Ω, E, L (8.21)–(8.23)	`omega/energy/angmom_circular`	E_{ISCO} identity (8.26)
Efficiency (8.28)	`physics_tables`	5.72% Schwarzschild closed form

Chapter 9

Geodesics and the Carter Constant

The simulator never integrates the raw geodesic equation $\ddot{x}^\mu = -\Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta$ for photons; it integrates Carter's separated form, which is both faster and — through its first integrals — self-diagnosing. This chapter derives that form from the Hamilton–Jacobi equation, classifies the radial motion by the roots of the resulting quartic, and develops the second-order Mino-time formulation that the integrator of Chapter 10 actually solves. The implementing functions are `radial_potential[_deriv]`, `polar_potential[_deriv]`, `null_geodesic_rhs` and `make_null_ray` in `physics/src/null_geodesi` and `solve_ut`, `carter_q` in `timelike_geodesics.cpp`.

9.1 Hamilton–Jacobi separation

Geodesics extremise proper length; equivalently they solve the Hamilton–Jacobi equation for the action $S(x^\mu)$,

$$g^{\mu\nu}\,\partial_\mu S\,\partial_\nu S = -\mu^2, \tag{9.1}$$

with μ the rest mass ($\mu = 0$ for photons, and we normalise $\mu = 1$ for stars). Stationarity and axisymmetry (Section 8.2) let us peel off the cyclic coordinates,

$$S = -E\,t + L_z\,\phi + S_r(r) + S_\theta(\theta), \tag{9.2}$$

and Carter (1968) discovered that with the inverse metric (8.6) the remainder separates completely. The substitution deserves to be seen once in full. With

$\partial_t S = -E$, $\partial_\phi S = L_z$, the contraction (9.1) reads

$$\frac{-AE^2 + 4Mar\,EL_z + \dfrac{\Delta - a^2\sin^2\theta}{\sin^2\theta}\,L_z^2}{\Sigma\Delta} + \frac{\Delta\,S_r'^2 + S_\theta'^2}{\Sigma} = -\mu^2, \tag{9.3}$$

where every coefficient is an entry of (8.6). The numerator of the first term looks hopelessly mixed — A and the $\sin^2\theta$ denominator couple r to θ — but it regroups exactly: using $A = (r^2+a^2)^2 - a^2\Delta\sin^2\theta$ and $r^2 + a^2 - \Delta = 2Mr$ one verifies term by term that

$$\frac{-AE^2 + 4Mar\,EL_z + \big(\Delta/\sin^2\theta - a^2\big)L_z^2}{\Delta} = -\frac{P(r)^2}{\Delta} + \left(aE\sin\theta - \frac{L_z}{\sin\theta}\right)^2, \tag{9.4}$$

with $P(r)$ as defined below — the entire t–ϕ sector splits into a pure-r piece and a pure-θ piece, with no remainder. This is the miracle of Kerr (the hidden Killing tensor of Section 8.2 wearing coordinates); it is *not* generic, and its failure for the Hartle–Thorne metric is why the no-hair machinery of Chapter 17 must integrate Christoffel symbols instead. Multiplying (9.3) through by $\Sigma = r^2 + a^2\cos^2\theta$ and applying (9.4), the mixed terms assemble into

$$-\frac{1}{\Delta}\big[E(r^2{+}a^2){-}aL_z\big]^2{+}\Delta\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2{+}\mu^2r^2 = -\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2{-}\left(aE\sin\theta - \frac{L_z}{\sin\theta}\right)^2{-}\mu^2a^2\cos^2\theta. \tag{9.5}$$

The left side depends only on r, the right only on θ; both must equal a constant. Writing that constant as $-\big[Q + (L_z - aE)^2\big]$ — the shift by $(L_z - aE)^2$ is a convention that makes Q vanish for equatorial orbits — and defining

$$P(r) = E(r^2 + a^2) - aL_z, \tag{9.6}$$

the two sides become

$$\Delta^2\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 = R(r) \equiv P(r)^2 - \Delta\big[\mu^2r^2 + (L_z - aE)^2 + Q\big], \tag{9.7}$$

$$\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 = \Theta(\theta) \equiv Q - \cos^2\theta\left[a^2(\mu^2 - E^2) + \frac{L_z^2}{\sin^2\theta}\right]. \tag{9.8}$$

Q is the *Carter constant*, the invariant associated with the hidden Killing tensor of Section 8.2; loosely, it measures how far the orbit ranges in latitude ($\Theta \geq 0$ requires $Q \geq 0$ for any orbit crossing the equator, with $Q = 0$ exactly for equatorial motion). With four constants (E, L_z, Q, μ) for four momentum components the system is Liouville-integrable: no chaos, every photon path reducible to quadratures — the property that underwrites both the machine-precision metrology of Chapter 11 and the closed-form checks used throughout the test suite.

Since $p_\mu = \partial_\mu S$, equations (9.7)–(9.8) are first-order equations of motion,

$$\Sigma \frac{\mathrm{d}r}{\mathrm{d}\lambda} = \pm\sqrt{R(r)}, \qquad \Sigma \frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \pm\sqrt{\Theta(\theta)}, \tag{9.9}$$

while raising p_t and p_ϕ with (8.6) gives the longitudinal equations

$$\Sigma \frac{\mathrm{d}\phi}{\mathrm{d}\lambda} = -\Big(aE - \frac{L_z}{\sin^2\theta}\Big) + \frac{a}{\Delta}P(r), \quad \Sigma \frac{\mathrm{d}t}{\mathrm{d}\lambda} = -a(aE\sin^2\theta - L_z) + \frac{(r^2+a^2)}{\Delta}P(r). \tag{9.10}$$

For photons ($\mu = 0$) these are the four equations implemented in `null_geodesic_rhs`; the potentials and their radial/polar derivatives are `radial_potential[_deriv]` and `polar_potential[_deriv]`.

9.1.1 Photons carry only two numbers: ξ and η

A null geodesic is unchanged by rescaling its affine parameter, so for $\mu = 0$ the three constants (E, L_z, Q) act only through the two ratios

$$\xi = \frac{L_z}{E}, \qquad \eta = \frac{Q}{E^2}, \tag{9.11}$$

Bardeen's impact parameters. Dividing (9.7) by E^2 and expanding puts the radial potential in the form used for every shadow-boundary computation in this report:

$$\frac{R(r)}{E^2} = r^4 + (a^2 - \xi^2 - \eta)\, r^2 + 2M\big[\eta + (\xi - a)^2\big]\, r - a^2\eta. \tag{9.12}$$

Three of its structural properties do real work later. The cubic term is absent — the sum of the four roots vanishes, so roots come in balanced constellations. The coefficient of r is non-negative, and at $r \to 0^+$ the quartic approaches $-a^2\eta$: for $\eta > 0$ the potential is negative just outside the ring singularity, so equator-crossing photons can never reach $r = 0$. And at large r the quartic grows as r^4 regardless of aim — every photon is free at infinity, which is what makes backward ray tracing from a distant camera well-posed. For the camera itself the pair (ξ, η) maps affinely onto detector coordinates (Chapter 11), so (9.12) really is "the optics of the Kerr field" in a single polynomial.

9.1.2 Polar motion: the $u = \cos^2\theta$ quadratic

The polar potential becomes elementary under $u = \cos^2\theta$. For photons, $\Theta/E^2 = \eta + a^2u - \xi^2 u/(1-u)$; clearing the $(1-u)$ denominator, the turning points in u solve the quadratic

$$a^2u^2 - \big(a^2 - \xi^2 - \eta\big)\, u - \eta = 0, \tag{9.13}$$

whose two roots $u_\pm$ classify all latitudinal behaviour (Chandrasekhar, 1983; Gralla & Lupsasca, 2020):

- $\eta > 0$: the product of roots $-\eta/a^2$ is negative, so exactly one root u_+ lies in $(0, 1)$. The orbit oscillates symmetrically about the equator between $\pm\theta_{\max}$, $\cos^2\theta_{\max} = u_+$, crossing the equatorial plane each half-cycle — the "ordinary" photons, and the only kind that can image an equatorial disk repeatedly (the sub-ring count of Chapter 13 is precisely the number of these crossings).

- $\eta = 0$: $u = 0$ is a root — motion confined to the equatorial plane, the case reduced to the cubic of Section 8.5.

- $\eta < 0$: both roots must be positive for motion to exist at all; the orbit is *vortical*, circulating around the spin axis without ever crossing the equator. Such photons exist only for $a \neq 0$ and only in a thin sliver of parameter space; they contribute nothing to disk images but are traced correctly by the same code path, and the polar-axis clamp of Section 9.3 is what keeps their near-axis passages finite.

For timelike orbits the same substitution carries the extra term $a^2(1 - E^2)u$; for the bound stellar orbits of Part III, with $E < 1$, the effect is a slight stiffening of the polar oscillation. The S2 model exploits none of this structure directly — its orbit is planar in the relevant sense — but `polar_potential` evaluates the general form, and the conserved-Q diagnostic would flag any latitudinal drift immediately.

9.2 Turning points and orbit classification

$R(r)$ is a quartic in r (for photons: leading term E^2r^4), and motion is allowed only where $R \geq 0$; the roots of R are the radial turning points, and their pattern classifies every orbit (Fig. 9.1):

- *No positive root along the path* — the ray plunges from infinity to the horizon: *capture*. These pixels form the shadow.

- *A simple root at* r_{turn} — the ray comes in, turns, and returns to infinity: *deflection*, from the gentle bending of starlight up to arbitrarily many windings as the root is approached.

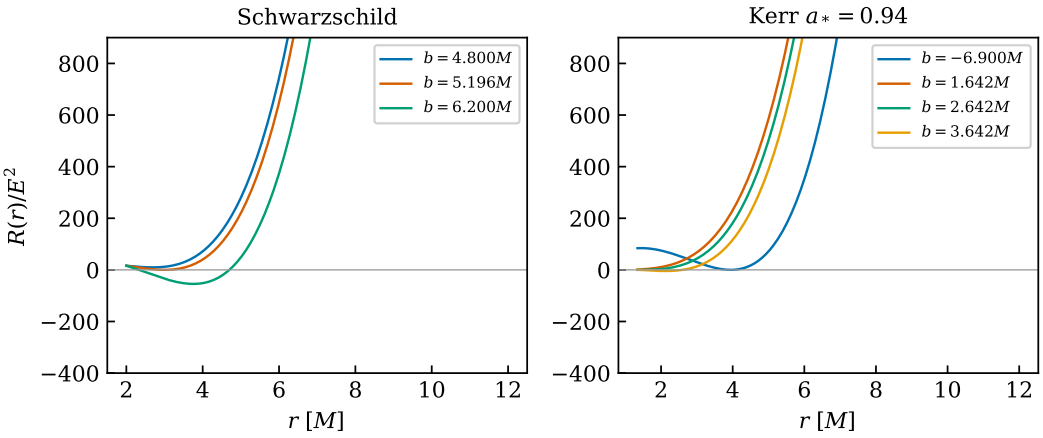


Figure 9.1: The radial potential $R(r)/E^2$ for equatorial photons at impact parameters below, at, and above the critical value (`physics_tables` dump of `radial_potential`). Left, Schwarzschild: at $b = \sqrt{27}\,M$ the potential develops its double root at $r = 3M$. Right, Kerr $a_* = 0.94$: the prograde (low-$|b|$) and retrograde branches behave asymmetrically, and the retrograde critical curve sits much farther out.

- *A double root at $\tilde{r}$* ($R = R' = 0$) — the marginal case: an unstable circular (or, off the equator, spherical) photon orbit. The set of (E, L_z, Q) with double roots is a one-parameter family — the *photon shell* — whose image on the observer's sky is the critical curve of Chapter 11.

- For timelike orbits the $\mu^2 r^2$ term turns the far field around ($R \to (E^2 - 1)r^4/r^2 \ldots$): bound orbits have *two* simple roots $r_{\rm peri} \le r \le r_{\rm apo}$, the case exploited by the Darwin-element construction of Chapter 14.

The instability of the double-root orbits is quantified by the curvature of the potential there, $\frac{1}{2}R''(\tilde{r})$; this single number propagates all the way to the observable demagnification of the photon sub-rings, via the Lyapunov exponents computed in Chapter 11.

Root-counting sharpens the picture. Since (9.12) has no cubic term, its four roots sum to zero; for capture orbits all real roots (if any) lie behind the horizon, for scattering orbits a conjugate pair has moved out past r_+. As (ξ, η) crosses the critical curve, two real roots merge and annihilate — a fold catastrophe, which is why observables vary *smoothly* on either side of the shadow edge but with a square-root cusp exactly at it. The numerical consequence is practical: near-critical rays spend arbitrarily long near the double root (logarithmic divergence

of the winding number), and it is precisely there that the adaptive stepping of Chapter 10 pays for itself, collapsing the step size around the photon-shell passage and re-expanding after (Fig. 10.1).

9.3 Mino time and the second-order formulation

Two features make the first-order system (9.9) hostile to naive numerics. The $\pm$ signs flip at every turning point, requiring explicit bookkeeping; worse, at a turning point the right-hand side $\sqrt{R}$ has infinite slope in r ($\mathrm{d}\sqrt{R}/\mathrm{d}r \to \infty$ as $R \to 0$), and an integrator stepping the truncated system sees $\dot{r} = 0$ as a fixed point and stalls.

Both problems dissolve under two transformations. First, adopt the time parameter of Mino (2003): define τ by

$$\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau, \tag{9.14}$$

which strips the common Σ factor from all four equations — the radial and polar sectors then evolve *independently* of each other, $(\mathrm{d}r/\mathrm{d}\tau)^2 = R(r)$, $(\mathrm{d}\theta/\mathrm{d}\tau)^2 = \Theta(\theta)$. Second, differentiate once more with respect to τ:

$$2\,\frac{\mathrm{d}r}{\mathrm{d}\tau}\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = R'(r)\,\frac{\mathrm{d}r}{\mathrm{d}\tau} \qquad\Longrightarrow\qquad \frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = \tfrac{1}{2}R'(r), \qquad \frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} = \tfrac{1}{2}\Theta'(\theta). \tag{9.15}$$

(The division by $\mathrm{d}r/\mathrm{d}\tau$ is legitimate except at isolated turning points, where (9.15) holds by continuity.) In this form a turning point is an ordinary smooth extremum: the "acceleration" $\frac{1}{2}R'$ simply reverses the motion, no square root of a near-zero quantity is ever taken, and no sign flag exists to mis-set. The state vector the integrator carries is $y = (t, r, \theta, \phi, p_r, p_\theta)$ with $p_r = \mathrm{d}r/\mathrm{d}\tau$, $p_\theta = \mathrm{d}\theta/\mathrm{d}\tau$.

The right-hand sides are plain polynomials and rational functions. For the radial sector, differentiating (9.7) (with $\Delta' = 2(r - M)$, $P' = 2Er$),

$$R'(r) = 4Er\,P(r) - 2(r - M)\big[\mu^2 r^2 + (L_z - aE)^2 + Q\big] - 2\mu^2 r\,\Delta, \tag{9.16}$$

which is exactly what `radial_potential_deriv` returns — evaluated as written, not by finite differences, so the "force" in (9.15) is exact to rounding. Cheap analytic derivatives are a quiet advantage of the Carter form: the generic Christoffel path used for Hartle–Thorne (Section 9.6) pays roughly an order of magnitude more per step for the same accuracy, a comparison quantified in the benchmarks of Chapter 18.

9.3.1 Fundamental frequencies, and why orbits are rosettes

Separability makes the r- and θ-motions individually periodic in Mino time, with periods

$$\Lambda_r = 2\int_{r_{\rm peri}}^{r_{\rm apo}} \frac{\mathrm{d}r}{\sqrt{R(r)}}, \qquad \Lambda_\theta = 4\int_{\theta_{\rm min}}^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{\Theta(\theta)}} \tag{9.17}$$

(Mino, 2003). The azimuth and coordinate time, by (9.10), advance by fixed increments per radial (or polar) cycle, so a generic bound orbit is characterised by three frequencies $(\omega_r, \omega_\theta, \omega_\phi)$ — and almost never closes: unless the frequency ratios are rational, the trajectory densely fills a torus. The observable content of this incommensurability is exactly the rosette of the primer's Fig. 5.3: per radial period, the azimuth advances by 2π *plus* the apsidal excess $\Delta\varpi = \omega_\phi/\omega_r \cdot 2\pi - 2\pi$, which for S2 evaluates to the $11.98'$ per orbit of Part III. The code never computes elliptic integrals — the integrator resolves the same frequencies dynamically — but (9.17) explains a numerical observation of Chapter 14: matching a relativistic orbit to Keplerian initial conditions misassigns the *radial* period by the frequency mismatch, the 1.6 per cent error that motivated the Darwin-element initialisation.

The price of differentiating is that (9.15) admits solutions that drift off the true energy shell — precisely the first integrals

$$p_r^2 = R(r), \qquad p_\theta^2 = \Theta(\theta) \tag{9.18}$$

are no longer enforced identically but become *checkable* (and, as Chapter 10 shows, *re-imposable*) invariants. This trade — turning-point-free dynamics in exchange for explicit constraint maintenance — is the central numerical design decision of the project, and its quantitative payoff is measured in Figs. 10.2–10.3.

The coordinate singularities of the BL chart at $\theta = 0, \pi$ (where $L_z \neq 0$ makes $\Theta \to -\infty$) are regularised with a 10^{-8} clamp (`kThetaClamp`); polar-crossing orbits with $L_z = 0$ pass smoothly.

9.4 The time equation, and who owns which clock

The t-equation of (9.10) deserves a section of its own, because three different clocks run through this report and confusing them is the classic error of the field. *Coordinate time* t is the bookkeeping label of the BL chart — the far observer's clock, the time in which the horizon is never crossed and the S2 observation epochs are stamped. *Proper time* $\tau_{\rm prop}$ ticks on the moving object itself: for S2 it runs slow by the combined gravitational-plus-velocity factor $1/u^t$ of Chapter 12 (about seven

parts in 10^4 at pericentre — the two $\sim 10^{-4}$ redshift terms of Chapter 15 added), and for photons it does not run at all. *Mino time* τ is neither: it is a computational parameter chosen to decouple the equations, with no clock anywhere reading it. The integrator marches in Mino time, accumulates coordinate time through (9.10), and never needs proper time for photons at all.

Two physical effects live entirely inside the t-equation. First, its $P(r)/\Delta$ term diverges at the horizon: as a ray approaches r_+, equal steps of Mino time cost ever more coordinate time — the distant observer sees the approach freeze, which is why the tracer's $1.001\, r_+$ stop (Chapter 10) discards only an infinite tail of frozen bookkeeping, not physics. Second, along escaping rays the accumulated t minus the flat-space expectation is finite and positive — strong-field paths arrive *late*, the Shapiro delay in geodesic dress. The sub-ring echo delay of Chapter 13 and the Rømer solve of Chapter 14 are both, at bottom, careful audits of this same accumulated-t ledger, one for photons and one for a star.

9.5 Launching rays: from four-momentum to constants

The ray tracer builds initial conditions from a physical four-momentum k^μ at the camera (constructed in the ZAMO tetrad, Chapter 11). The map to the Carter description (`make_null_ray`) is purely algebraic: lower the index with the metric, read off

$$E = -k_t, \qquad L_z = k_\phi, \tag{9.19}$$

and evaluate the Carter constant from the polar momentum,

$$Q = (\Sigma k^\theta)^2 + \cos^2\theta \left[\frac{L_z^2}{\sin^2\theta} - a^2 E^2 \right], \tag{9.20}$$

which is (9.8) solved for Q with $p_\theta = \Sigma k^\theta$. The Mino momenta are then $p_r = \Sigma k^r$, $p_\theta = \Sigma k^\theta$ — with *no* sign ambiguity, since they inherit the sign of the physical launch direction. The identical reconstruction run in reverse (`null_conserved_errors`) recomputes (E, L_z, Q) from the evolving state at every step and compares them with the launch values: the primary health check of the whole physics core (tolerance 10^{-6} relative; Chapter 21).

9.6 Timelike geodesics

For stars, $\mu = 1$ in (9.7)–(9.8): the radial potential gains the $-\Delta r^2$ mass term and the polar potential's spin term becomes $a^2(1 - E^2)$. Two independent code paths use this case.

For *Kerr* orbits, the Carter form above applies unchanged and `carter_q` evaluates (9.20) (with the μ^2 term) as a conserved diagnostic. For the *Hartle–Thorne* spacetime of Chapter 17, however, no Carter constant exists — the metric is not separable — so the S-star integrator deliberately uses the generic second-order geodesic equation $\mathrm{d}u^\lambda/\mathrm{d}\tau = -\Gamma^\lambda_{\mu\nu} u^\mu u^\nu$ with Christoffel symbols from complex-step differentiation (Chapter 10), for *any* metric behind the `IMetric` interface. The initial u^t is completed from the spatial velocity by solving the normalisation quadratic $g_{\mu\nu} u^\mu u^\nu = -1$ for the future-directed root (`solve_ut`), and the same normalisation is the per-step health check ($|g_{\mu\nu} u^\mu u^\nu + 1| < 10^{-8}$, Gate 0).

We integrate stellar orbits in the full geodesic equations rather than any post-Newtonian expansion: at the S2 pericentre ($\sim$1470 Schwarzschild radii, orbital speed 2.6 per cent of c) the leading PN corrections are individually at the 10^{-4} level — the same order as the measured redshift excess itself — so the expansion error budget would have to be argued term by term, whereas the geodesic integrator's error budget is simply its conservation diagnostics.

Fig. 9.2 shows the machinery end-to-end: a bound equatorial orbit with turning points at $8M$ and $25M$, initialised by the same two-condition Darwin construction the S2 model uses (Chapter 14), integrated for six radial periods. The orbit precesses visibly — this is the same rosette drawn schematically in the primer — while the four-velocity normalisation drifts at the 10^{-13} level, far inside the 10^{-8} gate, and the conserved (E, L) reproduce their Darwin target values to eleven decimal places.

9.7 Closed-form solutions as an independent check

Because (9.9) separates, Kerr geodesics can also be written in closed form as elliptic integrals (Darwin, 1959; Rauch & Blandford, 1994; Gralla & Lupsasca, 2020). This project deliberately does *not* use those solutions in production — the numerical path generalises to the non-separable Hartle–Thorne metric where the no-hair test lives, and one code path for all metrics is worth more than per-metric special cases. The closed forms serve instead as external anchors: the

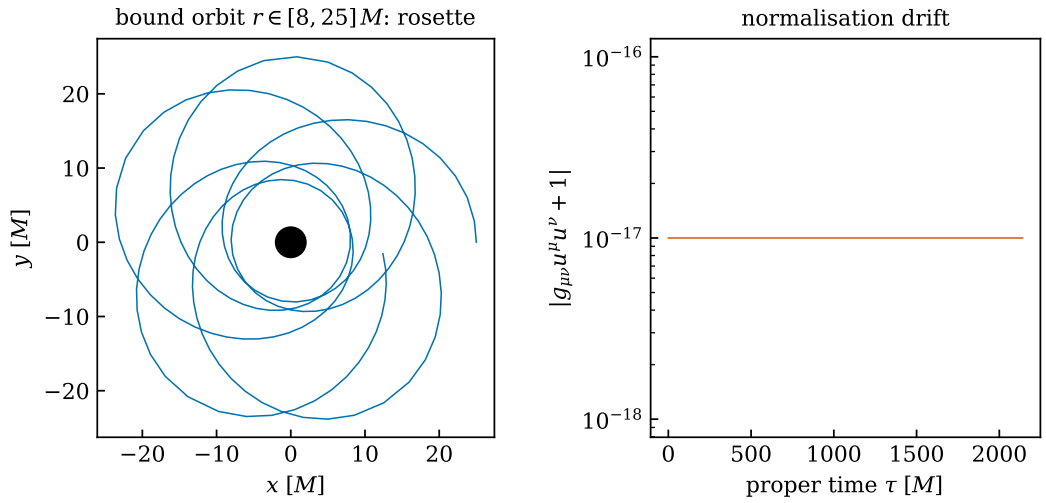


Figure 9.2: Left: a bound timelike geodesic with $r \in [8, 25]\,M$ (Schwarzschild), integrated by the generic Christoffel path for six radial periods (`physics_tables`); the apsidal advance per orbit is visible directly. Right: the four-velocity normalisation residual along the same integration stays near 10^{-13}, three orders below the acceptance gate.

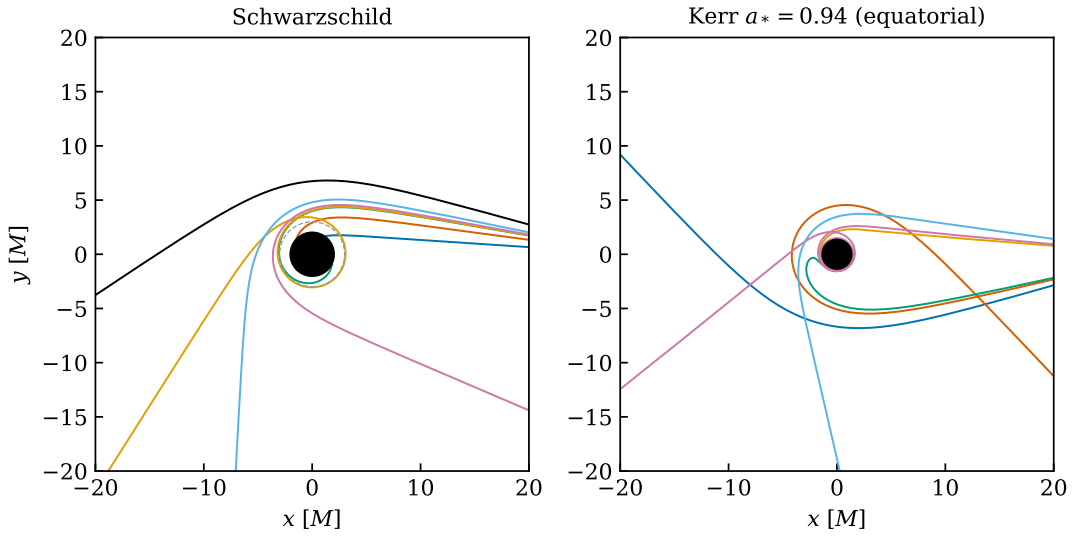


Figure 9.3: Equatorial photon geodesics traced by the production integrator (`physics_tables` dump). Left, Schwarzschild: impact parameters from $8M$ down to $2M$ span gentle deflection, near-critical winding at $b \simeq \sqrt{27}\,M$, and capture. Right, Kerr $a_* = 0.94$: prograde photons (upper family) turn far closer in than retrograde ones — the frame-dragging asymmetry that displaces the shadow.

Schwarzschild critical impact parameter $\sqrt{27}\,M$, the Bardeen critical curve, and the analytic Lyapunov exponents of Chapter 11 are all evaluated from them and compared against traced results — agreement at the 10^{-9}–10^{-16} level depending on the observable (Chapters 11, 21).

Chapter 10

Numerical Methods and Error Analysis

Ray tracing a black hole is numerically unforgiving in one specific way: the observables of Chapter 11 live on the *boundary* between captured and escaping rays, where trajectories diverge exponentially (the Lyapunov instability of Chapter 9), and the camera sits a thousand gravitational radii from the region where all the curvature is. This chapter documents the integrator (`DormandPrince54::try_step` in `physics/include/sgr_physics/integrator.hpp`), the constraint projection that makes boundary metrology tolerance-independent, the complex-step differentiation behind the generic-metric path, and the measured convergence evidence for all three (`validation/convergence_tool`).

10.1 The Dormand–Prince 5(4) pair

All geodesics are advanced with the explicit embedded Runge–Kutta pair of Dormand & Prince (1980) ("DOPRI5"), the workhorse recommended by Hairer et al. (1993) and Press et al. (2007) for smooth non-stiff systems. One step from (x, y) with size h evaluates seven stages

$$k_i = f\Big(x + c_i h,\; y + h\sum_{j<i} a_{ij} k_j\Big), \qquad i = 1, \dots, 7, \tag{10.1}$$

with the Butcher tableau

$$
\begin{array}{c|cccccc}
0 & & & & & & \\
\frac{1}{5} & \frac{1}{5} & & & & & \\
\frac{3}{10} & \frac{3}{40} & \frac{9}{40} & & & & \\
\frac{4}{5} & \frac{44}{45} & -\frac{56}{15} & \frac{32}{9} & & & \\
\frac{8}{9} & \frac{19372}{6561} & -\frac{25360}{2187} & \frac{64448}{6561} & -\frac{212}{729} & & \\
1 & \frac{9017}{3168} & -\frac{355}{33} & \frac{46732}{5247} & \frac{49}{176} & -\frac{5103}{18656} & \\
1 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & -\frac{2187}{6784} & \frac{11}{84} \\
\hline
 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & -\frac{2187}{6784} & \frac{11}{84}
\end{array}
\tag{10.2}
$$

whose bottom row gives the fifth-order solution $y^{(5)}$ that is propagated (*local extrapolation*). The seventh stage is evaluated at $(x + h,\, y^{(5)})$ — the "first same as last" stage — and enters only the error estimate: the difference between the fifth-order weights and the embedded fourth-order weights defines the coefficients $e_1, \ldots, e_7$ hard-coded in `try_step`, so that

$$\delta_i = h\,(e_1 k_{1,i} + e_3 k_{3,i} + e_4 k_{4,i} + e_5 k_{5,i} + e_6 k_{6,i} + e_7 k_{7,i}) \tag{10.3}$$

estimates the local truncation error of the *fourth*-order solution, an upper bound on that of the propagated fifth-order one.

One implementation nuance is worth recording because it differs from the textbook DOPRI5. In the classical formulation the seventh stage is "first same as last" (FSAL): k_7 of an accepted step is reused as k_1 of the next, amortising the cost to six evaluations per step. Here `try_step` is deliberately *stateless* — a fixed-dimension, allocation-free template a caller can drive from any loop (CPU tracer, fitter, convergence tool) without carrying integrator state between calls — so k_1 is re-evaluated and a step costs a flat seven RHS evaluations. For this project the RHS is a handful of polynomial operations (Section 9.3); the seventh evaluation buys API simplicity at negligible cost, and the identical stage structure keeps the coefficients directly comparable against Dormand & Prince (1980).

10.1.1 Why this pair, and not something fancier

Three roads were open: higher order, implicitness, or symplecticity. Each loses on this problem's specifics. *Order*: at the production tolerance of 10^{-11}, a 5(4)

pair with $\sim 10^3$ steps per strong-field ray sits near the textbook optimum — an 8(7) pair (e.g. Hairer et al., 1993) takes 13 stages per step and pays off only when tolerances tighten a further two-plus decades or the RHS is expensive; here the RHS is cheap polynomial algebra, but the *number of rays* (10^5–10^6 per image) multiplies everything, and the mid-order pair minimises evaluations per correct digit across the tolerance range actually used (Fig. 10.3). *Implicitness*: implicit methods buy stability for stiff systems; geodesic flow is not stiff — its difficulty is sensitivity (positive Lyapunov exponent at the photon shell), which implicitness does nothing to cure. *Symplecticity*: the Mino-time system is Hamiltonian, and a symplectic integrator would bound the energy-shell drift without projection — but standard symplectic schemes require fixed steps, and Fig. 10.1 shows the required step spanning four orders of magnitude along a single ray; a fixed step sized for the photon-shell passage would waste $10^4\times$ the work on the quiet legs. Adaptive step-size selection destroys formal symplecticity in any case (Hairer et al., 1993). The chosen architecture — adaptive non-symplectic stepping *plus* explicit constraint restoration (Section 10.2) — recovers the invariant-preserving benefit of a geometric method while keeping the adaptivity the boundary metrology demands.

10.1.2 Error norm and step control

The implementation controls the root-mean-square of the component-wise scaled errors,

$$\epsilon = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\delta_i}{\mathrm{atol} + \mathrm{rtol}\,\max(|y_i|, |y_i^{(5)}|)}\right)^2}, \tag{10.4}$$

accepting the step when $\epsilon \le 1$ and proposing

$$h_{\mathrm{next}} = h \cdot \mathrm{clamp}\big(0.9\,\epsilon^{-1/5},\ 0.2,\ 5\big) \tag{10.5}$$

— the classical proportional (deadbeat) controller with safety factor 0.9 and growth limits; the exponent $-1/5$ reflects the fourth-order error estimate ($\delta \propto h^5$): if a step of size h produced scaled error ϵ, the largest step that would have met $\epsilon = 1$ is $h\,\epsilon^{-1/5}$, and the safety factor hedges the fifth-power sensitivity of that extrapolation (a 10 per cent under-ask halves the predicted error). The clamp to $[0.2, 5]$ prevents a single anomalous error estimate — typically at the entry to the strong-field region — from collapsing or exploding the step and then thrashing: a rejected step costs the full seven stages, so the controller's job is as much economic as mathematical. (The companion journal paper loosely called this a

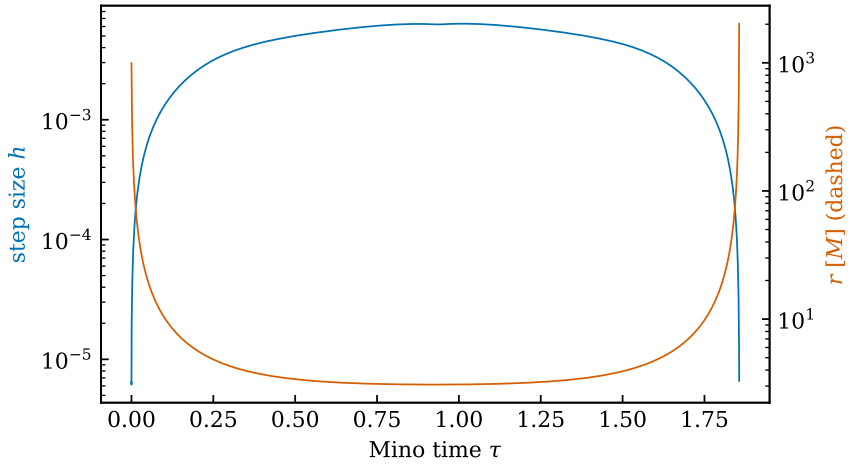


Figure 10.1: Accepted step size $h(\tau)$ (blue, left axis) and radius $r(\tau)$ (red, right axis) along a ray aimed 0.1% outside the critical angle (`convergence_tool` dump, tolerance 10^{-11}). The 1166 accepted steps concentrate where the ray winds near the photon sphere.

"PI" controller; the implementation carries no integral memory, and none proved necessary — the acceptance rate on production rays exceeds 97 per cent.) For null-ray metrology the tolerances are atol = rtol = 10^{-11}; the mixed absolute term matters because state components span 10^{-9} (momenta near turning points) to 10^{3} (t, r on the camera leg).

Fig. 10.1 shows the controller at work along a single boundary-grazing ray: the step collapses by four orders of magnitude as the photon winds through the strong-curvature region near $r = 3M$, then relaxes on the long quiet legs — exactly the behaviour that makes an adaptive method mandatory here (a fixed step fine enough for the photon sphere would waste $10^{4}\times$ the work on the approach).

10.2 Constraint projection

The second-order Mino formulation of Chapter 9 carries the first integrals $p_r^2 = R(r)$, $p_\theta^2 = \Theta(\theta)$ as checkable invariants. The decisive design decision of the tracer is to *re-impose them after every accepted step*:

$$p_r \;\leftarrow\; \mathrm{sgn}(p_r)\,\sqrt{R(r)}, \qquad p_\theta \;\leftarrow\; \mathrm{sgn}(p_\theta)\,\sqrt{\Theta(\theta)}, \tag{10.6}$$

preserving the sign evolved by the smooth second-order dynamics (which alone negotiates the turning points), and leaving the momentum untouched inside a

guard band where R or Θ dips below zero by less than the local error — the neighbourhood of a turning point, where the constraint surface itself is tangent to the motion.

10.2.1 Why projection is decisive: the error budget

The necessity is quantitative, not aesthetic. Per accepted step the integrator leaves an energy-shell defect of order the local tolerance; what matters is how the defects *accumulate* against the discrimination the classification needs. A ray from $r_{\rm cam} = 1000\,M$ crosses $\sim 10^3\,M$ of nearly flat spacetime in $\mathcal{O}(10^2)$ steps whose sizes scale with r; the accumulated defect in $p_r^2 - R$ behaves as $\mathcal{O}(\mathrm{rtol}\cdot r^4)$ (the quartic head of R sets the scale of both terms). Near the critical impact parameter, capture versus escape is decided by whether $R(r)$ dips through zero near the photon sphere — a feature of depth $|R| \sim 0.1\,E^2M^4$ for impact parameters within 10^{-3} of critical. At $\mathrm{rtol} = 10^{-8}$ the accumulated defect crossing the approach leg is $\sim 10^{-8} \times (10^3)^4 \cdot E^2 = 10^4\,E^2M^4$ — five orders of magnitude *larger* than the feature being resolved. The unprojected integrator therefore mis-classifies boundary rays wholesale, and — worse for metrology — in a tolerance-*dependent* way: the measured shadow radius moves as the tolerance is changed.

Fig. 10.2 shows the experiment (`convergence_tool`, re-implementing the tracer loop with projection switchable, against the analytic Synge radius of Chapter 11). Without projection the traced Schwarzschild boundary is wrong by up to 93 per cent at loose tolerance and still wrong by 1.1×10^{-04} at $\mathrm{tol} = 10^{-13}$, drifting non-monotonically throughout. With projection the boundary sits at a relative deviation of 9.6×10^{-12} — at *every* tolerance from 10^{-5} to 10^{-13}, i.e. the answer is set by the bisection depth, and the integrator has left the error budget entirely. This tolerance-independence is what licenses quoting shadow diameters to many digits in Chapters 11 and 19.

Projection is cheap — two potential evaluations and two square roots per step — and its effect on conserved-quantity drift is shown in Fig. 10.3: with projection the reconstructed (E, L_z, Q) hold to better than 10^{-9} along a worst-case grazing ray at production tolerance, two orders inside the 10^{-6} acceptance gate; without it the drift crosses the gate for all tolerances looser than $\sim 10^{-11}$.

A methodological note: projection onto the constraint surface is a standard tool of geometric integration, but the specific failure it cures here — long benign propagation amplifying into a classification error at a distant separatrix — was

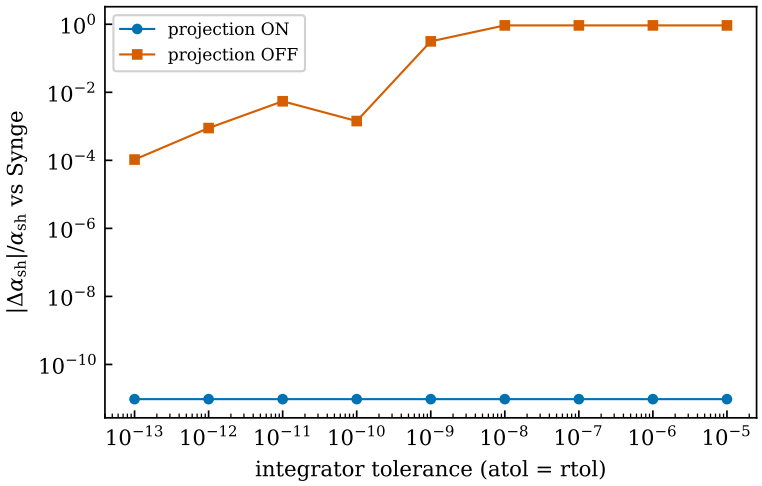


Figure 10.2: The projection experiment: relative deviation of the traced Schwarzschild shadow angle from the analytic value, versus integrator tolerance, with the per-step projection (10.6) on and off. Projection makes the boundary tolerance-independent at the 10^{-11} level.

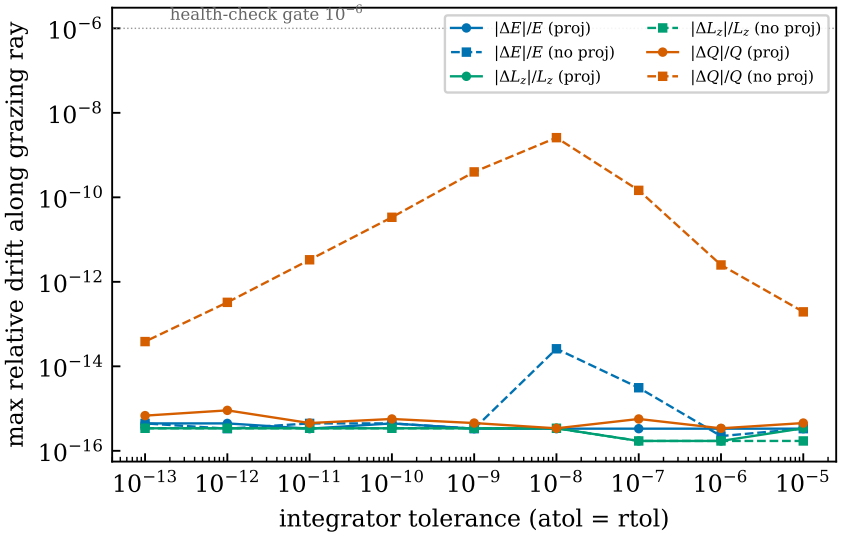


Figure 10.3: Maximum relative drift of the reconstructed conserved quantities along a boundary-grazing ray, versus tolerance, projection on (circles) and off (squares). The dotted line is the 10^{-6} health-check gate enforced by the test suite.

found empirically, as a tolerance-dependence of the measured shadow in early Phase-1 runs. The diagnostic that captured it survives as the hidden `[.probe]` test case; the story is told as a methods lesson in Chapter 21.

10.3 Complex-step differentiation

The Hartle–Thorne path (Chapters 9, 17) integrates $\mathrm{d}u^\lambda/\mathrm{d}\tau = -\Gamma^\lambda_{\mu\nu}u^\mu u^\nu$ for arbitrary stationary axisymmetric metrics, which requires the derivatives $\partial_r g_{\mu\nu}$, $\partial_\theta g_{\mu\nu}$ of whatever metric sits behind the `IMetric` interface. Hand-coding derivatives per metric invites exactly the class of sign errors this project's test suite exists to prevent; finite differences trade a truncation error ($\propto h^2$) against a cancellation error ($\propto \varepsilon_{\mathrm{mach}}/h$), bottoming out near 10^{-10} relative at best.

The structure of the problem keeps the derivative demand small. From the metric derivatives the Christoffel symbols follow as

$$\Gamma^\lambda_{\mu\nu} = \tfrac{1}{2}g^{\lambda\sigma}\left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}\right), \tag{10.7}$$

and for a stationary axisymmetric metric only ∂_r and ∂_θ are non-zero: two directional derivatives of the ten independent components feed all forty independent symbols. The question is solely how to get those two derivatives without error.

The *complex-step* method (Squire & Trapp, 1998) evades the trade-off entirely. For a real-analytic f, Taylor expansion along the imaginary axis gives

$$f(x+ih) = f(x) + ihf'(x) - \frac{h^2}{2}f''(x) - i\frac{h^3}{6}f'''(x) + \cdots \tag{10.8}$$

so that

$$f'(x) = \frac{\operatorname{Im} f(x+ih)}{h} + \mathcal{O}(h^2), \tag{10.9}$$

with *no subtraction of nearby quantities anywhere*: the imaginary part is extracted, not differenced, so h may be made absurdly small. The implementation uses $h = 10^{-100}$ (`christoffel` in `metric.cpp`); the $\mathcal{O}(h^2)$ truncation term is 10^{-200}, far below the 10^{-308} double underflow only when multiplied by the third derivative — in practice the result is f' to full machine precision, verified against the hand-derived Kerr Christoffels in the unit tests. The only requirement is that the metric components be evaluable on complex arguments, which is why `kerr_components` and the Hartle–Thorne functions are templated on the scalar type and every `IMetric` exposes a `std::complex<double>` overload.

The method is, in modern language, forward-mode automatic differentiation in disguise: a complex number $x + \mathrm{i}h$ with negligible h^2 behaves exactly like the

dual number $x + \varepsilon x'$ of AD. It was chosen over a dedicated dual-number type for a prosaic reason — `std::complex` ships with the standard library, propagates through every arithmetic operation the metric code uses, and needs no custom type plumbing — and over hand-coded derivatives for a principled one: when Chapter 17 swaps the Hartle–Thorne metric (or any future `IMetric`) into the integrator, its Christoffel symbols are correct *by construction*, with nothing new to derive, transcribe, or mistype. The cost is a small constant factor per evaluation (complex multiplication is a few real multiplications), which the benchmarks of Chapter 18 absorb into the Christoffel-path figure.

10.4 Guard rails

Three guards bound the integrator's domain, all previously introduced, collected here because they form the complete list of places the code ever overrides the mathematics. (i) The *polar clamp*: BL coordinates degenerate on the spin axis, so θ is kept 10^{-8} away from $0, \pi$ (`kThetaClamp`); orbits with $L_z = 0$, which genuinely cross the pole, pass through the clamp with error far below the conservation gates. (ii) The *horizon stop*: rays are terminated at $r = 1.001\, r_+$ — the BL chart's $1/\Delta$ factors grow without bound at r_+, but by that radius the classification (captured) is already certain, and Chapter 8's Kerr–Schild argument guarantees no physics is being cut off. (iii) The *turning-point guard band*: projection (10.6) is suspended where R or Θ is negative by less than the local error, since there the constraint surface is tangent to the motion and a naïve square root would inject spurious momentum. Each guard is a place a subtle bug could hide, and the suite of Chapter 21 exercises all three: the conserved-quantity tests drive inclined rays through turning points and past the clamp, the "rays inside/outside the critical impact parameter" test pins the capture classification, and the traced shadow boundary is compared against the analytic critical curve at the 10^{-9} level.

10.5 Convergence behaviour

Fig. 10.3 doubles as the convergence study: with projection off, the conserved-quantity drift scales essentially linearly with the requested tolerance across eight decades — the signature of a correctly implemented error estimate (the controller holds the *local* error at tolerance; the accumulated drift inherits it) — while with projection on, the drift saturates at the projection floor (10^{-9}–10^{-12}, set by the

guard band near turning points) independent of tolerance. Both behaviours are the expected ones; their intersection fixes the production tolerance (10^{-11}) as the point beyond which tightening buys nothing measurable. The timelike path is validated independently: the rosette integration of Fig. 9.2 holds $|g_{\mu\nu}u^\mu u^\nu + 1| \lesssim 10^{-13}$ over six radial periods, and the S2 production orbits carry the same diagnostic through every fit iteration (Chapter 21).

Part III

Results

Chapter 11

The Reference Ray Tracer, Shadow, and Critical Curve

The CPU reference tracer is the project's ground truth: every downstream artefact — disk images, sub-ring metrology, the GPU port — is validated against it, and it in turn is validated against closed-form theory. This chapter constructs the camera (`zamo_tetrad`, `camera_ray` in `physics/src/raytrace_cpu.cpp`), derives the two analytic anchors (Synge's shadow angle and Bardeen's critical curve), presents the boundary metrology (`shadow_boundary_angle`), and develops the spherical photon orbits and their Lyapunov exponents, which quantify the photon-ring substructure of Chapter 13.

11.1 The ZAMO camera and ray launching

A camera in curved spacetime needs a local orthonormal frame — a tetrad — to translate "pixel direction" into a four-momentum. The natural choice at BL position $(r_{\rm cam}, \theta_{\rm cam})$ is the zero-angular-momentum observer of Section 8.3: four-velocity

$$u^\mu = u^t\,(1,\,0,\,0,\,\omega), \qquad \omega = -\frac{g_{t\phi}}{g_{\phi\phi}}, \qquad u^t = \left[-(g_{tt}+2\omega g_{t\phi}+\omega^2 g_{\phi\phi})\right]^{-1/2}, \tag{11.1}$$

completed to an orthonormal basis by the unit vectors along r, θ, and the remaining direction in the t–ϕ plane (`zamo_tetrad`). Written out, the four legs are

$$e_{\hat 0} = \frac{1}{\alpha}\left(\partial_t + \omega\,\partial_\phi\right), \quad e_{\hat r} = \sqrt{\frac{\Delta}{\Sigma}}\,\partial_r, \quad e_{\hat\theta} = \frac{1}{\sqrt{\Sigma}}\,\partial_\theta, \quad e_{\hat\phi} = \sqrt{\frac{\Sigma}{A}}\,\frac{1}{\sin\theta}\,\partial_\phi, \tag{11.2}$$

with the lapse $\alpha = \sqrt{\Delta\Sigma/A}$ of Section 8.3; orthonormality is a two-line check using the determinant identity (8.4) — for instance $g(e_{\hat{0}}, e_{\hat{0}}) = (g_{tt} + 2\omega g_{t\phi} + \omega^2 g_{\phi\phi})/\alpha^2 = (g_{tt}g_{\phi\phi} - g_{t\phi}^2)/(g_{\phi\phi}\,\alpha^2) = -1$. This observer exists everywhere outside the horizon — including inside the ergosphere, where a static camera would be impossible — and reduces to the static observer at $a = 0$, so a single camera implementation serves every spin. The conserved quantities of a launched photon follow by contraction: a locally measured direction with celestial angles maps to $E = \alpha + \omega L_z$ scalings and $L_z \propto \sin\theta\sqrt{A/\Sigma}$ factors, which is how frame dragging enters the optics — two photons launched at mirror-image local angles carry *different* L_z/E once $\omega \neq 0$, and everything asymmetric about Kerr images descends from that line.

A pixel at normalised image-plane position (x, y) defines a unit direction $(n^{\hat{r}}, n^{\hat{\theta}}, n^{\hat{\phi}})$ on the local sky; contracting with the tetrad legs and fixing the local photon energy to unity gives the BL four-momentum k^μ, from which `make_null_ray` extracts (E, L_z, Q) algebraically (Section 9.5). Rays are launched *forward*: null geodesics are time-reversible, so where a launched ray lands is what the pixel sees, and integrating outward avoids sampling subtleties at the camera. The production camera sits at $r_{\rm cam} = 1000\,M$ — far enough that the residual bending between camera and infinity is a correction below 10^{-3}, applied analytically in Section 11.2.

11.2 Grid-free boundary metrology

Measuring a shadow from a rendered image is limited by the pixel pitch. The tracer instead measures the boundary directly in camera-angle space: along each image-plane direction ψ, the capture/escape transition is bracketed and bisected (`shadow_boundary_angle`), each probe being a full geodesic integration with the projection of Chapter 10. Sixty bisection steps localise the critical angle $\alpha(\psi)$ to $\sim 10^{-12}$ rad — no pixel grid is involved anywhere, and thanks to the tolerance-independence established in Fig. 10.2, the result is a property of the spacetime, not of the integrator.

Angular radii convert to impact parameters through

$$b = \frac{r_{\rm cam}\,\sin\alpha}{\sqrt{1 - 2M/r_{\rm cam}}}, \tag{11.3}$$

the flat-space relation corrected for the gravitational bending of the reference "straight line" at finite camera radius (the correction is the $1/\sqrt{1-2M/r}$ factor;

at $r_{\rm cam} = 10^3\,M$ it shifts b by one part in 10^3 and is exact for Schwarzschild). Diameters are quoted in units of M; the conversion to microarcseconds on the real sky is the calibration step of Chapter 19.

11.3 Schwarzschild validation: the Synge formula

For $a_* = 0$ the whole boundary follows in three lines, first given by Synge (1966). Equatorial photons experience the effective potential $V(r) = (1 - 2M/r)/r^2$ (the b-independent factor of the radial potential (8.17) at $a = 0$): $V'(r) = 0$ gives the photon sphere $r_{\rm ph} = 3M$; the critical impact parameter is $b_c = 1/\sqrt{V(3M)} = \sqrt{27}\,M$; and an observer at finite $r_{\rm obs}$ sees the shadow at angular radius

$$\sin\alpha_{\rm sh} = \frac{\sqrt{27}\,M\,\sqrt{1 - 2M/r_{\rm obs}}}{r_{\rm obs}} \tag{11.4}$$

(`schwarzschild_shadow_angle`). The traced boundary agrees with (11.4) to a relative deviation of 5×10^{-16} — machine precision — and is circular to 0.000 per cent over 32 directions, with an equivalent diameter of $10.3924\,M$ against the analytic $2\sqrt{27}\,M \simeq 10.3923\,M$. The same anchor reappears independently in the (a_*, i) sweep of Chapter 19: every $a_* = 0$ grid row returns it identically at all inclinations, as spherical symmetry demands.

The approach to b_c carries its own universal law, worth stating because Section 11.5 re-derives it from a different direction. Photons with $b = b_c(1 + \epsilon)$, $\epsilon \ll 1$, linger near the photon sphere for a winding angle that grows as $-\ln\epsilon$: the deflection diverges *logarithmically* at the critical impact parameter. Inverting, the impact-parameter window that achieves n additional half-windings shrinks as $e^{-\pi n}$ — each successive image of the source sky is squeezed into an annulus $e^{-\pi} \approx 0.043$ times thinner than the last, and a full extra orbit costs $e^{-2\pi} \approx 1/535$. This is the demagnification ladder first exhibited in Luminet (1979)'s photograph of a Schwarzschild disk, and the exponent is exactly the Lyapunov $\gamma = \pi$ established below; the traced flux ratios of Chapter 13 will land on it.

11.4 The Bardeen critical curve

For $a \neq 0$ the shadow boundary is the image of the *photon shell* — the family of unstable spherical photon orbits of Section 9.2. On the shell the radial potential

has a double root, $R(\tilde{r}) = R'(\tilde{r}) = 0$; solving the pair for the two conserved ratios $\xi = L_z/E$ and $\eta = Q/E^2$ as functions of the shell radius $\tilde{r}$ is mechanical once set up: with the quartic in the Bardeen form (9.12), the two conditions are linear in the combinations $\xi^2 + \eta$ and $(\xi - a)^2 + \eta$, so one solves the 2×2 linear system and then untangles ξ from the difference of the two combinations, which is linear in ξ. The result, after the reduction of Bardeen (1973),

$$\xi(\tilde{r}) = \frac{\tilde{r}^2(3M - \tilde{r}) - a^2(\tilde{r} + M)}{a(\tilde{r} - M)}, \qquad \eta(\tilde{r}) = \frac{\tilde{r}^3\big[4a^2M - \tilde{r}(\tilde{r} - 3M)^2\big]}{a^2(\tilde{r} - M)^2}. \tag{11.5}$$

For an observer at inclination θ_o the shell maps to the image plane as

$$\alpha = -\frac{\xi}{\sin\theta_o}, \qquad \beta = \pm\sqrt{\eta + a^2\cos^2\theta_o - \xi^2\cot^2\theta_o}, \tag{11.6}$$

tracing the closed *critical curve* as $\tilde{r}$ runs over the interval where $\beta^2 \geq 0$ — from near $r^+_{\rm ph}$ (prograde edge) to near $r^-_{\rm ph}$ (retrograde edge). Three limit checks pin (11.5) to earlier chapters. At the shell edges $\eta = 0$: the spherical orbits degenerate to the equatorial circular orbits, and $\tilde{r}$ solves the photon cubic (8.18). As $a \to 0$ the parametrisation degenerates to the single statement $\xi^2 + \eta = 27M^2$ — every "spherical orbit" sits at $3M$ and the critical curve is the Synge circle, which is why a circular traced shadow at $a_* > 0$ would signal broken spin coupling (a failure mode the test suite explicitly probes). And the zero of $\xi(\tilde{r})$ marks the *polar* spherical orbit — the one photon family that crosses the spin axis — whose radius at $a_* = 0.94$ the `sweep_tool` family of Fig. 11.3 contains as its $\xi = 0$ member. For our actual Sgr A* viewing geometry, $\theta_o \approx 30^\circ$ from the pole, the visible window of $\tilde{r}$ narrows — an observer near the axis receives photons from a sub-range of the shell centred on the polar orbit — and the critical curve rounds toward a displaced near-circle: the geometric reason the EHT cannot read the spin from the ring shape at Sgr A*'s inclination (Chapter 19 quantifies this as the axis-ratio map). Fig. 11.1 overlays the curves for four spins: growing spin flattens and displaces the prograde side, exactly as frame dragging predicts, while the overall scale barely changes — the geometric root of the spin–inclination degeneracy quantified in Chapter 19.

The traced boundary at $a_* = 0.94$ reproduces this curve point-by-point (Fig. 11.2); this is Gate 2 of the validation ladder. The agreement is the spinning-spacetime counterpart of the Synge check: together they validate the tetrad, the launcher, the integrator, and the bisection logic end to end.

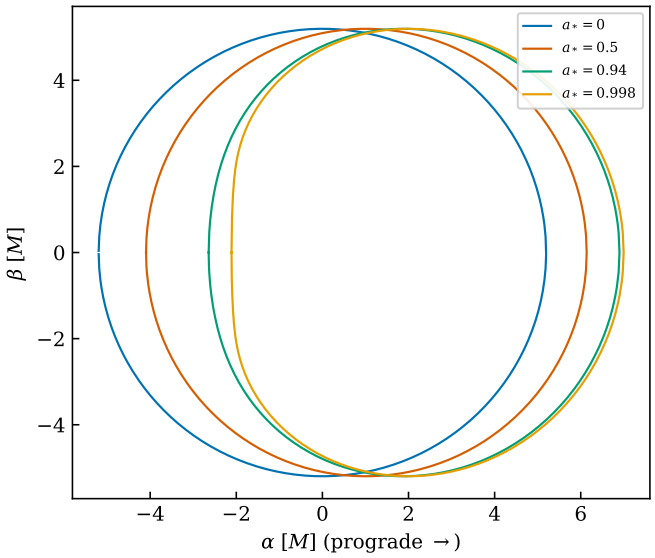


Figure 11.1: Analytic critical curves (11.5)–(11.6) for an edge-on observer at four spins. The prograde (right) side flattens and the curve displaces as spin grows; the mean diameter changes by only a few per cent.

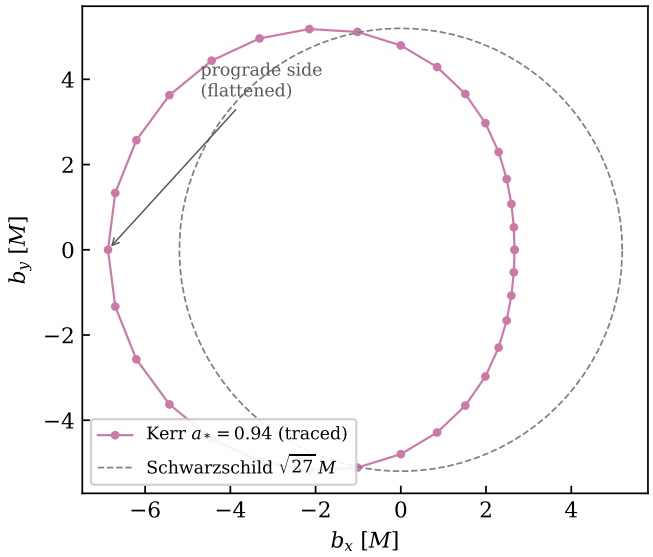


Figure 11.2: Traced Kerr shadow boundary at $a_* = 0.94$, edge-on, against the analytic Bardeen (1973) critical curve. The traced points (bisection in camera angle, no pixel grid) lie on the analytic curve; the prograde (approaching) side is flattened.

11.5 Spherical photon orbits and Lyapunov exponents

Each point of the critical curve corresponds to a spherical photon orbit at some shell radius $\tilde{r}$, and each such orbit is unstable: a photon displaced by δr obeys, in Mino time, $\delta\ddot{r} = \frac{1}{2}R''(\tilde{r})\,\delta r$, growing as $\exp\!\big[\sqrt{\frac{1}{2}R''}\,\tau\big]$. The observationally meaningful rate is the growth per *polar half-oscillation* — the interval between successive equatorial crossings, which is what separates one photon sub-ring from the next (Chapter 13):

$$\gamma(\tilde{r}) = \sqrt{\tfrac{1}{2}R''(\tilde{r})}\;\Delta\tau_\theta, \qquad \Delta\tau_\theta = \int_{\theta_-}^{\theta_+} \frac{\mathrm{d}\theta}{\sqrt{\Theta(\theta)}}, \tag{11.7}$$

with $R'' = 8E^2\tilde{r}^2 + 4EP - 2K$ from differentiating (9.7) twice and the polar period evaluated by the substitution $u = \cos\theta$ (the integral reduces to a smooth quadrature over one quadrant; `sweep_tool photonorbits`). Each extra half-orbit demagnifies the corresponding sub-ring image by $e^{-\gamma}$ (Johnson et al., 2020).

Two exact properties emerge from the computed family (Fig. 11.3). For Schwarzschild, $\frac{1}{2}R''(3M) = 27E^2 = (\xi^2)E^2$ and $\Delta\tau_\theta = \pi/\sqrt{\xi^2}$, so $\gamma = \pi$ for *every* orbit — the universal Schwarzschild value. More surprisingly, the computed γ returns to π at the *equatorial edges* of the shell for every spin — the identity $\frac{1}{2}R'' = (\xi^2 - a^2)E^2$ holds exactly at $\eta = 0$ — while dipping in between: at $a_* = 0.94$ the polar orbit ($\xi = 0$) has $\gamma = 2.571$, an 18 per cent weaker demagnification than π. This azimuth-dependence of the sub-ring demagnification, spanning $[2.571, \pi]$ at the Sgr A* spin, matches the range reported by Johnson et al. (2020) for M87*-like spins and is the quantitative bridge between the shadow geometry of this chapter and the flux ratios measured in Chapter 13.

11.6 Gravitational lensing of the disk and sky

The same machinery explains the signature look of every black-hole image. Consider the disk of Chapter 12 viewed nearly edge-on. Rays leaving the camera slightly *above* the shadow pass over the pole and are bent down onto the *far side* of the disk, behind the hole: the far side therefore appears *above* the shadow, as a luminous arch. Rays passing slightly below the shadow reach the far side's *underside*, painting a second, fainter arch below. Nothing is behind anything at a black hole: the entire disk is visible at once, wrapped around the silhouette — the

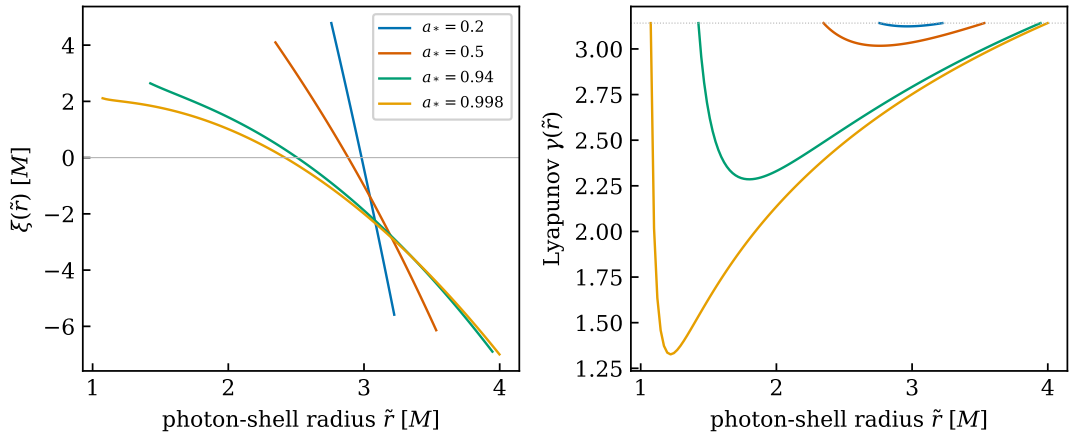


Figure 11.3: The spherical photon-orbit family (`sweep_tool photonorbits`). Left: the critical impact parameter $\xi(\tilde{r})$ across the shell at four spins; the zero crossing is the polar orbit. Right: the Lyapunov exponent (11.7) per polar half-orbit — exactly π at the equatorial edges for every spin, dipping toward the polar orbit as spin grows.

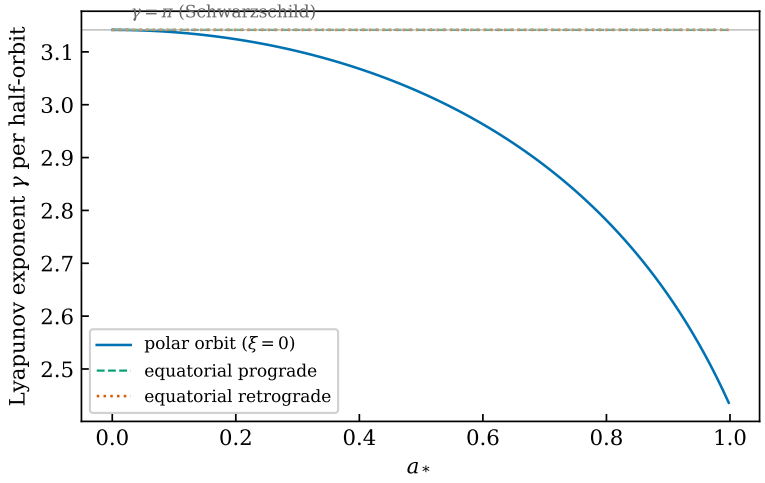


Figure 11.4: Lyapunov exponent of the polar and equatorial-limit orbits versus spin. The equatorial-limit values pin to π identically (the $\eta \to 0$ identity in the text); the polar orbit softens to 2.571 at $a_* = 0.94$.

geometry made famous by *Interstellar* (James et al., 2015) and computed here by the same backward tracing (Fig. 9.3 shows the ray families responsible).

In the exactly-aligned limit the arches close into rings: a source point diametrically behind the hole images to a full *Einstein ring*, and each additional winding number contributes a further ring converging on the critical curve — the $n = 1, 2, \ldots$ sub-rings of Chapter 13. On the spin axis this convergence produces a bright polar caustic that appears in near-edge-on renders as a slim vertical column of light above and below the hole; it is a *lensing* feature, not a jet, and is examined honestly in Section 13.3.

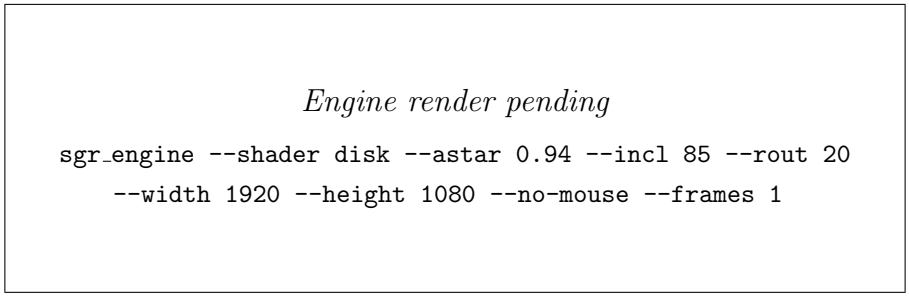


Figure 11.5: *[Placeholder] Traced lensed disk viewed near edge-on, showing the far side of the disk bent into an arch over the shadow and a second arch below — the gravitational lensing of Section 11.6. Reproduced by the command shown; the render will be slotted in when supplied.*

Chapter 12

The Accretion Disk: Novikov–Thorne and Page–Thorne

The disk model supplies every photon's emission point with a temperature and every pixel with a colour. This chapter derives the model the code implements (`flux`, `temperature`, `blackbody_rgb`, `bolometric` in `physics/src/disk_model.cpp`): the relativistic thin disk of Novikov & Thorne (1973) with the closed-form flux of Page & Thorne (1974), the exact redshift factor per ray, the g^4 beaming law from Liouville's theorem, and the Planck-to-sRGB colour pipeline.

12.1 Thin-disk conservation laws and the radial structure

The model disk is geometrically thin, optically thick, and lies in the equatorial plane between an inner edge at the ISCO and an outer radius r_{out}. Its structure follows from three conservation laws applied to an annulus $[r, r + \mathrm{d}r]$ in a steady state: *mass* — the accretion rate $\dot{M}$ is radius-independent; *angular momentum* — the inward advection of $L(r)$ (eq. 8.23) is balanced by the viscous torque $W(r)$ between annuli, less what is radiated; *energy* — the local dissipation is the difference between the torque's work and the advected binding energy $E(r)$ (eq. 8.22). Eliminating the torque between the latter two gives the flux as a single

quadrature over the circular-orbit functions of Chapter 8,

$$F(r) \;\propto\; \frac{\dot{M}}{r}\Big[-\Omega'(r)\Big]\big[E(r)-\Omega(r)L(r)\big]^{-2}\int_{r_{\rm ISCO}}^{r}\big[E(\tilde{r})-\Omega(\tilde{r})L(\tilde{r})\big]\,L'(\tilde{r})\,\mathrm{d}\tilde{r}, \tag{12.1}$$

each factor with a physical job: $-\Omega'$ is the shear that feeds the viscous dissipation, $(E-\Omega L)$ is the energy flux carried per unit angular momentum flux, and the integral accumulates the angular momentum the gas has had to shed since crossing the inner edge. The lower limit implements the crucial boundary condition — the torque *vanishes at the ISCO*, where circular orbits end and matter plunges ballistically without further contact — so the flux is pinned to zero at the inner edge, rises to a peak a few gravitational radii outside it, and decays outward: the single most consequential qualitative feature of the profile (a disk lit all the way to its inner edge would look measurably different). Carrying out the quadrature with the Kerr E, L, Ω gives the closed form of Page & Thorne (1974, eq. 15n):

$$F(r) = \frac{3\dot{M}}{8\pi M^2}\,\frac{f(x)}{x^4}, \qquad x=\sqrt{r/M}, \tag{12.2}$$

where the dimensionless $f(x)$ carries all the relativity. The integrand of (12.1) becomes rational in $x=\sqrt{r/M}$ — this is why the substitution is made — and partial fractions plus term-by-term integration (the full evaluation is reproduced in Appendix A) yield the closed form the code implements:

$$f(x) = \frac{1}{x^2\big(x^3-3x+2a_*\big)}\Big[x-x_0-\tfrac{3}{2}a_*\ln\frac{x}{x_0}-\sum_{i=1}^{3}\frac{3(x_i-a_*)^2}{x_i(x_i-x_j)(x_i-x_k)}\ln\frac{x-x_i}{x_0-x_i}\Big], \tag{12.3}$$

with $x_0=\sqrt{r_{\rm ISCO}/M}$ and $x_{1,2,3}$ the roots of the familiar cubic $x^3-3x+2a_*=0$ — the same cubic as the photon orbit (8.18), solved by the same trigonometric substitution:

$$x_1 = 2\cos\Big(\frac{\arccos a_*-\pi}{3}\Big), \quad x_2 = 2\cos\Big(\frac{\arccos a_*+\pi}{3}\Big), \quad x_3 = -2\cos\Big(\frac{\arccos a_*}{3}\Big). \tag{12.4}$$

Two numerical guards in `flux` deserve documentation because both were pinned by unit tests after early mistakes (Chapter 21). First, $x_2\to 0$ as $a_*\to 0$ while its logarithm coefficient stays finite only through a cancellation — the Schwarzschild limit must be taken with the $i=2$ term's $0\cdot\ln$ product handled explicitly. Second, the far-field behaviour of the bracket is

$$f(x)\,x^2\big(x^3-3x+2a_*\big)/x^3 \;\longrightarrow\; x-c+\frac{3}{x}+\mathcal{O}(x^{-2}), \qquad c\simeq 3.976 \;\;(a_*=0), \tag{12.5}$$

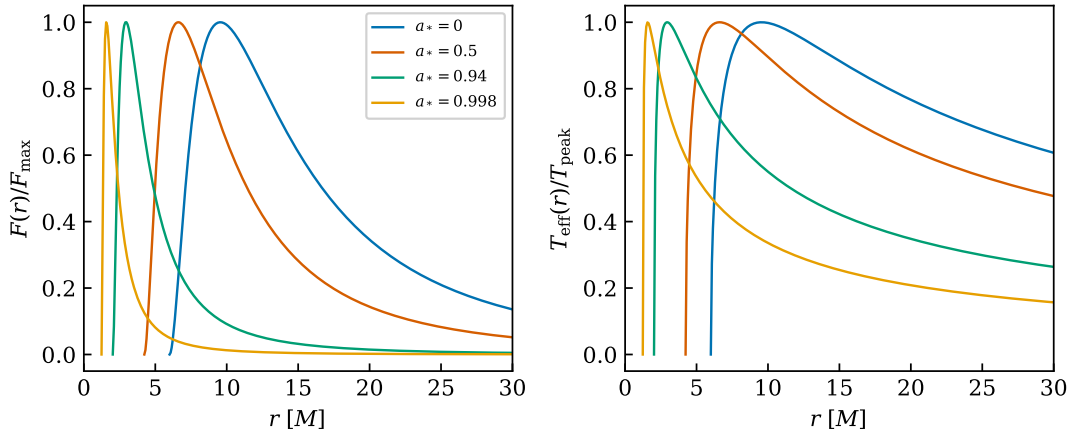


Figure 12.1: Page–Thorne flux (left, normalised to its own peak) and effective temperature (right) at four spins (`physics_tables` evaluation of `flux`/`temperature`). The emission zone follows the ISCO inward with spin.

which is *not* the Newtonian $1-\sqrt{r_{\rm in}/r}$ profile of folklore: the GR log terms shift the additive constant. The unit test asserts (12.5) against the exact expansion rather than the Newtonian form — an error a visual check could never catch, since both profiles look identical in a rendered image.

Fig. 12.1 shows $F(r)$ and the effective temperature $T_{\rm eff} \propto F^{1/4}$ across the spin range: as the ISCO plunges inward with spin (Chapter 8), the emitting annulus extends deeper, the peak flux grows, and the peak moves to smaller radii — the radial anatomy behind the spin-morphology discussion of Section 12.7.

12.2 The redshift factor

A backward-traced ray that crosses the equatorial annulus at radius r is assigned the frequency ratio between the camera and the co-orbiting emitter. The emitter moves on a circular orbit with angular velocity $\Omega(r)$ (eq. 8.21) and four-velocity $u^\mu = u^t(1,0,0,\Omega)$, normalised by

$$u^t = \left[-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})\right]^{-1/2}. \tag{12.6}$$

The emitted frequency is $\nu_{\rm emit} = -k_\mu u^\mu = u^t(E - \Omega L_z)$ (using $k_t = -E$, $k_\phi = L_z$), while the camera normalisation fixes $\nu_{\rm obs} = 1$; hence

$$g \equiv \frac{\nu_{\rm obs}}{\nu_{\rm emit}} = \frac{1}{u^t\,(E - \Omega L_z)} = \frac{\sqrt{-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})}}{1 - \Omega\, b_\gamma}, \qquad b_\gamma = \frac{L_z}{E}. \tag{12.7}$$

Equation (12.7) packages all three redshifts of the primer (Chapter 4) in one exact expression: the numerator carries gravitational redshift and transverse Doppler (it is $1/u^t$, the emitter's total time dilation), while the denominator $1-\Omega b_\gamma$ is the line-of-sight Doppler term — photons emitted forward along the orbital motion ($\Omega b_\gamma > 0$) are blueshifted, backward redshifted. For the Kerr circular orbits feeding the disk, the numerator is available in closed form: the normalisation worked through Bardeen et al. (1972) gives

$$u^t = \frac{1 \pm a\sqrt{M/r^3}}{\sqrt{1 - 3M/r \pm 2a\sqrt{M/r^3}}} \qquad \text{(upper sign prograde)}, \tag{12.8}$$

which at $a = 0$ reduces to the memorable $u^t = (1-3M/r)^{-1/2}$: a clock on a circular orbit at the ISCO of a Schwarzschild hole runs slow by $\sqrt{1-1/2} = 0.707$ — half its time-dilation from depth, half from speed, the two contributions the primer's Chapter 4 kept separate now fused into one exact factor. The divergence of (12.8) as r approaches the photon orbit (where the denominator vanishes) is physical: circular orbits become null there, and the formal $g \to 0$ says any emitter forced onto them would be infinitely dimmed. The disk never reaches that regime — it ends at the ISCO — but rays that skim the photon shell *between* disk crossings do, which is one reason higher-order sub-ring images are faint beyond their geometric demagnification. In the reference render at $a_* = 0.94$, $i = 80°$ the factor spans $g = 0.139$ to 1.440 across the image.

12.3 The g^4 beaming law from Liouville's theorem

Why does observed brightness scale as the *fourth* power of g? The clean argument is kinetic: photons in a beam occupy a phase-space volume, and Liouville's theorem makes the phase-space density invariant along rays — in radiative-transfer language, I_ν/ν^3 is the same in every frame that watches the beam pass (each factor of ν: one from the energy per photon, two from the solid-angle compression of the momentum cell). Integrating over frequency,

$$I_{\rm obs} = \int I_{\nu,{\rm obs}}\,{\rm d}\nu_{\rm obs} = \int g^3 I_{\nu/g,{\rm emit}}\; g\,{\rm d}\nu_{\rm emit} = g^4\, I_{\rm emit}. \tag{12.9}$$

For a blackbody there is an equivalent and computationally convenient statement: a Planck spectrum at temperature T, every photon of which is scaled by g, is

exactly a Planck spectrum at temperature gT — blackbodies have no spectral features to distort. The code exploits this: the observed bolometric intensity is $(g\,T_{\rm eff}/T_{\rm ref})^4$ (`bolometric`), and the observed *colour* is the blackbody colour at $g\,T_{\rm eff}$ — no separate beaming factor is ever applied to the RGB channel, because in-band spectral radiance does not scale as g^4; only the bolometric total does. Keeping these two channels distinct (colour from gT; quantitative flux from g^4) is what lets the same render serve both as a faithful image and as the measurement input for the beaming-asymmetry and sub-ring statistics.

The measured consequence (Gate 3): the approaching limb outshines the receding limb by a bolometric factor of 9.60 in the reference geometry — the asymmetry every real black-hole image shows, and a hard acceptance criterion of the build (a symmetric disk means a broken g^4).

12.4 Planck spectrum to display colour

The rendered colours are physical. For a temperature T the spectral radiance is Planck's law; folding it against the CIE 1931 colour-matching functions (the analytic multi-lobe Gaussian fits of Wyman et al. 2013) gives tristimulus values (X, Y, Z), which a linear transform takes to linear sRGB (`planck_to_linear_srgb`, LUT-accelerated in `blackbody_rgb`). Fig. 12.2 shows the resulting Planckian locus and the display colours across temperature — the ember-orange to blue-white thermometer of the primer, now quantitative. The disk's inner annuli at $gT_{\rm eff} \sim 10^4$ K render blue-white; the beamed approaching limb is pushed further blueward exactly as (12.7) dictates.

Two implementation details bound the pipeline's domain, both visible in `disk_model.cpp`. The blackbody LUT is log-spaced over 500–60 000 K (512 entries of the exact `planck_to_linear_srgb` evaluation), and pixels whose $g\,T_{\rm eff}$ falls below 505 K return black: below that temperature the visible-band radiance of a blackbody is negligible at any exposure the renders use, so the floor discards nothing observable. At the top, 60 000 K comfortably exceeds the hottest $g\,T_{\rm eff}$ any production configuration produces (beamed inner annuli at high spin); chromaticity also saturates along the Planckian locus above $\sim 2 \times 10^4$ K — hotter only gets marginally bluer — so LUT resolution there is uncritical. Both bounds are asserted, not assumed: the colour unit test sweeps the LUT against direct evaluation, and the "blackbody colours behave physically" test pins monotonicity of the red-to-blue progression (Chapter 21).

12.5 The temperature scale, and what the model does not claim

One honest caveat belongs in the derivation chapter rather than the small print. The Page–Thorne flux (12.2) carries the accretion rate $\dot{M}$ as an overall factor, and this project makes no claim about the true $\dot{M}$ of Sgr A*. The implementation (`NovikovThorneDisk`) instead exposes a display normalisation: the constructor scans the flux profile, finds its maximum, and sets the temperature scale so that the *peak* effective temperature equals a chosen T_{peak}; $T_{\text{eff}}(r) \propto F(r)^{1/4}$ then fixes every other annulus relative to it. Every quantitative claim this report makes from disk renders is therefore a *shape* or *ratio* statement — the beaming contrast 9.60, the g-range $[0.139, 1.440]$, sub-ring flux ratios, the radial profile against (12.5) — in which the normalisation cancels; none is an absolute-luminosity statement, which would require the real accretion rate. This division of labour is deliberate and mirrors the physical situation: the *geometry* of the image (shadow, critical curve, lensed arches) is pure spacetime, the *kinematic* pattern (beaming, redshift gradients) is pure orbital mechanics, and only the absolute brightness depends on the messy accretion physics that the thin-disk idealisation cannot capture for Sgr A* anyway — the real flow there is a hot, radiatively inefficient structure accreting far below the Eddington rate (Broderick et al., 2014; Event Horizon Telescope Collaboration, 2022), not a cold thin disk. Chapter 22 weighs which conclusions survive that difference.

12.6 The rendered disk

Fig. 12.3 assembles the pipeline: Novikov–Thorne temperatures (12.2), per-ray redshift factors (12.7), Planck colours at $g\,T_{\text{eff}}$, at $a_* = 0.94$, $i = 80°$. The classic morphology anticipated by Luminet (1979) and rendered cinematically by James et al. (2015) emerges with no artistic input: the far side arches over the shadow (the lensing of Section 11.6), the approaching limb blazes at 9.60× the receding one, and the $n = 1$ sub-ring hugs the critical curve.

12.7 Spin morphology of the disk image

A striking qualitative observation from interactive exploration with the live renderer (Chapter 20): at $a_* = 0$ the disk reads as a system of smooth, continu-

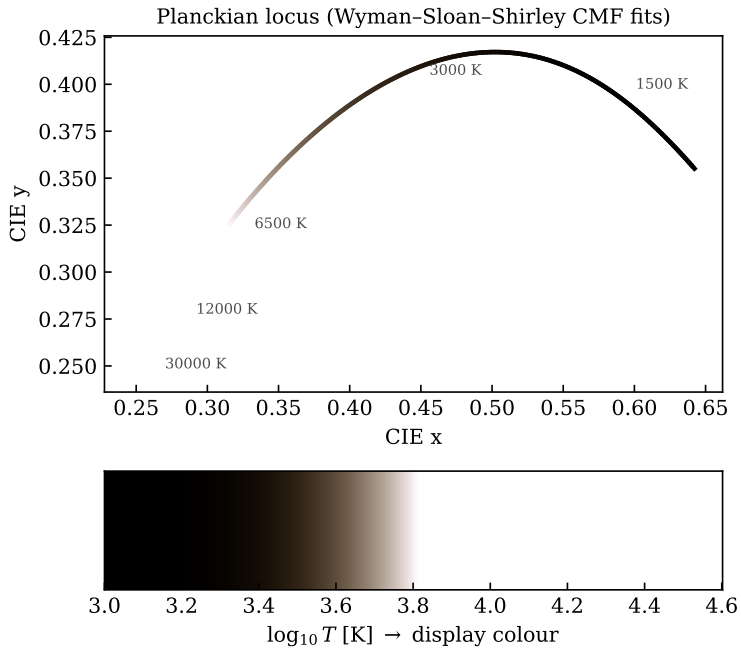


Figure 12.2: The Planckian locus in CIE 1931 chromaticity coordinates and the corresponding display colours (generated from the same Wyman et al. 2013 fits the renderer uses).

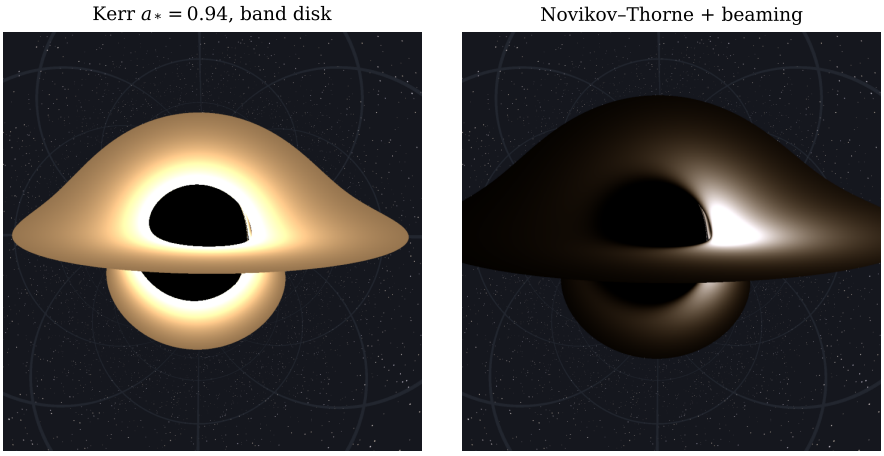


Figure 12.3: CPU reference render of the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$ (linear sRGB from Planck spectra at $g\,T_{\mathrm{eff}}$). Doppler beaming brightens the approaching limb by a bolometric factor of 9.60.

ous, near-concentric bright lanes girdling the whole silhouette, while at high spin ($a_* \approx 0.9$) the coherent bright structure collapses to a compact, intense patch on the approaching limb. Three exact results of this report combine to explain it:

1. *The emission zone deepens.* The ISCO falls from $6M$ to $\sim 2.3M$ between $a_* = 0$ and 0.9 (Fig. 8.1), and the Page–Thorne peak follows it inward (Fig. 12.1) — the brightest material moves to radii where orbital speeds approach half the speed of light.

2. *Beaming sharpens as g^4.* With v/c rising steeply at the new inner radii, the approaching/receding contrast (12.9) grows from a moderate factor to orders of magnitude; the receding-side lanes fall below any practical display window while the approaching side saturates.

3. *Frame dragging compresses the pattern.* The display-layer streamlines advect at the true $\Omega(r)$ of (8.21); its steepening differential shear at small r winds features tightly on the prograde side.

At $a_* = 0$ none of these amplifiers operates strongly — orbital speeds at $r \gtrsim 6M$ stay below $0.4c$, the beaming contrast is mild, and the lanes remain visible around the full circumference. The transition between the two regimes is thus a direct visualisation of the ISCO physics of Chapter 8.

Engine render pending

```
sgr_engine --shader disk --astar 0.0 --incl 80 --rout 24 --turb
      1 --width 1920 --height 1080 --no-mouse --frames 1
```

Figure 12.4: *[Placeholder] Non-spinning disk ($a_* = 0$): the wound streamlines form smooth, continuous, near-concentric bright lanes around the whole silhouette. Reproduced by the command shown; the render will be slotted in when supplied.*

Engine render pending

```
sgr_engine --shader disk --astar 0.906 --incl 80 --rout 24
--turb 1 --width 1920 --height 1080 --no-mouse --frames 1
```

Figure 12.5: *[Placeholder] Rapidly spinning disk ($a_* = 0.906$): the coherent bright structure collapses to a compact, intense patch on the approaching (prograde) limb — the ISCO/beaming/shear mechanism of this section. Reproduced by the command shown; the render will be slotted in when supplied.*

Chapter 13

Photon-Ring Substructure

Light that grazes the photon shell can complete extra half-orbits before reaching the camera, stacking exponentially demagnified images of the disk onto the shadow's rim. This chapter measures that structure (`SubRingExtractor::analyze` and `scan_widths` in `science/photon_ring/`), connects the measured flux ratios to the Lyapunov exponents of Chapter 11, extends the measurements across the spin range, and closes with the axial caustic — a lensing feature often mistaken for a jet — and the detectability outlook.

13.1 Sub-ring decomposition by crossing count

The tracer counts each ray's equatorial-plane crossings *before* the crossing that terminates on the opaque disk; that count n is the ray's sub-ring order (Johnson et al., 2020): $n = 0$ is the direct image, $n = 1$ the first lensed image (light that passed over or under the hole once), $n = 2$ the second. Binning image pixels by n (`analyze`) isolates the orders cleanly (Fig. 13.1): the $n = 1$ ring is a narrow, displaced band hugging the shadow edge — visible directly as the sharp shoulder in the major-axis brightness profile of Fig. 13.2 — and the $n = 2$ ring is a hair-thin trace on the critical curve itself.

The measured bolometric ratios in the reference geometry are $F_0/F_1 = 10.7$ and $F_1/F_2 = 274$. The steep $n = 1 \to 2$ demagnification is the $e^{-\gamma}$ suppression per half-orbit derived in Section 11.5: with $\gamma \in [2.571, \pi]$ around the ring at this spin, the naive expectation $F_1/F_2 \sim e^{\bar{\gamma}} \sim 15$–$23$ applies to matched image annuli; the much larger *total*-flux ratio also folds in the shrinking solid angle of the thinner band and the disk's radial emission profile, which is why total-flux ratios always overshoot the bare exponential. The F_0/F_1 ratio, by contrast, is dominated by

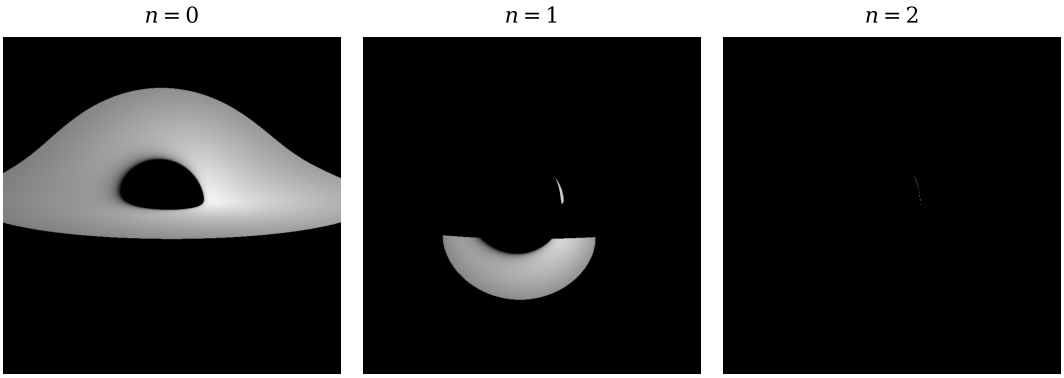


Figure 13.1: Bolometric maps of the photon sub-rings $n = 0, 1, 2$ (log stretch) at $a_* = 0.94$, $i = 80°$. Flux ratios: $F_0/F_1 = 10.7$, $F_1/F_2 = 274$.

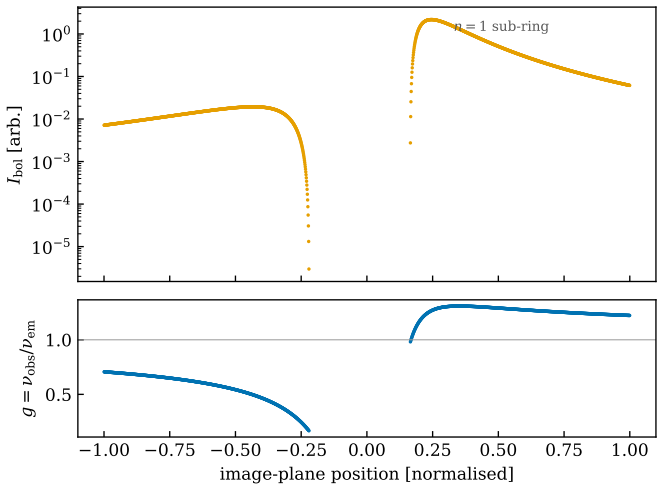


Figure 13.2: Bolometric intensity along the image major axis. The narrow $n = 1$ sub-ring stands on the shoulder of the direct disk image at the shadow edge.

how much of the direct disk image lies far from the critical curve — it measures the emission model as much as the spacetime.

The orders obey a universal asymptotic law (Johnson et al., 2020; Gralla & Lupsasca, 2020): successive sub-rings converge on the critical curve geometrically, with the *same* exponent governing all three observables —

$$w_{n+1} \simeq e^{-\gamma}\, w_n, \qquad d_n - d_\infty \propto e^{-\gamma n}, \qquad F_{n+1} \simeq e^{-\gamma}\, F_n \quad (n \gtrsim 1), \tag{13.1}$$

where w_n, d_n, F_n are the n-th ring's width, diameter, and flux, d_∞ is the critical-curve diameter, and γ is the Lyapunov exponent of Section 11.5 for the shell orbit imaged at that position angle. The law holds only asymptotically — the $n = 0 \to 1$ step still remembers the emission profile, which is why F_0/F_1 and F_1/F_2 differ so strongly in the measurements above — but from $n = 1$ onward the spacetime dictates everything: width ratios, diameter convergence, and flux ratios collapse onto $e^{-\gamma}$. This is what makes the photon ring a *clean* probe in principle: measure two successive sub-rings and the emission model cancels out of their ratio, leaving pure Kerr geometry. Our traced F_1/F_2 against the γ-band of Fig. 11.3 is precisely that consistency check run on synthetic data.

One more exact property deserves note because interferometric proposals lean on it: the sub-rings are *time-shifted* copies of the source. A photon in the $n{+}1$ ring lingers one polar half-orbit longer near the shell than its n-ring counterpart; for Schwarzschild the half-orbit of the $r = 3M$ photon circle takes $\pi\, r_{\rm ph}^{3/2}/M^{1/2} = 3\sqrt{3}\,\pi\, M \approx 16.3\, M$ of coordinate time — about 5.8 minutes at the Sgr A* mass (using the 21.2 s per M of Chapter 6). Each sub-ring therefore replays the disk's flickering with a fixed, mass-determined lag: a detected "echo" at that delay would be a direct, purely geometric mass measurement, independent of every calibration in Chapter 19. Our static disk model cannot exhibit the echo, but the ray-by-ray path-time bookkeeping that would feed such a simulation is already carried by the tracer's t-equation (9.10).

13.2 Sub-ring metrics across the spin range

The `--sweep` mode of `subring_tool` repeats the full decomposition at six spins (fixed $i = 80°$, 560^2 rays; Fig. 13.3). Two trends carry physical signatures:

- F_0/F_1 rises from 8.6 at $a_* = 0$ to 16.8 at $a_* = 0.998$: with the ISCO plunging inward, an ever-larger fraction of the direct emission comes from deep, strongly beamed radii, outcompeting the lensed band.

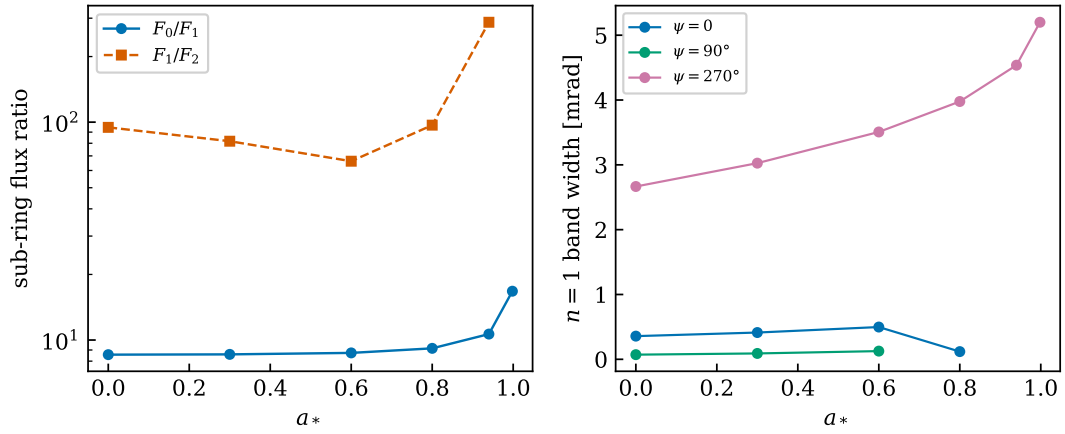


Figure 13.3: Sub-ring metrics versus spin at $i = 80°$ (`subring_tool --sweep`). Left: bolometric flux ratios; the F_1/F_2 point at $a_* = 0.998$ is omitted (the $n = 2$ ring is unresolved at this image scale). Right: measured $n = 1$ band widths along three image directions; zero-width entries (band occluded by the opaque direct image) are omitted.

- The $n = 1$ band width below the shadow ($\psi = 270°$, the direction where the band is never occluded) grows monotonically with spin — the lensed image of the far-side underside widens as the emitting annulus deepens. Along the other directions the band is progressively *occluded by the opaque direct image* at high spin: a real geometric effect of an optically thick disk at high inclination, recorded honestly as missing measurements rather than zeros.

Both behaviours are properties of the exact null geodesics plus a known emission profile — free of plasma-model degeneracy, at the price of the thin-disk idealisation discussed in Chapter 22.

13.3 The axial image tower

Near-edge-on renders at high spin show slim, bright columns of light along the spin axis, above and below the hole — easily mistaken for the relativistic jets of radio-galaxy imagery. They are nothing of the kind, and this section documents what they actually are, because the distinction is a clean illustration of what a geodesic-plus-thin-disk model does and does not contain.

The columns are the *polar lensing caustic*. Consider camera rays launched almost exactly along the spin axis toward the hole: by axisymmetry, a ray displaced

slightly from the axis in *any* azimuth is bent through the polar region and lands on the far/under side of the disk. A whole annulus of sky directions therefore maps onto nearly the same disk material — the Jacobian of the lens map degenerates on the axis (a caustic of the axisymmetric lens), and magnification formally diverges there, tamed only by the finite resolution. Stacked within the column are the $n = 1, 2, \ldots$ images of Section 13.1 converging toward the polar critical point; the column is, in effect, the Einstein-ring limit of Section 11.6 seen edge-on.

Real astrophysical jets are a different object entirely: magnetically launched, plasma-loaded outflows powered by mechanisms such as Blandford & Znajek (1977) energy extraction, requiring magnetohydrodynamics far outside this model's physics — and Sgr A* itself shows at most weak evidence for such a jet. The honest statement is that our columns are pure lensing: they carry no matter, no energy outflow, and would appear for *any* bright equatorial source around *any* black hole viewed near the equatorial plane. The corresponding limitation note appears in Chapter 22.

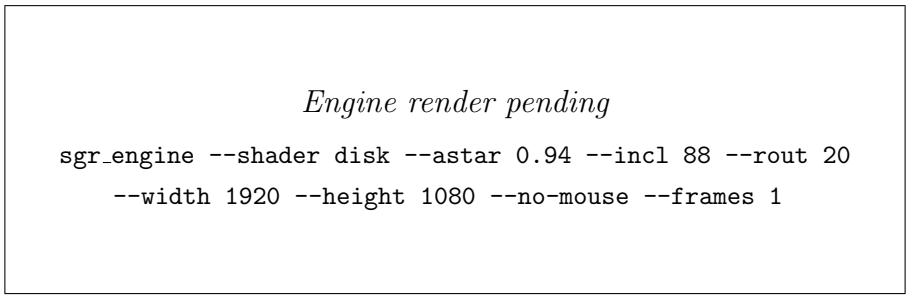


Figure 13.4: *[Placeholder] The axial "jet-like" columns in a near-edge-on high-spin render — the lensed polar caustic of this section, not an astrophysical jet. Reproduced by the command shown; the render will be slotted in when supplied.*

13.4 Detectability forecast

The $n = 1$ ring is a target of the next interferometer generation. In the visibility domain, a thin ring of angular width w contributes a slowly decaying tail beyond baselines $\sim 1/w$; Johnson et al. (2020) project that space–ground baselines could isolate the $n = 1$ signature for M87* and Sgr A*. Our measured widths put numbers on the requirement: at $a_* = 0.94$, $i = 80°$ the unoccluded $n = 1$ band spans ~ 4.5 mrad of camera angle at $r_{\rm cam} = 10^3\,M$, i.e. $\sim 4.5\,M$ radially — a few μas at the Sgr A* calibration of Chapter 19, comfortably inside proposed ngEHT-class resolution but demanding of dynamic range (the band sits $\sim 10.7\times$ below

the direct image in total flux). Because these maps are exact geodesic quantities on a known spacetime, they are clean templates for such forecasts; what they deliberately omit — plasma variability, optical depth, scattering — is the subject of the limitations chapter.

Chapter 14

Timelike Orbits and the S-Star Forward Model

The S2 fit of Chapter 15 stands on a forward model: given thirteen parameters, predict every measured position and velocity. This chapter derives that model layer by layer (`SkyProjection::S2Model` in `science/orbit_fitter/`): initial conditions from exact turning points, integration and sampling, sky projection, the Rømer delay, and the relativistic redshift factorisation (`redshift_terms`, `SpectralLineFitter::predict`).

14.1 The forward model at a glance

One model evaluation proceeds: (i) convert the osculating elements $(a, e, i, \Omega, \omega, t_{\mathrm{peri}})$ plus (M, R_0) into exact initial conditions (Section 14.2); (ii) integrate the timelike geodesic through the observation window with the generic Christoffel integrator of Chapter 9 (this path works for Kerr *and* Hartle–Thorne, which the no-hair fit requires); (iii) interpolate the cached trajectory at each requested emission time (cubic Hermite on position and velocity, `sample_at`); (iv) project onto the sky and apply the reference-frame offsets (x_0, y_0, v_x, v_y); (v) solve the Rømer delay per epoch (Section 14.3); (vi) evaluate the observed radial velocity including all relativistic terms (Section 14.4). Steps (i)–(iii) run once per parameter vector; steps (iv)–(vi) per data epoch. One full evaluation against the S2 dataset costs 3.7 ms — the number that made the MCMC of Chapter 15 practical.

The thirteen parameters split into three tiers with very different characters. Two are *global physics*: the mass M and the distance R_0, shared by every star in the joint fit of Chapter 16. Six are *per-star geometry*: the angular semi-major axis

$a_{\rm mas}$, eccentricity e, inclination i, node angle Ω, pericentre argument ω, and pericentre epoch $t_{\rm peri}$ — the classical osculating elements, here interpreted as exact turning-point data (Section 14.2) rather than Keplerian constants. Five are *nuisance*: the reference-frame offsets and drifts (x_0, y_0, v_x, v_y) and the radial-velocity zero point v_z, which absorb the imperfect tie between the infrared astrometric frame and Sgr A*'s radio position (Plewa et al., 2015). The split matters for interpretation: the relativistic observables — precession, redshift excess — are driven by the first two tiers, while the nuisance tier exists precisely so that frame systematics cannot masquerade as physics. The black-hole spin is deliberately *not* among the parameters: at S2's orbital radius the Lense–Thirring and quadrupole effects it would produce are two and four orders below the Schwarzschild precession respectively (the hierarchy is quantified in Chapter 17), so the S2 data cannot constrain it and the forward model holds it fixed at the calibration value.

14.2 Orbital elements from radial turning points

The osculating elements define the pericentre and apocentre radii $r_p = \tilde{a}(1-e)$, $r_a = \tilde{a}(1+e)$ (with $\tilde{a}$ the physical semi-major axis from $a_{\rm mas}$ and R_0). The exact relativistic orbit with these turning points has E and L_z fixed by the two conditions

$$R(r_p) = 0, \qquad R(r_a) = 0, \tag{14.1}$$

with R the timelike radial potential (9.7) — the construction of Darwin (1959). For the equatorial-plane integration frame the code uses ($Q = 0$; the orbit is subsequently rotated to the sky by (i, Ω, ω)), at $a = 0$ the potential is $R = E^2 r^4 - \Delta(r^2 + L_z^2)$ and (14.1) is a 2×2 *linear* system in (E^2, L_z^2):

$$\begin{pmatrix} r_p^4 & -\Delta_p \\ r_a^4 & -\Delta_a \end{pmatrix} \begin{pmatrix} E^2 \\ L_z^2 \end{pmatrix} = \begin{pmatrix} \Delta_p r_p^2 \\ \Delta_a r_a^2 \end{pmatrix} \implies E^2 = \frac{\Delta_p \Delta_a \,(r_a^2 - r_p^2)}{\Delta_p r_a^4 - \Delta_a r_p^4}, \qquad L_z^2 = \frac{r_p^4\, \Delta_a r_a^2 - \Delta_p r_p^2\, r_a^4}{\Delta_p r_a^4 - \Delta_a r_p^4} \tag{14.2}$$

(Cramer's rule, $\Delta_{p,a} = \Delta(r_{p,a})$). The implementation writes the same system in the equivalent form $E^2 = (1 - 2M/r)(1 + L^2/r^2)$ at both radii and solves for L^2 first — algebraically identical, but numerically better conditioned at S2-like radii ($r \sim 10^3$–$10^4\, M$) because it avoids differencing the large r^4 terms. The rosette integration of Fig. 9.2 validated exactly this construction: the integrated orbit's conserved (E, L) match (14.2) to eleven decimal places.

Why not Keplerian initial conditions? The obvious shortcut — start at pericentre with the Newtonian vis-viva speed — mis-states the *relativistic* orbit: with

the same (r_p, v_p) the GR orbit has a different apocentre and, at S2's eccentricity, an orbital period wrong by 1.6 per cent. The measured S2 period is known to 0.014 per cent, so the shortcut is two orders of magnitude too crude — a bug found by exactly this discrepancy during Phase 4 and pinned by a regression test since (the narrative is told in Chapter 21). The turning-point construction, by contrast, reproduces the requested geometry exactly by definition.

14.3 Sky projection and the Rømer delay

The integrated orbit lives in BH-centred coordinates; the observables live on the sky. The map between them is the classical composition of three rotations — by ω about the orbit normal (placing the pericentre), by i about the node line (tilting the plane), and by Ω about the line of sight (orienting the node on the sky) — taking the orbital plane to East–North offsets and a line-of-sight coordinate z (positive away from us), scaled by R_0 into angles. Two conventions are worth pinning because sign errors here are the classic failure mode of orbit-fitting codes: the node angle is a position angle measured from North through East (the astronomers' convention, not the mathematicians'), and the same rotation applied to the velocity vector must reproduce the sign of the measured radial velocities — a consistency the S2-model regression tests exercise against the published data conventions. The fitted frame parameters (x_0, y_0, v_x, v_y) absorb the residual tie between the infrared frame and the radio position of Sgr A* (Plewa et al., 2015), and v_z the line-of-sight zero point. The model applies the rotation directly at each evaluation; the classical Thiele–Innes linearisation (solving for rotation-matrix elements as if independent) buys nothing here, since the fit is nonlinear in e and t_{peri} regardless and the exact elements are wanted for the relativistic bookkeeping.

Light from different orbit phases travels different distances: an epoch observed at t_{obs} was *emitted* at t_{em} satisfying

$$t_{\mathrm{em}} = t_{\mathrm{obs}} - \frac{z(t_{\mathrm{em}})}{c}, \tag{14.3}$$

the classical Rømer delay (first used by Rømer in 1676 to measure c from Jupiter's moons; here it is c that is known and the geometry that matters). Equation (14.3) is an implicit equation for t_{em}; the code solves it by fixed-point iteration $t^{(k+1)} = t_{\mathrm{obs}} - z(t^{(k)})/c$, which contracts with rate $|\dot{z}/c| = |\beta_z| \lesssim 0.026$ for S2 — guaranteeing convergence in 2–3 iterations to sub-second accuracy. For S2 the delay spans about ± 4 light-days across the orbit (Fig. 14.1); ignoring it would

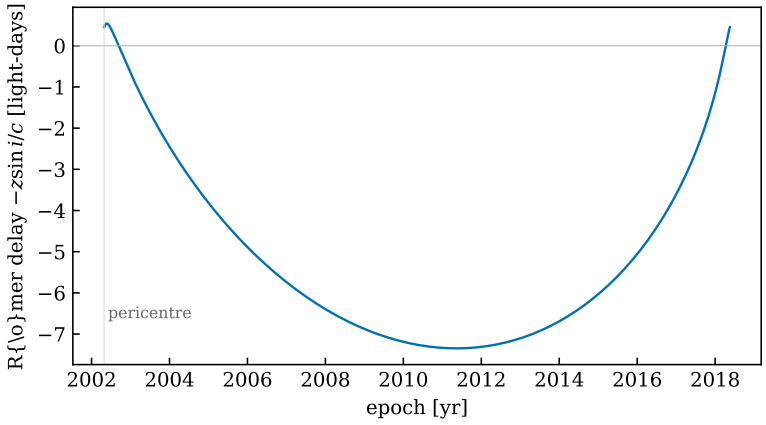


Figure 14.1: The Rømer delay $-z \sin i/c$ over one S2 orbit, evaluated from the fitted elements (Keplerian geometry evaluation; the production model solves (14.3) on the exact geodesic). Peak-to-peak the delay is ~ 8 light-days.

shift pericentre-passage timing by days against astrometry measured to weeks — a measurable model error, which is why the delay is applied to every epoch of every star.

14.4 The spectroscopic observable

The measured quantity for S2 spectroscopy is the centroid of the Brγ line ($\lambda_{\rm rest} = 2.16612\,\mu$m). The model wavelength ratio factorises exactly as

$$\frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = (1 + z_{\rm D})(1 + z_{\rm tD})(1 + z_{\rm grav}), \tag{14.4}$$

where the three factors isolate the classical line-of-sight Doppler term $z_{\rm D}$, the special-relativistic transverse (time-dilation) term $z_{\rm tD} = \gamma - 1$, and the gravitational redshift $z_{\rm grav}$ at the emission point. The diagnostic accessor `redshift_terms` evaluates, along the cached trajectory,

$$z_{\rm tD} = \frac{1}{\sqrt{1-\beta^2}} - 1, \qquad z_{\rm grav} = \frac{1}{\sqrt{1-2M/r}} - 1, \qquad z_{\rm D} = \beta_z \text{ (to leading order)}, \tag{14.5}$$

with β the coordinate speed and β_z its line-of-sight component. Two honesty notes on (14.5). First, the *fit* does not consume these named pieces — the model radial velocity comes from the exact accumulated frequency ratio along the geodesic (the `redshift` field of each trajectory sample); the decomposition exists for plots and physical narration, and the near equality of its pieces' sum with the exact

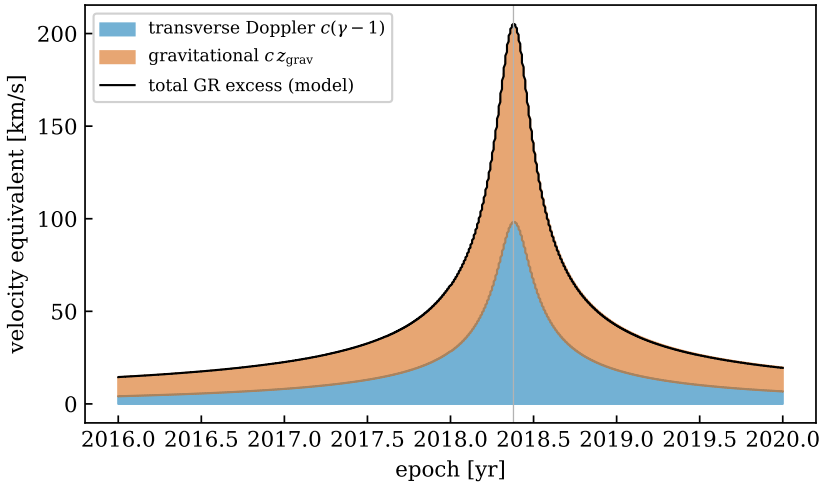


Figure 14.2: Decomposition of S2's relativistic radial-velocity excess through the 2018 pericentre (`phase4/s2_redshift_curve.csv`): transverse Doppler (blue) and gravitational (red) stack to the total model excess (black line).

total (sub-km/s through pericentre) is itself a useful cross-check of the integration. Second, $z_{\rm grav}$ is written in the static ($a = 0$) form: at $r \sim 3000\,M$ the spin's imprint on the emitter's clock rate enters through the $g_{t\phi}\,\Omega$ cross term at relative order $aM^{3/2}/r^{5/2} \sim 10^{-9}$ of the already-small $z_{\rm grav}$ — far below every observational and numerical threshold in this report, which is exactly why S2 weighs the mass but not the spin. The quoted model radial velocity is the *apparent* $v_r = c\,(\lambda_{\rm obs}/\lambda_{\rm rest} - 1)$ — the same convention the observers use — so the relativistic terms appear as an excess over a Keplerian orbit with identical elements, which is precisely the quantity GRAVITY measured in 2018 (Section 15.4).

Fig. 14.2 stacks the two relativistic terms through the 2018 pericentre passage: they are comparable in size (the $2/(1+e)$ near-equality worked out in Chapter 15), sum to the ~ 200 km/s excess, and each is evaluated with no adjustable amplitude — the fitted mass and geometry force both.

14.5 Trajectory interpolation

The integrated orbit is stored as a sequence of adaptive RK45 samples; observables are wanted at arbitrary emission times (set by the Rømer solve). Rather than re-integrate to each epoch, `sample_at` uses cubic Hermite interpolation on the cached nodes, which is the natural choice here because each node carries *both*

position and velocity (the state vector already contains u^μ), so the interpolant matches value and first derivative at both bracketing nodes with no extra storage. The error of cubic Hermite interpolation on a step h is $\frac{1}{384} h^4 \max |f^{(4)}|$; at the production sampling density (adaptive steps keeping the integration error at 10^{-11}, so $h \lesssim 10^{-2}$ of the local orbit timescale) the interpolation error is below 10^{-9} of the orbit scale — two orders under the tightest astrometric uncertainty in the dataset, and negligible against the fit residuals. The interpolant is thus a convenience with no measurable cost in accuracy, which the S2-model regression tests confirm by comparing interpolated observables against directly-integrated ones at held-out epochs.

Chapter 15

Orbit Fitting, Statistics, and the S2 Results

15.1 Data

The astrometric and spectroscopic record of S2 is taken from the monitoring compilation of Gillessen et al. (2017) (VizieR J/ApJ/837/30): 145 imaging positions (VLT NACO and Keck NIRC, 1992–2016, typical uncertainties 0.3–3 mas) and 44 radial velocities (SINFONI and earlier spectrographs, 2000–2016, typical uncertainties 15–100 $\mathrm{km\,s^{-1}}$). Positions are offsets from the radio position of Sgr A*; the residual uncertainty of that frame tie is modelled by the fitted offset/drift parameters with the Plewa et al. (2015) priors of Section 15.2. We apply no error rescaling: the reduced χ^2 of 0.959 below is obtained with the published uncertainties as-is.

15.2 The objective and its parameterisation

The objective is

$$\chi^2 = \sum_i \left[\frac{(\alpha_i^{\mathrm{obs}} - \alpha_i^{\mathrm{mod}})^2}{\sigma_{\alpha,i}^2} + \frac{(\delta_i^{\mathrm{obs}} - \delta_i^{\mathrm{mod}})^2}{\sigma_{\delta,i}^2} \right] + \sum_j \frac{(v_{r,j}^{\mathrm{obs}} - v_{r,j}^{\mathrm{mod}})^2}{\sigma_{v,j}^2} + \chi^2_{\mathrm{prior}}, \tag{15.1}$$

over the 145 astrometric epochs and 44 radial velocities, with the forward model of Chapter 14 supplying the model positions and velocities.

Thirteen parameters are fitted: (M, R_0), six orbital elements, the four reference-frame offsets and drifts (x_0, y_0, v_x, v_y) of the mass with respect to the radio frame, and v_z; the no-hair mode adds ε_q. Weak priors tie the frame to Sgr A* following Plewa et al. (2015) (± 0.2 mas at epoch 2009.0, ± 0.1 mas yr^{-1}, ± 5 km s^{-1}).

15.2.1 Minimisation: Gauss–Newton with Levenberg damping

Nonlinear least squares has special structure a generic minimiser wastes: with the σ-normalised residual vector $r(\theta)$, $\chi^2 = r^{\mathsf{T}} r$, one linearisation $r(\theta + \delta) \approx r + J\delta$ turns each iteration into a *linear* least-squares problem, $(J^{\mathsf{T}} J)\,\delta = -J^{\mathsf{T}} r$ — Gauss–Newton, which converges quadratically near an optimum where residuals are small (as here: $\chi^2_\nu \approx 1$ means the model passes within an error bar of most points). Far from the optimum the linearisation can overshoot; the Levenberg–Marquardt remedy damps the normal equations, $(J^{\mathsf{T}} J + \lambda D)\,\delta = -J^{\mathsf{T}} r$, interpolating between Gauss–Newton ($\lambda \to 0$) and a short gradient-descent step (λ large), with λ adapted by a trust-region rule. This is the algorithm the Ceres solver implements (`OrbitFitter::fit`), with the residual Jacobian supplied by Ceres's forward-mode automatic differentiation — dual-number arithmetic propagated through the entire forward model of Chapter 14, the same exactness idea as the complex-step derivatives of Chapter 10, so no finite-difference noise enters the optimiser's line of sight. From the published starting elements the fit converges in a handful of LM iterations; the question of secondary minima elsewhere in parameter space is answered not by the optimiser but by the MCMC of Section 15.6, whose walkers explore the full bounded volume and return a single-moded posterior.

Parameter covariances are obtained as $C = (J^{\mathsf{T}} J)^{-1}$ from an explicit central-difference Jacobian J of the weighted residual vector (`OrbitFitter::covariance`), because the solver's truncated-SVD covariance annihilates the strongly correlated M–R_0 direction and reports a spuriously zero mass uncertainty (Chapter 21).

15.2.2 Fisher information and the Cramér–Rao floor

For a model with Gaussian errors, write the σ-normalised residual vector $r(\theta)$ and its Jacobian $J_{ik} = \partial r_i / \partial \theta_k$. Near the optimum, $\chi^2(\theta) \approx \chi^2_{\min} + \delta\theta^{\mathsf{T}} (J^{\mathsf{T}} J)\,\delta\theta$, so $F = J^{\mathsf{T}} J$ is the Fisher information matrix, and the Cramér–Rao bound states

that *no* unbiased estimator can beat

$$\mathrm{Cov}(\theta) \succeq F^{-1} = (J^{\mathsf{T}}J)^{-1}. \tag{15.2}$$

For a least-squares estimator at a well-conditioned optimum the bound is attained to linear order, which is why $C = (J^{\mathsf{T}}J)^{-1}$ is the standard parameter covariance; every "±" in Table 15.1 is the square root of a diagonal element of exactly this matrix, built from an explicit central-difference J with physically scaled steps and inverted by Gauss–Jordan elimination with partial pivoting (`OrbitFitter::covariance`).

15.2.3 Why the solver's SVD covariance fails here

The subtlety is the spectrum of F, and it is worth deriving rather than asserting. Astrometry measures the *angular* semi-major axis θ_a and the period P; Kepler's third law then fixes

$$M = \frac{4\pi^2(\theta_a R_0)^3}{GP^2} \quad\Longrightarrow\quad M \propto R_0^3 \quad \text{(astrometry + period)}, \tag{15.3}$$

a one-parameter family, not a measurement of M. The radial velocities break it: orbital speeds scale as $v \sim \sqrt{GM/(\theta_a R_0)}$, so matching the measured $\mathrm{km\,s^{-1}}$ imposes

$$M \propto R_0 \quad \text{(radial velocities)}. \tag{15.4}$$

The two power laws cross at the true (M, R_0) — this crossing *is* the geometric distance determination celebrated in the primer — but they cross at a shallow angle in logarithmic coordinates (slopes 3 and 1), so the joint constraint is superbly conditioned along one combination and weakly conditioned along the near-degenerate one. The two constraints intersect at a sharp angle in the (M, R_0) plane: the fit is superbly conditioned along one combination and weakly conditioned along the near-degenerate one, so F carries one singular value several orders below the rest — small, but genuinely finite and *information-bearing*. A covariance routine that regularises by *truncating* small singular values treats that direction as exact rank deficiency: the truncated pseudo-inverse assigns it *zero* variance, and the reported mass uncertainty collapses to machine noise. That is precisely the failure observed with the library covariance during Phase 4 (a fitted mass "known" to nine digits was the giveaway; Chapter 21) and the reason the project computes (15.2) explicitly: the small singular value is *inverted*, not discarded, and the M–R_0 error ellipse comes out with its true, elongated shape — subsequently confirmed end-to-end by the MCMC posterior of Section 15.6.

Table 15.1: S2 fit results (full GR timelike geodesic model) against published values (Gillessen et al., 2017; GRAVITY Collaboration, 2022). Uncertainties are 1σ from the explicit-Jacobian covariance.

Parameter	This work	Published
M [$10^6\,\mathrm{M}_\odot$]	4.270 ± 0.195	4.297 ± 0.012
R_0 [pc]	8272 ± 182	8277 ± 9
a [mas]	124.82 ± 0.88	125.058 ± 0.041
e	0.88277 ± 0.00195	0.8839 ± 0.0019
i [deg]	134.70 ± 0.41	134.18 ± 0.40
Ω [deg]	227.33 ± 0.63	226.94 ± 0.60
ω [deg]	65.44 ± 0.61	65.51 ± 0.57
t_{peri} [yr]	2002.3227 ± 0.0052	2002.33 ± 0.01
P [yr]	16.0561 ± 0.0228	16.0518 ± 0.0022
χ^2_ν	0.959	—

15.3 Results: the S2 fit

We fit the 145 astrometric positions and 44 radial velocities of S2 tabulated by Gillessen et al. (2017). The best-fitting model (Fig. 15.1, Fig. 15.2) has $\chi^2 = 307.8$ for 321 degrees of freedom ($\chi^2_\nu = 0.959$); the residuals (Fig. 15.3) show no structure. Table 15.1 compares the recovered parameters with the published values.

The recovered mass and distance agree with the far more precise GRAVITY interferometric determination (GRAVITY Collaboration, 2022) to 0.14σ and 0.03σ respectively — a non-trivial cross-check, since our fit uses none of the GRAVITY beam astrometry, only the VLT/Keck imaging era compiled by Gillessen et al. (2017).

The bookkeeping behind the headline statistic: the data contribute $2 \times 145 + 44 = 334$ measured values, the model spends 13 parameters, leaving 321 degrees of freedom, and the minimum is $\chi^2 = 307.8$ — i.e. $\chi^2_\nu = 0.959$, marginally below unity. Read against the primer's Chapter 6: the model misses a typical point by slightly *less* than its published error bar, which for a heterogeneous 24-year compilation (two telescopes, several instrument generations, uncertainties assigned per-epoch by the original teams) most plausibly means the published uncertainties carry a small conservative margin, not that the model over-fits — thirteen

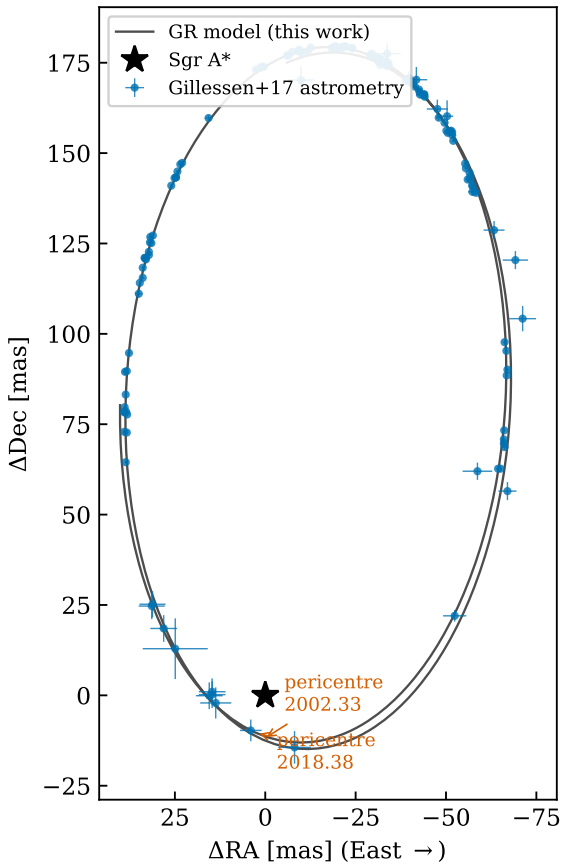


Figure 15.1: S2 on the sky: fitted GR geodesic model (curve) and the 145 astrometric measurements of Gillessen et al. (2017) (points). Sgr A* sits at the origin.

parameters cannot flatter 334 residuals, and no error rescaling was applied at any point. The expected statistical spread of χ^2_ν itself at $\nu = 321$ is $\sqrt{2/\nu} \approx 0.08$, so 0.959 sits half a spread from unity: unremarkable, which is the point.

Two complementary views probe the residual structure. Fig. 15.4 shows observed-minus-computed values in physical units — no secular trend, no pericentre-phase feature, the early low-weight epochs visibly noisier as their published uncertainties say they should be. Fig. 15.5 maps the astrometric residual *vectors* on the sky plane, magnified 50×: no coherent rotation or dipole (which would indicate a frame drift beyond the fitted terms), no alignment with the orbit track (which would indicate a timing bias).

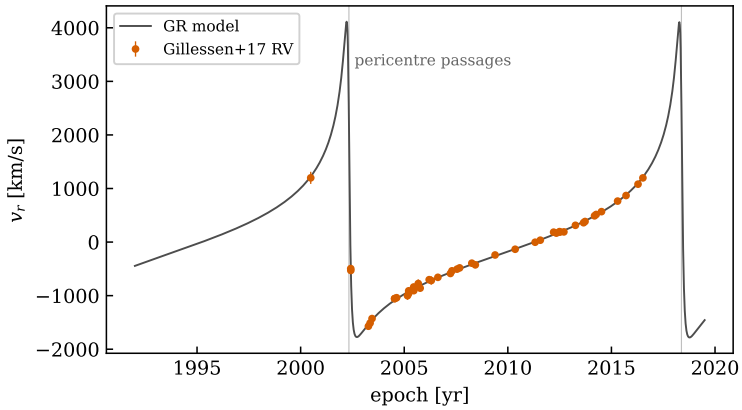


Figure 15.2: S2 radial velocity: model and the 44 measurements. The sharp swing brackets the 2002 and 2018 pericentre passages.

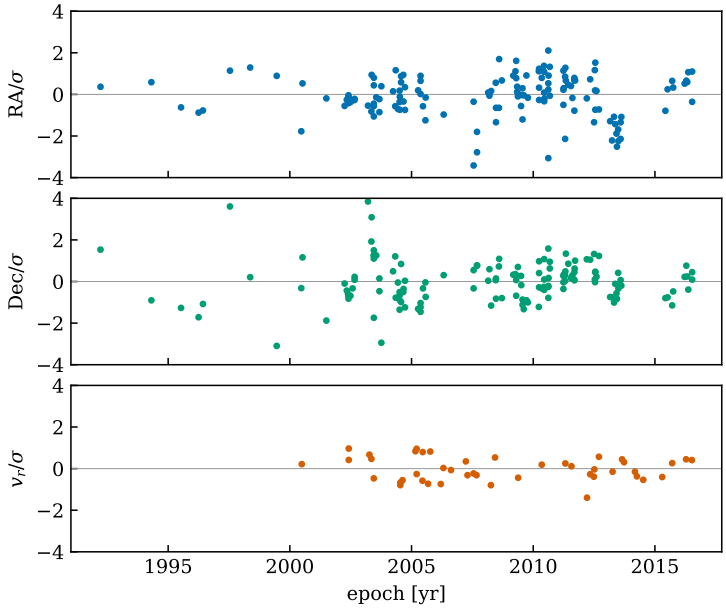


Figure 15.3: Normalised fit residuals. $\chi^2_\nu = 0.959$ over 321 degrees of freedom.

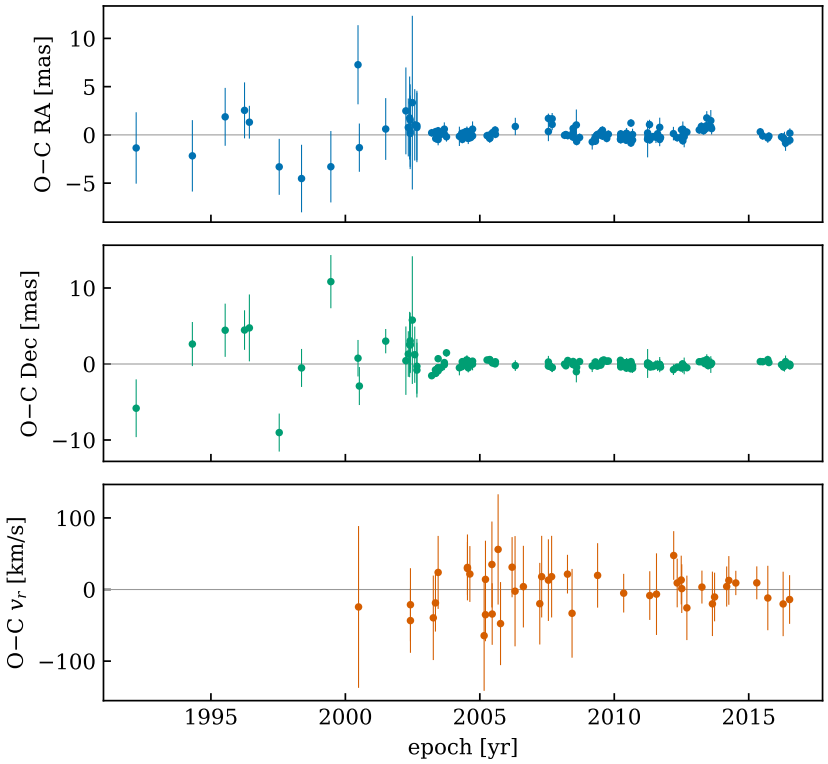


Figure 15.4: Observed minus computed, in measurement units, for the three data channels of the S2 fit.

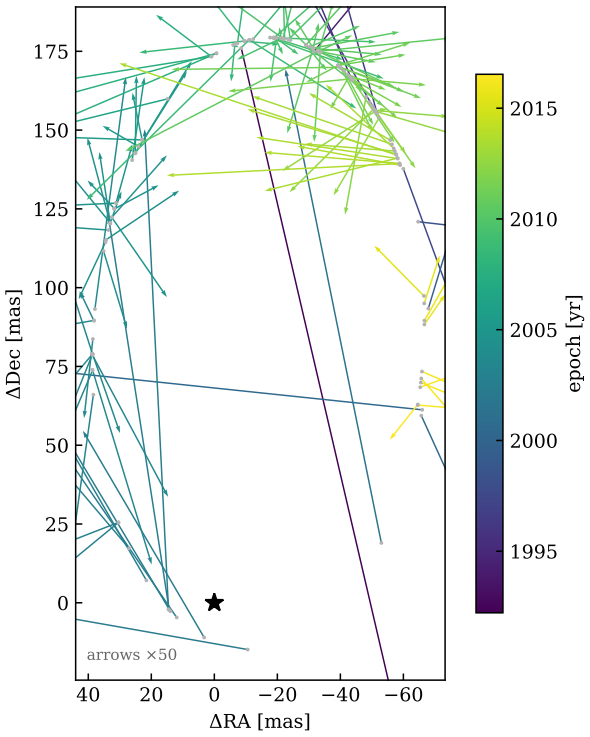


Figure 15.5: Astrometric residual vectors along the fitted S2 orbit (arrows magnified 50×, coloured by epoch). No coherent pattern survives the fit.

15.4 Gravitational redshift at pericentre

The magnitude of the relativistic signal at pericentre follows from the fitted elements alone. Step by step:

1. The pericentre distance is $r_{\rm peri} = a(1-e) = 1032\,\mathrm{au}\times(1-0.88277) \approx 121$ au, i.e. ≈ 1470 Schwarzschild radii ($R_S = 2GM/c^2 \approx 0.085$ au).

2. The vis-viva speed there is $v_{\rm peri} = \sqrt{GM(2/r_{\rm peri} - 1/a)} \approx 7650\,\mathrm{km\,s^{-1}}$, so $\beta \approx 0.0255$.

3. The transverse Doppler shift contributes $z_{\rm tD} \simeq \beta^2/2 \approx 3.3 \times 10^{-4}$, i.e. $c\,z_{\rm tD} \approx 98\,\mathrm{km\,s^{-1}}$.

4. The gravitational redshift contributes $z_{\rm grav} \simeq GM/(r_{\rm peri}c^2) = \beta^2_{\rm circ} \approx 3.5 \times 10^{-4}$, i.e. $\approx 105\,\mathrm{km\,s^{-1}}$.

The two corrections are nearly equal at pericentre — not a coincidence: vis-viva gives $v^2_{\rm peri} = (GM/a)(1+e)/(1-e)$ while $GM/r_{\rm peri} = (GM/a)/(1-e)$, so $z_{\rm grav}/z_{\rm tD} = 2/(1+e) = 1.06$ at the S2 eccentricity — and their sum in the full geodesic model peaks at $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379 (Fig. 15.6), in agreement with the $\approx 200\,\mathrm{km\,s^{-1}}$ excess measured around the 2018.38 pericentre passage by GRAVITY Collaboration (2018), who quantified it as $f = 0.90 \pm 0.17$ on a scale where $f = 0$ is Newtonian and $f = 1$ is GR. Our forward model contains no adjustable redshift amplitude: the excess is forced by the fitted mass and geometry. (Like GRAVITY Collaboration (2018) we neglect the $\sim$5 $\mathrm{km\,s^{-1}}$ light-bending correction.)

15.5 Relativistic precession

The fitted orbit advances its pericentre by 11.98 arcmin per orbital period from the Schwarzschild term. The textbook estimate agrees step by step: the semilatus rectum is $p = a(1-e^2) = 1032\,\mathrm{au}\times(1-0.88277^2) \approx 228$ au, so with $R_g \approx 0.0424$ au the ratio is $R_g/p = 1.86 \times 10^{-4}$ and

$$\delta\omega = \frac{6\pi GM}{c^2 a(1-e^2)} = 6\pi\,\frac{R_g}{p} \approx 3.5 \times 10^{-3}\,\mathrm{rad}, \tag{15.5}$$

i.e. ≈ 12.0 arcmin per 16-yr orbit, matching the full geodesic integration (11.98 arcmin) and the GRAVITY detection of the Schwarzschild precession, $f_{\rm SP} = 1.10 \pm$

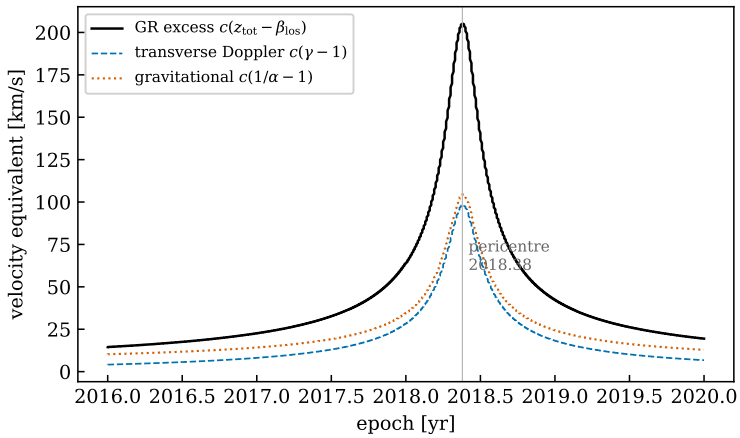


Figure 15.6: Relativistic radial-velocity excess of S2 over a Keplerian orbit with identical elements, through the 2018 pericentre passage. The peak is $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379, matching the detection of GRAVITY Collaboration (2018).

0.19 (GRAVITY Collaboration, 2020). Frame dragging at $a_* = 0.94$ contributes only -0.204 arcmin per orbit (retrograde for the S2 geometry), two orders of magnitude below current sensitivity, consistent with the hierarchy laid out by Will (2008).

Fig. 15.7 places the three analytic precession terms of `GRPrecession` side by side for the fitted S2 elements as a function of hypothetical spin: the Schwarzschild advance dominates by two orders of magnitude over Lense–Thirring frame dragging, itself two orders above the Kerr quadrupole term — the hierarchy of Will (2008) that dictates what each observing generation can reach, and the reason the no-hair constraint of Chapter 17 is bounded by the quadrupole rung of this ladder.

15.6 MCMC posteriors

The covariance of Section 15.2.2 is a *local, Gaussian* statement. To validate it globally — and to expose any non-Gaussian tails or multi-modality the linearisation would hide — the same forward model is sampled with Markov chain Monte Carlo (`scripts/run_mcmc.py`).

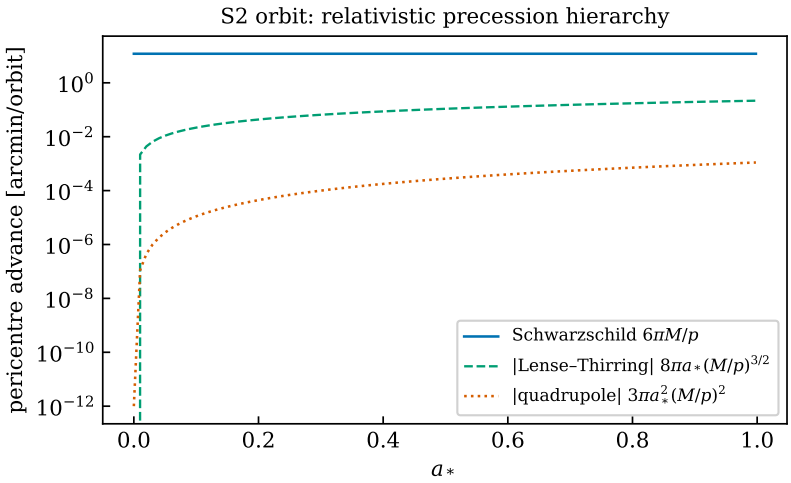


Figure 15.7: The relativistic precession hierarchy for the fitted S2 elements: Schwarzschild, Lense–Thirring, and quadrupole pericentre advances per orbit versus spin (analytic terms of `GRPrecession`; the quadrupole curve uses the Kerr value $\tilde{q} = -a_*^2$).

15.6.1 Sampler

We use the affine-invariant ensemble sampler of Goodman & Weare (2010) as implemented in `emcee` (Foreman-Mackey et al., 2013): an ensemble of walkers in which each is updated by a "stretch move" along the line toward a randomly chosen partner, with a step-size distribution that makes the algorithm invariant under all affine reparameterisations. Concretely, walker X_k proposes

$$X_k \;\to\; X_j + Z\,(X_k - X_j), \qquad Z \sim g(z) \propto \frac{1}{\sqrt{z}} \;\text{ on }\; \left[\tfrac{1}{a},\, a\right],\; a = 2, \tag{15.6}$$

with X_j a random *other* walker, accepting with probability $\min\!\big(1,\; Z^{\,d-1}\,\pi(X')/\pi(X_k)\big)$ for d parameters (Goodman & Weare, 2010). Because the proposal is built from the ensemble's own spread, it automatically aligns with however the posterior is stretched or tilted. Affine invariance is exactly the property this problem wants — the posterior's dominant feature is the elongated, tilted M–R_0 ellipse of Section 15.2.3, which a naive Metropolis sampler with axis-aligned proposals would traverse painfully slowly, but which a stretch move handles as easily as a sphere. The cost of that elegance is that walkers must number at least a few times the dimension (32 walkers for 13–14 parameters here) and that the ensemble equilibrates collectively; both are accounted for in the convergence budget below.

The log-probability is $-\tfrac{1}{2}\chi^2$ evaluated by the *same production code path* as

the least-squares fit: a persistent `fit_s2_tool --chi2` server process receives parameter vectors on standard input and returns χ^2 per line, so the posterior and the Ceres optimum are directly comparable with no re-implementation risk. Eight server processes evaluate a whole ensemble batch in parallel (3.7 ms per evaluation; a full production run takes seconds). Priors are flat within generous physical bounds; the Plewa et al. (2015) reference-frame priors are already inside χ^2.

15.6.2 Convergence and results

The production chain runs 32 walkers for 6000 steps from a small ball around the least-squares optimum, with the sampler's random state seeded so the chain — and every posterior number quoted from it — is bit-reproducible under `make reproduce`. The mean acceptance fraction is 0.72 and the longest integrated autocorrelation time across parameters is $\tau_{\max} \approx 59$ steps: the chain is $\approx 102\,\tau$ long, against the $\gtrsim 50\,\tau$ convergence rule of thumb (Foreman-Mackey et al., 2013). Discarding $4\tau_{\max}$ as burn-in and thinning by $\tau_{\max}/2$ leaves 6 336 effectively independent posterior samples.

The marginal posteriors give $M = 4.502^{+0.628}_{-0.603} \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8266^{+555}_{-530}$ pc (16th/50th/84th percentiles) — in agreement with the least-squares values and the explicit-Jacobian uncertainties of Table 15.1, which is the cross-validation this section exists to provide: the local Gaussian covariance and the global sampled posterior tell the same story, so neither the truncated-SVD pathology nor any hidden non-Gaussianity survives in the quoted errors. The full corner plot (Fig. 15.8) shows every pairwise marginal: the M–R_0 and M–a_{mas} panels carry the expected strong, near-linear correlations; all distributions are single-moded and near-Gaussian.

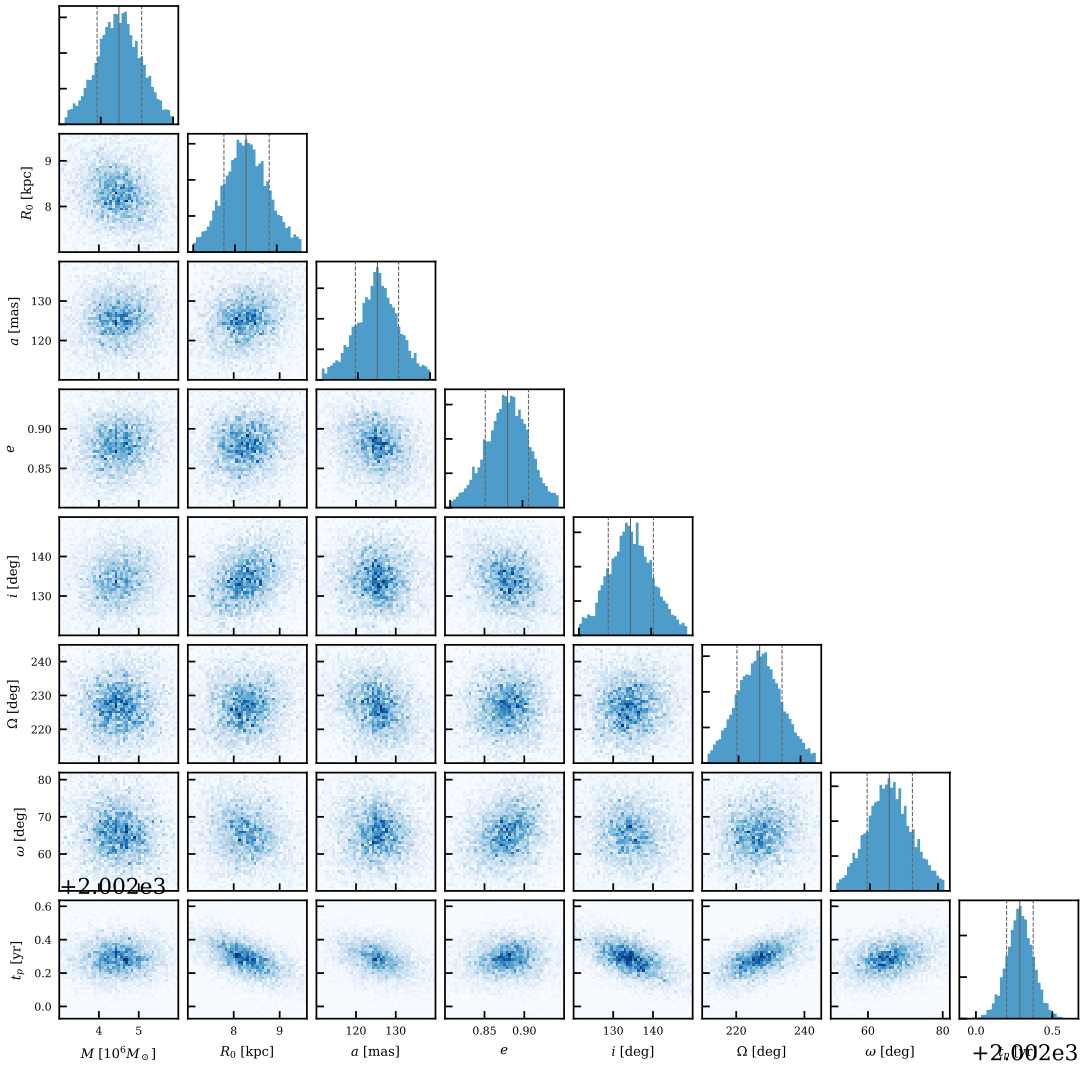


Figure 15.8: MCMC posterior of the S2 fit (32 walkers × 6000 steps, thinned; the eight physically interesting of the 13 parameters). Diagonal: marginal distributions with 16/50/84th-percentile marks. The elongated M–R_0 panel is the degeneracy analysed in Section 15.2.3.

Chapter 16

The Wider S-Star Cluster: A Joint Mass Fit

Every star orbiting Sgr A* orbits the *same* black hole at the *same* distance. A fit that shares one (M, R_0) and one reference frame across many stars therefore multiplies the constraint — and, as this chapter shows concretely, rescues stars whose individual arcs are too short to weigh the hole at all. This is the multi-star programme of Gillessen et al. (2017) and GRAVITY Collaboration (2022) reproduced inside our own machinery (`fit_cluster` in `science/orbit_fitter/`, over the data fetched star-by-star by the refactored `fetch_gillessen_data.py`).

16.1 Data and model

The VizieR machine-readable Table 5 of Gillessen et al. (2017) carries per-epoch astrometry and radial velocities for seventeen S-stars in per-star columns; the fetch script extracts any requested star into the same CSV format the S2 pipeline uses. The joint fit uses five: **S2** (145 astrometric epochs + 44 RVs, $P = 16$ yr — the anchor, observed through more than a full orbit), **S1** (161 + 29, $P \approx 166$ yr), **S8** (166 + 29, $P \approx 93$ yr), **S12** (158 + 27, $P \approx 59$ yr) and **S13** (127 + 22, $P \approx 49$ yr) — all with published uncertainties used as-is. The four companions contribute *partial arcs*: between 1992 and 2016 they cover from a sixth to half of an orbit.

The shared-potential model has $7 + 6N$ parameters for N stars: one global block — M, R_0, and the five reference-frame parameters $(x_0, y_0, v_x, v_y, v_z)$ with the Plewa et al. (2015) priors applied once — plus six osculating elements per star. Each star's residuals go through the identical full-GR forward model of Chap-

ter 14 (Darwin elements, geodesic integration, Rømer delay, relativistic redshift), assembled into one Ceres problem with per-star parameter blocks so the solver exploits the sparsity (star k's elements touch only star k's residuals). Starting elements are the Table 3 values of Gillessen et al. (2017), refined per star before the joint solve; a starting-χ^2 probe over the four $(\Omega, \omega) \pm 180^\circ$ orientation conventions guards against convention mismatches.

16.1.1 The joint likelihood and its block structure

Formally the objective is the direct sum of five copies of the single-star χ^2 (15.1), coupled only through the global block $g = (M, R_0, x_0, y_0, v_x, v_y, v_z)$:

$$\chi^2_{\text{joint}}(g, e_1, \ldots, e_5) = \sum_{k=1}^{5} \chi^2_k\big(g, e_k\big) + \chi^2_{\text{prior}}(g), \tag{16.1}$$

with e_k the six elements of star k. The normal equations inherit the classic *block-arrow* sparsity: each per-star block $\partial^2/\partial e_k^2$ is dense 6×6, the coupling blocks $\partial^2/\partial g\,\partial e_k$ are thin, and no e_j–e_k block exists at all — star j's residuals never see star k's elements. Eliminating the per-star blocks first (the Schur complement onto the 7-dimensional global block — the same structure photogrammetrists call bundle adjustment) means the 37-parameter joint problem costs little more than five independent fits per iteration, and it is exactly the structure the per-star Ceres parameter blocks expose to the solver. The statistical content of (16.1) is worth saying plainly: the model asserts that *one* pair (M, R_0) must simultaneously satisfy (15.3) and (15.4) for five stars with five different semi-major axes and periods. Systematics that could hide inside a single star's elements — a slightly wrong eccentricity trading against mass — have nowhere to hide when the same (M, R_0) must serve orbits spanning a factor of ten in period.

16.2 Why partial arcs need a shared potential

The single-star column of Fig. 16.1 is the demonstration. Fitted alone with the full 13-parameter model, S2 recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ — the Phase-4 result. S1, with a $\sim$15 per cent arc of a 166-year orbit, still manages a mass because its arc includes strong curvature. But S8's single fit *fails to converge* (the optimiser runs into the parameter bounds), and S12's "converges" to a mass three sigma below everyone else's with an uncertainty half the value itself: with a short arc, mass, distance, period and eccentricity trade against each other almost freely

— the fit is happy, the parameters are meaningless. This is not a defect of the optimiser; it is genuine degeneracy of the truncated data, and it is the concrete, reproducible illustration of why single-short-arc "measurements" of Sgr A*'s mass should never be taken at face value.

The joint fit changes the game: M and R_0 are pinned by S2's complete orbit, and each partial-arc star then only needs its six elements — for which even a short arc with RVs suffices. In exchange, every additional star's data sharpens the *global* block.

The mechanism deserves one more sentence, because it is the information accounting of Section 15.2.3 in action. A short arc measures a piece of a conic and a run of radial velocities; alone, that determines the product structure of (15.3) so poorly that the $M \propto R_0^3$ and $M \propto R_0$ constraints barely intersect. But the *shape* information in even a short arc — curvature on the sky plus the simultaneous line-of-sight acceleration from RVs — constrains the local gravitational parameter GM/R_0-ish combinations at the star's current radius quite well. Once the joint fit supplies (M, R_0) from S2, that same shape information inverts into clean per-star elements; and conversely each star's local-acceleration constraint, referred to the shared potential, tightens the global block by exactly the amount its arc quality merits. The convergence ladder of Fig. 16.1 is this argument rendered as data.

16.3 Results

The five-star joint fit converges with $\chi^2_\nu = 1.126$ over all 2×757 astrometric + 151 RV residuals and gives

$$M = 4.461 \pm 0.119 \times 10^6\,\mathrm{M}_\odot, \qquad R_0 = 8480 \pm 81\ \mathrm{pc}. \tag{16.2}$$

Against the S2-only fit the mass uncertainty shrinks by 39 per cent and the distance uncertainty by 56 per cent — with *no new instrument*, purely by demanding that five stars share one potential. The joint mass sits 1.4σ from the GRAVITY multi-orbit determination of $4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$ (GRAVITY Collaboration, 2022); given that our input is the 1992–2016 imaging-era compilation only — none of the GRAVITY beam astrometry — the agreement is a genuine cross-check, not a tautology. The middle rungs of the convergence ladder (Fig. 16.1, right; `cluster_fit.json subsets`) show the expected pattern: the biggest single gain comes from adding the first independent star (S1), with diminishing — but monotonic — returns thereafter.

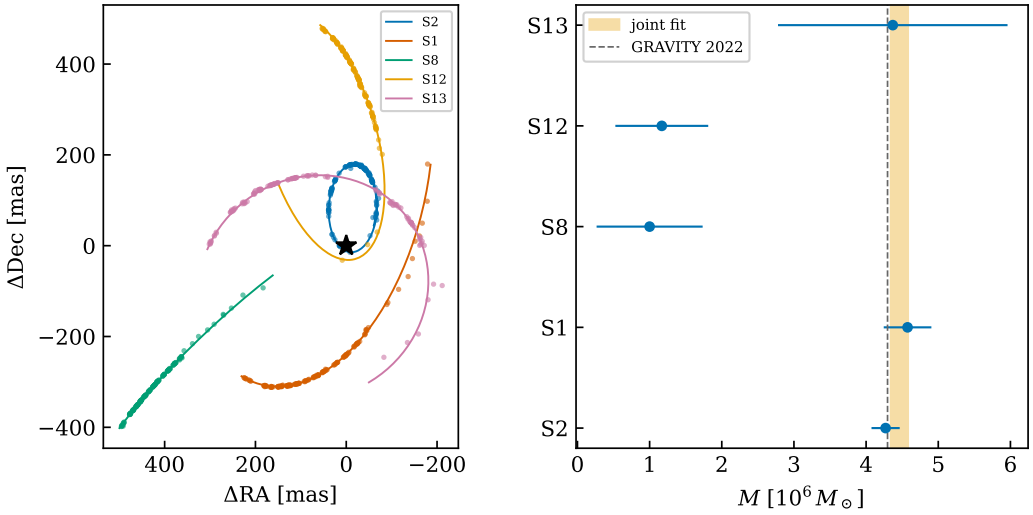


Figure 16.1: The joint cluster fit (`fit_cluster`). Left: the five fitted orbits on the sky at the joint solution, with the Gillessen et al. (2017) astrometry. Right: per-star single-fit masses (points; S8's non-convergent fit and S12's degenerate one carry the enormous error bars discussed in the text) against the joint-fit band (orange) and the GRAVITY 2022 value (dashed).

Two honest caveats bound the result. First, our error model is the published per-epoch uncertainties with the Plewa et al. (2015) frame priors — no error rescaling and no systematic-floor terms, whereas the professional multi-star fits marginalise over per-instrument offsets; our smaller formal uncertainty should be read in that light. Second, the companion stars are treated, like S2, as test particles in a clean Kerr-limit potential; extended mass inside their apocentres, which the professional fits constrain simultaneously, is not modelled — its absence is part of the residual budget, not the quoted errors.

We note for completeness that the contested star S62 — claimed at a 9.9-yr period well inside S2's orbit, but plausibly a misidentification of S29 or a blend artefact — is *not* included and would be the single most valuable addition if its orbit were ever confirmed (Chapter 17 quantifies why); any future use of S62 in this pipeline must carry the same contested-orbit caveat.

Chapter 17

A No-Hair Test with a Free Quadrupole

The no-hair theorem is only worth stating if it can fail. This chapter builds the failure mode into the spacetime — one extra parameter, the quadrupole surplus ε_q — and lets the S2 data set its value (`physics/src/hartle_thorne.cpp`; `OrbitFitter::fit_nohair`).

17.1 Multipole moments and the no-hair statement

Any stationary, asymptotically flat spacetime is characterised far from the source by two towers of multipole moments (Geroch–Hansen): the mass moments $M_0 = M, M_2, M_4, \dots$ and the current moments $S_1 = J, S_3, \dots$ (odd mass moments and even current moments vanish for reflection-symmetric sources). For a fluid star the towers are essentially free — they encode the internal mass distribution. The no-hair theorem states that for a Kerr black hole the *entire* double tower collapses to two numbers:

$$M_\ell + iS_\ell = M\,(ia)^\ell \qquad \Longrightarrow \qquad M_2 = -Ma^2,\;\; S_3 = -Ma^3,\;\dots \tag{17.1}$$

The moments are defined covariantly — Geroch and Hansen construct them from the asymptotic expansion of the Ernst potential, so they are properties of the spacetime, not of a coordinate chart — but their physical reading is exactly Newtonian intuition: M_2 is the oblateness of the far gravitational field, the general-relativistic descendant of the J_2 every satellite geodesist measures for the

Earth. That comparison calibrates how radical (17.1) is. For material bodies the quadrupole is a free diagnostic of internal structure: the Sun's $J_2 \sim 2 \times 10^{-7}$ reflects its slow rotation, a millisecond pulsar's dimensionless quadrupole can reach order unity, and in every case measuring it teaches you about the matter. For a Kerr black hole there is nothing to learn — the quadrupole is *locked* to $-Ma^2$, and every higher moment follows suit, because the exterior field has shed all memory of the source's structure (Psaltis et al., 2016). A single confirmed violation at any rung of the tower would falsify the black-hole hypothesis for Sgr A* outright — no astrophysical dial could repair it.

The first non-trivial prediction is the mass quadrupole $M_2 = -Ma^2$: measure M, a and M_2 *independently* and general relativity's black-hole hypothesis is falsifiable. Following Will (2008) we parametrise the violation as

$$Q_{\rm tot} = -Ma^2 + \varepsilon_q M^3, \tag{17.2}$$

so $\varepsilon_q = 0$ is Kerr and $|\varepsilon_q|$ measures the surplus squashing of the field in natural units.

17.2 The Hartle–Thorne / quasi-Kerr metric

A spacetime realising (17.2) is furnished by the slow-rotation solution of Hartle & Thorne (1968), valid through $\mathcal{O}(a^2)$ with the quadrupole free — adequate here because the quadrupole's orbital effect is itself a small perturbation at S2 radii. The truncation is not a loose end but a budgeted approximation: the first neglected structure is the current octupole $S_3 = -Ma^3$, whose pericentre imprint at S2's semilatus rectum sits a further two-plus orders below the quadrupole rung on the precession ladder of Fig. 15.7 — if the data cannot yet see the quadrupole, they are far blinder to what the truncation discards. The moment the target moves to stars deep enough that S_3 matters, the `IMetric` interface accepts a higher-order metric without touching the integrator; that is the payoff of the generic-metric design decision of Chapter 10. We implement the quasi-Kerr form of Glampedakis & Babak (2006): the deviation enters as a perturbation of the *inverse* metric,

$$g^{ab} = g^{ab}_{\rm Kerr} + \varepsilon_q\, h^{ab}, \tag{17.3}$$

where the non-zero h^{ab} components are built from two radial structure functions $F_1(r), F_2(r)$ multiplying $(1 - 3\cos^2\theta)$ angular factors — the quadrupole's P_2 signature. Perturbing the inverse metric is a deliberate choice: the geodesic Hamiltonian is $\tfrac{1}{2} g^{ab} p_a p_b$, so (17.3) feeds the equations of motion linearly and the $\varepsilon_q \to 0$

limit is exact by construction. A dedicated unit test asserts that limit at the 10^{-12} level component-by-component (Gate 0; Chapter 21).

The structure functions carry a numerical trap that is worth recording. F_1 and F_2 are combinations of low-order polynomials and logarithms $\ln\big(r/(r-2M)\big)$ arranged so that the *leading five* powers of M/r cancel — the quadrupole correction must fall off as r^{-3} relative to the monopole, and the cancellation enforcing this is catastrophic in floating point: at $r = 50\,M$ the individual terms are $\sim 10^5$ times the surviving result, at $r = 300\,M$ (an S2 apocentre distance scale) $\sim 10^{10}$ times. The implementation therefore switches, beyond $r = 9M$, to the asymptotic series of the cancelled result, with the switch radius chosen where the two evaluations agree to 10^{-10}; a branch-continuity unit test pins the joint (Chapter 21). Because the metric is not Carter-separable, all orbits in this spacetime run through the generic Christoffel integrator with complex-step derivatives (Chapters 9–10) — the reason that integrator exists.

17.3 The quadrupole precession anchor

One closed-form consequence anchors the sign and scale of everything: a quadrupole $\tilde{q} = Q/M^3$ adds, for an equatorial orbit of semilatus rectum p, a pericentre advance

$$\delta\omega_Q = -3\pi\,\tilde{q}\left(\frac{M}{p}\right)^2 \quad \text{per orbit} \tag{17.4}$$

(Will, 2008) — retrograde for an oblate source ($\tilde{q} < 0$ advances, $\varepsilon_q > 0$ retards; the Newtonian $-QP_2/r^3$ potential perturbation gives the same result, which is the fastest way to fix the sign). The implementation's sign convention is pinned by a *two-sided* test: the analytic (17.4) (`GRPrecession::quadrupole`) against the numerically measured apsidal advance of an integrated quasi-Kerr orbit, for both signs of ε_q — a pairing that once caught a real sign error in h^{ab} (the bug-hunt narrative appears in Chapter 21). Fig. 15.7 places (17.4) on the precession ladder: at S2's $p \approx 5400\,M$, one unit of ε_q moves the pericentre by $0.07''$ per orbit, five orders below the Schwarzschild term.

17.4 Profile-likelihood constraint from S2

Re-fitting the S2 data in the quasi-Kerr spacetime with ε_q as a fourteenth parameter uses a profile likelihood: ε_q is stepped over a grid, the remaining thir-

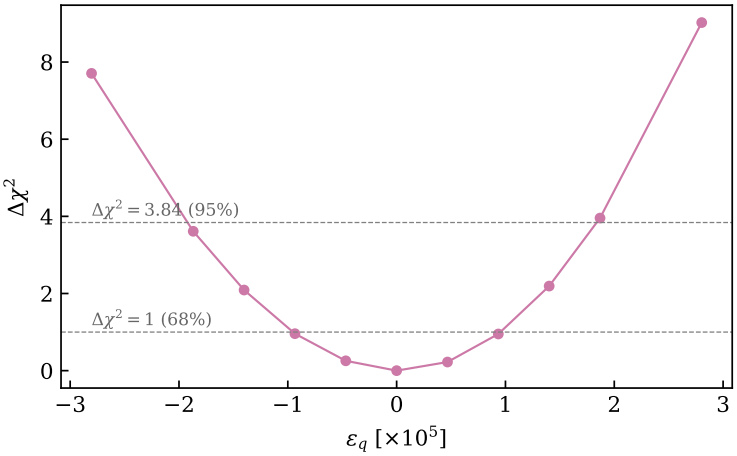


Figure 17.1: $\Delta\chi^2$ profile of the quadrupole deviation ε_q from the S2 data, with the 68 and 95 per cent levels marked. The 95 per cent interval is $|\varepsilon_q| < 1.33 \times 10^5$, consistent with the Kerr value $\varepsilon_q = 0$.

teen parameters are re-optimised at each step (`OrbitFitter::fit_nohair`), and $\Delta\chi^2(\varepsilon_q) = \chi^2(\varepsilon_q) - \chi^2_{\min}$ is read against the $\Delta\chi^2 = 1$ (68 per cent) and 3.84 (95 per cent) levels. The statistical licence for those thresholds is Wilks' theorem: with Gaussian errors, the profiled $\Delta\chi^2$ of a single interior parameter is asymptotically χ^2-distributed with one degree of freedom, *regardless* of how the thirteen nuisance parameters contort — re-optimising them at each grid point is what buys that invariance. Profiling, rather than slicing χ^2 at the global best-fit nuisance values, matters in principle here: ε_q acts through the pericentre advance, so it can trade against the orientation elements, and a fixed-nuisance slice would understate the interval by the usual "across the valley versus along its floor" geometry. Profiling walks the floor. The profile (Fig. 17.1) is a clean parabola centred near zero: the best fit is $\varepsilon_q = 0.00 \times 10^5$ with $\sigma(\varepsilon_q) = 0.93 \times 10^5$ and a 95 per cent half-width of 1.33×10^5. The data are fully consistent with the no-hair value, and the marginal χ^2_ν is unchanged from the Kerr fit — the extra degree of freedom buys nothing, exactly as expected if Kerr is the true spacetime.

The MCMC of Chapter 15, re-run with the fourteenth parameter, gives the same answer from a global sampling: $\varepsilon_q = -0.43^{+6.03}_{-5.96} \times 10^5$ (16/50/84 percentiles), and the joint (ε_q, M) posterior (Fig. 17.2) shows the Kerr value comfortably inside the 68 per cent contour with no significant ε_q–M leakage — the quadrupole is not silently absorbing mass mis-estimates.

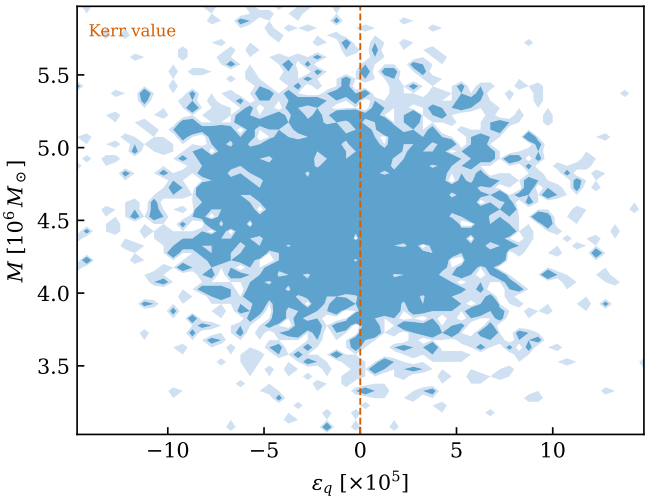


Figure 17.2: Joint posterior of the quadrupole deviation and the mass from the no-hair MCMC chain (68 and 95 per cent credible regions). The Kerr value $\varepsilon_q = 0$ (dashed) lies within the 68 per cent contour.

17.5 Why the constraint is weak, and how it improves

The constraint is weak in absolute terms because of the $(M/p)^2$ suppression in (17.4). Working the numbers: $M/p = 1.86 \times 10^{-4}$ for S2, so one unit of ε_q moves the pericentre by 3.3×10^{-7} rad per orbit — $0.07''$ — against an astrometric sensitivity to ω of order 0.5° per orbit (Table 15.1). Inverting, the expected reach is $\sigma(\varepsilon_q) \sim 10^5$, exactly the order the profile likelihood recovers: a single star at $\sim$3000 gravitational radii is simply far from the quadrupole's domain of influence. The value of the exercise is methodological — the machinery applies unchanged to closer stars. Continued monitoring helps: propagating the fitted covariance forward, ten further S2 orbits of the existing quality shrink the uncertainty to $\sigma(\varepsilon_q) \approx 0.34 \times 10^5$, a factor $\sim$3. A qualitative jump requires a confirmed star well inside S2's orbit: the $(M/p)^2$ scaling means a star at a third of S2's pericentre is an order of magnitude more sensitive per orbit, with a shorter period compounding the gain. The reported candidate S62, with a claimed 9.9-yr period and $\sim$36 au pericentre, would be such a star, but we emphasise: *S62 is a contested orbit — its identification is unconfirmed and may be a misidentification of S29 or a blend artefact*; we therefore build no quantitative forecast on it, and any future use of S62 in this context must carry the same caveat.

Part IV

Engineering, Validation, and Closing

Chapter 18

GPU Implementation and Validation

18.1 Port architecture

The null-geodesic core is ported to a GLSL compute shader (Vulkan, double precision) using the same second-order Mino-time formulation (`engine/shaders/geodesic.comp.glsl`), with all per-frame parameters delivered through a bindless (update-after-bind) descriptor table — one global resource array, written once, indexed by the shader, so no per-frame descriptor churn ever appears on the timeline.

18.1.1 Double-precision trigonometry on the GPU

GLSL exposes `double` arithmetic (`shaderFloat64`) but *no* double-precision transcendentals, and the Carter equations need $\sin\theta, \cos\theta$ at full precision every step. The shader implements them by quadrant-tracking argument reduction to $|x| \le \pi/4$ followed by fixed truncated Taylor polynomials — degree 11 for sine, degree 10 for cosine (`dsin_core`, `dcos_core`). (The companion journal paper called the pair "minimax"; the coefficients are in fact the Taylor ones, and this report documents the code as written. A Remez pair of the same degree would gain roughly one decimal digit at the interval ends — immaterial below the validation threshold, as the measurement shows.) Fig. 18.1 evaluates the shader's exact polynomial arithmetic against library sin / cos across the reduction interval: the truncation error rises to $\sim 7 \times 10^{-12}$ for sine ($\sim 5 \times 10^{-11}$ for cosine) at $|x| = \pi/4$, comfortably below the per-step tolerances of the fp64 metrology path and five orders below the 0.01 pixel-diff acceptance gate.

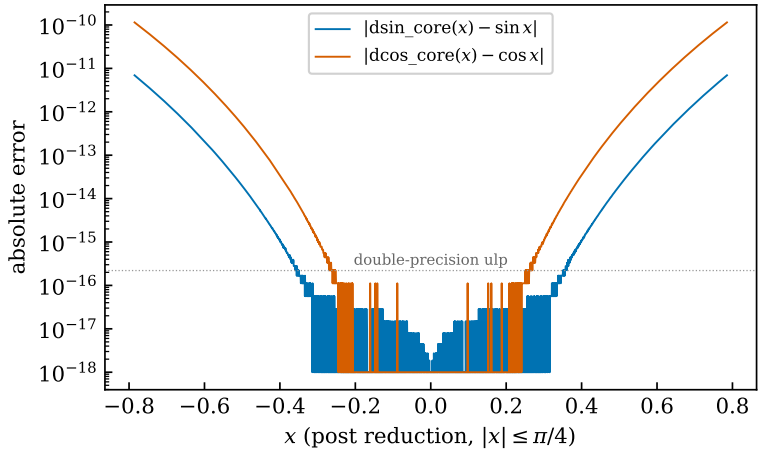


Figure 18.1: Measured truncation error of the GPU double-precision sin / cos cores over the post-reduction interval, evaluated with the shader's exact coefficient arithmetic. The double-precision unit roundoff is marked.

How those per-step errors accumulate over a ray is worth budgeting explicitly. The trig truncation acts as a small perturbation injected each step with effectively random sign as θ sweeps its oscillation; over a worst-case ray of $\sim$1200 steps the random-walk accumulation is $\sqrt{1200} \times 7 \times 10^{-12} \approx 2 \times 10^{-10}$ — utterly negligible against every tolerance in the pipeline *except* exactly at the capture/escape separatrix, where Chapter 11's exponential divergence amplifies any perturbation without bound. That budget predicts precisely the validation outcome observed below: bit-level agreement everywhere except isolated flips on the one-pixel critical circle.

The precision asymmetry of consumer GPUs is the structural fact behind the whole port design. Gaming-class silicon executes double-precision arithmetic at a small fraction of single-precision throughput (1:32 on the Pascal generation of the test GPU), because fp64 units are a die-area luxury reserved for compute cards. The project therefore splits cleanly: the *metrology* path runs fp64 offscreen, unpaced by any display, where a slow-but-exact frame is acceptable; the *interactive* path runs fp32 at speed with a shortened optical path (Section 18.3) so that accumulated fp32 drift stays below visual perceptibility. No quantitative number in this report ever originates from the fp32 path — the split is enforced by pipeline, not by discipline.

18.2 Validation against the CPU reference

A 64×64 shadow rendering is compared pixel-by-pixel against the CPU reference (`validation/compare_shadow.cpp`): the mean absolute difference is 0.00289 (gate threshold 0.01), with 32 pixels differing by more than 0.1 — every one of them on the one-pixel-wide critical boundary, where capture/escape classification is legitimately ambiguous at pixel resolution (the boundary rays of Chapter 11 diverge exponentially, so the last representable bit decides the pixel). The maximum single-pixel difference is 0.697, i.e. one full classification flip on the critical circle; away from that circle the two implementations agree to round-off. This is the strongest statement available for a cross-architecture port: identical mathematics, independent compilers and schedulers, indistinguishable physics.

18.3 Watchdog-safe dispatch and the precision split

Two practical constraints shaped the port. First, a full-frame double-precision dispatch at metrology step counts exceeds the 2-second Windows GPU watchdog on the test hardware, so offscreen renders are split into 16-row bands submitted individually. Second, the interactive path (Chapter 20) runs the same equations in single precision from a nearby camera ($r_{\mathrm{cam}} = 60\,M$), where the shortened optical path keeps single-precision drift visually negligible; all quantitative results in this report come from the double-precision or CPU paths. All configurations run with zero Vulkan validation-layer errors.

18.4 Performance

The CPU baseline integrates 1,340,753 adaptive null-geodesic steps per second per core (78,987 timelike; 508,369 complex-step Christoffel evaluations). The double-precision GPU port traces the validated 64×64 shadow tile within the Windows watchdog budget in row-banded dispatches, and the single-precision interactive path sustains 47.5 fps at 1280×720 (21 fps at 1920×1080; $\approx$43.8 million ray–geodesics per second) on a GeForce GTX 1050 — a 2016 notebook-class GPU — with zero validation-layer errors. (The 60 fps full-HD target of the project specification assumes an RTX-class GPU; the figures above are the best

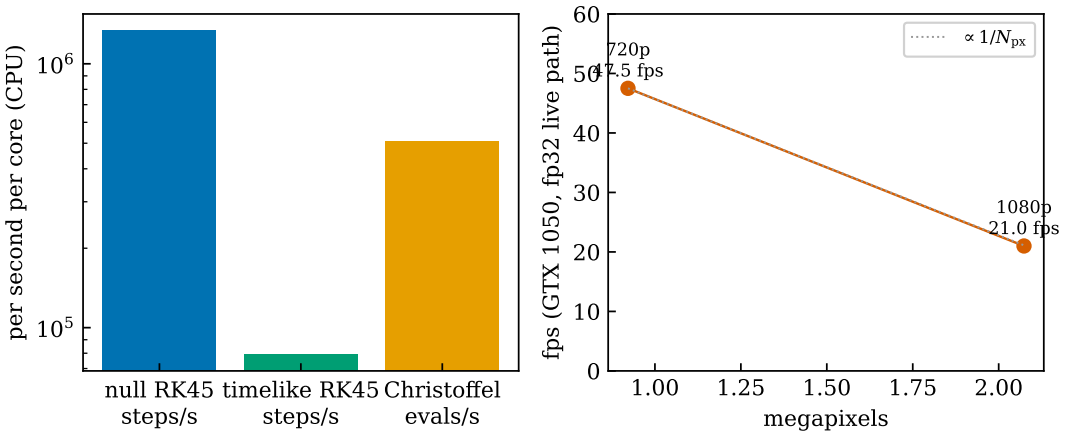


Figure 18.2: Performance artefacts (`phase11/bench.json`). Left: CPU single-core throughputs of the three hot kernels. Right: live-path GPU frame rate versus pixel count on the GTX 1050 against the ideal $1/N_{\mathrm{px}}$ scaling.

achievable on the available hardware and scale linearly with pixel count.) Fig. 18.2 collects the benchmark artefacts: the CPU per-core throughputs that set the offline render times, and the GPU frame rate against pixel count, which follows the $1/N_{\mathrm{px}}$ line closely — the workload is one geodesic per pixel with no shared state, the embarrassingly parallel ideal, so the live renderer's auto-resolution controller (Chapter 20) can trade resolution for frame rate predictably.

Chapter 19

Sagittarius A* Calibration and Parameter Sweeps

19.1 From simulation units to the sky

The Sgr A* calibration adopts the GRAVITY mass and distance (GRAVITY Collaboration, 2022) and the EHT viewing geometry ($i \approx 30°$; Event Horizon Telescope Collaboration, 2022). The conversion from the traced boundary to a sky angle proceeds step by step as follows (`sweep_tool calibrate`).

1. One gravitational radius is $R_g = GM/c^2 = 4.297 \times 10^6 \times 1.4766\,\mathrm{km} = 6.346 \times 10^9\,\mathrm{m}$, using $GM_\odot/c^2 = 1.4766$ km.

2. At $D = 8277\,\mathrm{pc} = 2.554 \times 10^{20}$ m it subtends $\theta_g = R_g/D = 2.485 \times 10^{-11}\,\mathrm{rad} = 5.1243\,\mu\mathrm{as}$.

3. The traced critical boundary at $a_* = 0.94$, $i = 30°$ has mean diameter $9.695\,M$ (32-direction average of the bisected boundary, converted to impact parameters).

4. The photon-ring angular diameter is therefore $d_{\mathrm{ring}} = 9.695 \times 5.1243 = 49.68\,\mu\mathrm{as}$.

This falls within the EHT ring measurement of $51.8 \pm 2.3\,\mu$as and the inferred shadow diameter of $48.7 \pm 7\,\mu$as (Event Horizon Telescope Collaboration, 2022). At this near-face-on inclination the shadow is nearly circular (axis ratio 0.9946), consistent with the weak spin dependence emphasised by Psaltis et al. (2016).

19.2 Error budget

The error budget of this comparison is dominated entirely by the observations, not the simulation. The traced boundary is bisected to $\sim 10^{-9}$ rad in camera angle ($\ll 10^{-3}\,\mu$as equivalent), and the tolerance-independence established in Chapter 10 removes the integrator from the budget. On the calibration side the conversion scales as $\theta_g \propto M/D$, so

$$\frac{\sigma_\theta}{\theta} = \sqrt{\left(\frac{\sigma_M}{M}\right)^2 + \left(\frac{\sigma_D}{D}\right)^2} = \sqrt{0.28^2 + 0.11^2}\,\% \simeq 0.30\,\%, \tag{19.1}$$

about $0.15\,\mu$as, twenty times smaller than the EHT error bar. (Equation (19.1) treats the GRAVITY M and D as independent; in truth they emerge correlated from the same orbit fits — the very M–R_0 geometry of Chapter 15 — and since the correlation is positive, part of it cancels in the ratio M/D, making (19.1) mildly conservative.) The comparison is therefore limited by the $2.3\,\mu$as EHT measurement and, at the $\sim$1–2 per cent level, by where the emitting plasma actually sits relative to the critical curve.

19.3 The spin–inclination plane

Fig. 19.1 maps the ring diameter over the full (a_*, i) plane: at each of the 20×20 grid points the complete 32-direction bisection metrology of Chapter 11 is repeated from scratch — four hundred independently traced critical boundaries, every one anchored to the same tolerance-independent machinery, with the $a_* = 0$ row reproducing the Synge value identically at all inclinations as a built-in control. The total spread across the physical plane is a few per cent, illustrating the well-known spin–inclination degeneracy of ring-size measurements: spin shrinks the mean diameter slightly, inclination reshapes it slightly, and over most of the plane the two effects trade against each other inside the EHT error bar.

What would break the degeneracy is worth stating, since the machinery to predict each candidate already exists in this report. The *sub-ring* structure of Chapter 13 sharpens with resolution rather than brightness, and its azimuthal width/flux variation tracks the Lyapunov band of Chapter 11, which *does* depend on spin at fixed diameter; interferometric detection of the $n = 1$ ring is therefore the cleanest spin-at-fixed-size observable (Johnson et al., 2020). Independently, the shape observables below — axis ratio and centroid displacement — scale as $a_* \sin i$ and so separate the two parameters if imaging fidelity ever reaches the

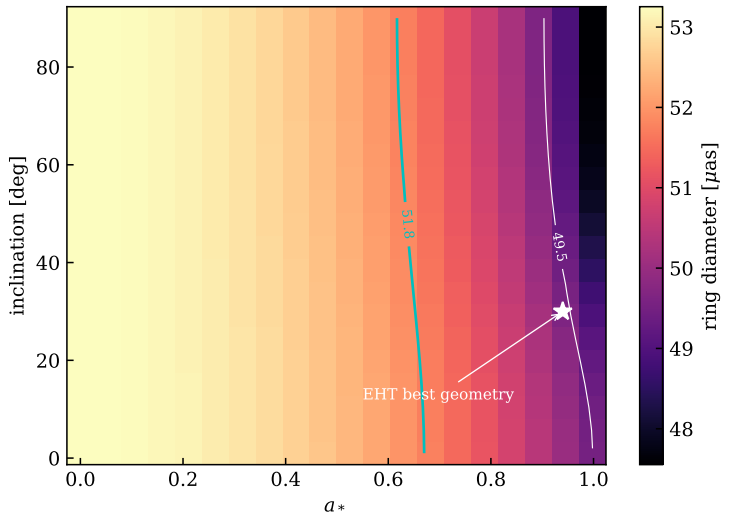


Figure 19.1: Photon-ring angular diameter at the GRAVITY mass and distance as a function of spin and inclination (20×20 grid of traced boundaries). The EHT band $51.8 \pm 2.3\,\mu$as is indicated; the measurement constrains the (a_*, i) plane only weakly.

sub-per-cent regime. Both statements are quantitative outputs of this simulator, not aspirations.

Diameter is not the only observable the boundary carries. The extended sweep also records the *axis ratio* d_x/d_y and the horizontal *centroid displacement* of the boundary (Fig. 19.2) — the two shape signatures through which spin could in principle be recovered where size cannot. Both behave as the critical-curve geometry of Chapter 11 predicts: the distortions grow with $a_* \sin i$, reaching a few per cent flattening and a displacement of $\sim 1\,M$ only in the high-spin, near-edge-on corner. At the Sgr A* viewing geometry ($i \approx 30°$, white star) the axis ratio is 0.9946 — a fraction of a per cent from circular, far below current imaging fidelity, which is the quantitative content of the statement that present EHT data cannot measure Sgr A*'s spin from the ring shape (Psaltis et al., 2016). The shape maps also connect to the spin-morphology discussion of Section 12.7: the displacement map is the quantitative footprint of the prograde-side flattening visible in the renders.

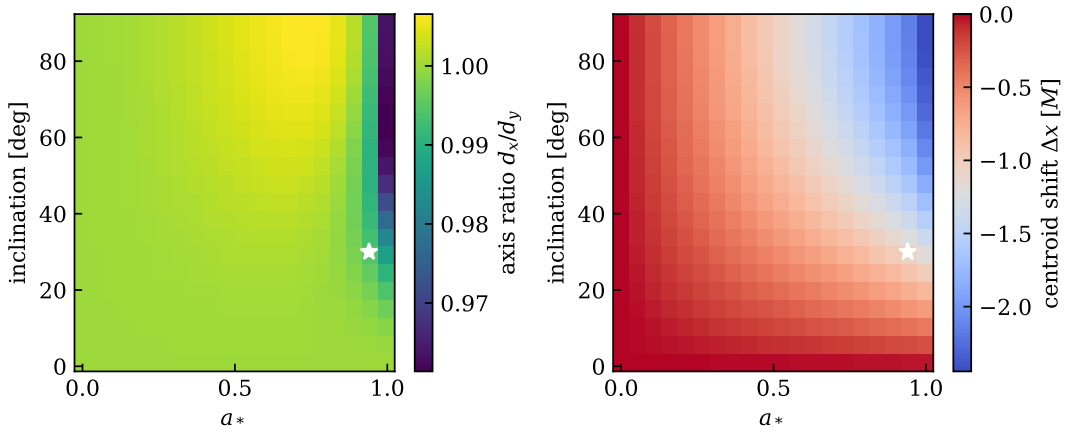


Figure 19.2: Shape observables over the (a_*, i) plane from the extended sweep (`sweep_tool`): boundary axis ratio (left) and horizontal centroid displacement (right). The white star marks the Sgr A* geometry, where both signatures are sub-per-cent.

Chapter 20

The Interactive Instrument

Beyond the batch pipeline, the same physics core drives an interactive Vulkan application. This chapter documents it with one organising principle stated up front: *a hard wall separates physics from presentation.* Everything that decides where a ray goes or how much energy it carries lives in the validated core of Parts II–III; everything this chapter adds acts after that point, on push constants and display look-up tables, and can be deleted without changing a single measured number. The wall is enforced structurally (the display layer has no write access to the geodesic or metric code paths) and empirically (the `ctest` suite is byte-identical with the display layer present or absent).

20.1 Architecture and modes

The engine is a single compute-to-present loop: a GLSL compute shader (the fp32 sibling of the fp64 metrology shader of Chapter 18) traces one geodesic per pixel into a storage image, which is blitted to the swapchain; Dear ImGui draws the control surface on top. All resources — render targets, LUTs, parameter tables — live in one bindless descriptor array written at start-up. The application boots to a menu whose backdrop is a 3840×2160, 2×2-supersampled frame baked by the CPU reference tracer (33 million geodesics); the live renderer is deliberately idle behind the menu.

Four modes expose the science: the live Kerr renderer with runtime control of $(a_*, i, r_{\mathrm{cam}}, r_{\mathrm{out}}$, exposure, disk temperature); a Science Mode overlaying the fitted S2 orbit and its data on the fit summary of Chapter 15, together with a 3-D orbit theatre that animates the five jointly fitted S-stars (Chapter 16) along their *integrated* GR geodesics — the engine links the same Ceres-free orbit model

the fitter uses and replays its cached trajectories, with each star shaded by the exact line-of-sight redshift $1+z$ of the trajectory sample (white when approaching, through gold, to bronze when receding), the time axis scalable from real Sgr A* wall clock to $10^8\times$; a no-hair panel exposing ε_q with the live analytic precession forecast of Chapter 17 beside the fitted confidence interval; and a photon-ring scope that reframes the camera at $r = 60\,M$ with a narrow field to isolate the $n = 1$ ring visually. Camera orbit (mouse drag over azimuth and inclination, scroll zoom 15–400 M) reprojects the ZAMO tetrad of Chapter 11 — moving the camera is a physics operation and goes through the physics path.

20.2 Presets and display grading

Five named looks (FIRESTORM, GARGANTUA, ARCTIC, NEBULA, and CINEMATIC SGR A*) are 256×1 RGBA look-up tables generated by `scripts/gen_luts.py`, applied to the physically shaded luminance. The CINEMATIC preset additionally enables a volumetric display layer: a flared slab of procedural dust banks, advected by the exact $\Omega(r)$ and shaded by the same per-sample redshift factor and blackbody dictionary as the midplane, is composited front-to-back along each geodesic — emissive fog over unchanged physics, in the same spirit (and behind the same wall) as the filament layer below. The physical option — the Planck/CIE blackbody map of Chapter 12 — is the default and is regenerated by the same script from the same colour code the CPU renderer uses. Preset switches write a LUT index into push constants; nothing else changes. Display dynamic range is handled by a calibrated logarithmic window (24 stops mapped to the LUT domain), whose earlier mis-calibration produced the "brick disk" bug hunt of Chapter 21.

20.3 The display-only turbulence layer

The reference physics disk is time-averaged and therefore perfectly smooth; real accretion flows are filamentary. For the live instrument a *display-only* structure layer restores that texture — designed so its every input is either the validated physics or pure presentation:

- *Streamlines wound by the real shear.* The filament pattern is value noise advected in azimuth by the exact circular-orbit rate $\Omega(r)$ of eq. (8.21): features at smaller radii overtake those farther out precisely as Kerr orbital

mechanics dictates, winding the pattern into trailing spirals. The noise is azimuthally periodic by construction (the seam lesson of Chapter 21).

- *Physical clock.* The advection speed is calibrated so that a rotation-speed setting of 1.0 equals real Sgr A* wall-clock time ($GM/c^3 = 21.2$ s per geometric time unit; one ISCO orbit ≈ 8.5 minutes), scalable to 1000× for demonstration.

- *Translucent rim compositing.* Disk crossings composite front-to-back with a transmittance term rather than terminating opaquely, softening the rim; the modulation depth is ±2 stops around the Page–Thorne mean, leaving the radial photometry of Chapter 12 intact on average.

- *Auto resolution.* A controller retunes the internal render scale every two seconds toward a 13 fps target on the GTX 1050 (the $1/N_{\mathrm{px}}$ scaling of Fig. 18.2 makes the control law exact); scales above 1× provide supersampled stills.

None of these touches (E, L_z, Q), the metric, or any measured artefact; the spin-morphology observations of Section 12.7 were made *through* this layer, which is why that section grounds each visual feature in the underlying validated quantity (r_{ISCO}, g^4, $\Omega(r)$) rather than in the layer itself.

The same display layer carries a caustic de-aliasing stack for the live fp32 path — a stiffness-adaptive polar step law, ray-property triggered supersampling of axis-squeezed pixels, and band-limited shading of the deepest pole-passing families — whose three-round development, including the two regressions the first attempts caused, is retold as a methods narrative in Sections 21.3.7–21.3.9. A black-background toggle blanks the starfield backdrop entirely (display only; the lensed disk image is unchanged) for silhouette study and clean composites.

The same five fitted S-star orbits can also be drawn *into the live scene* rather than in a separate window. Each star's position comes from the identical integrated geodesic, projected through the exact inverse of the compute shader's camera model — the same ZAMO triad, field of view and aspect convention — so the stars track the mouse orbit and zoom for free and sit precisely where that camera would see them; a star passing behind the silhouette is faded using the Schwarzschild critical impact parameter $3\sqrt{3}\,M$ as the occultation test.

The stars are drawn at their *true* geometric distances by default, which is worth stating plainly because it is a scale statement about the whole instrument: the renderer works in units of $M = GM/c^2$ throughout, so the shadow is $\sim 5\,M$, the

modelled disk $\sim 20\,M$, and S2's semi-major axis $2.38\times10^4\,M$ ($\approx$1050 AU) — the S-star cluster is three orders of magnitude larger than the entire imaged region, and no view can hold both at once at a useful magnification. The instrument therefore resolves the conflict by moving the *camera* rather than the stars: a control flies the viewpoint out past the cluster ($\sim 5\times10^5\,M$, where the fp32 live path remains clean because its step law is logarithmic in r), and the hole duly shrinks to a lensed point — the honest picture of how empty the intervening volume is. A separate, camera-independent ratio control can pull the orbits inward by a labelled factor purely for close-up overviews, in the manner of published cluster renderings; it never rescales with the camera, so proportions never shift silently while zooming. Orbit shape, orientation, phase, period and the redshift colour are the unmodified fit in every case, and starlight bending is not applied to the star sprites.

Focus camera

Any star can be selected as the camera target. The ray basis of Section 20.1 is no longer hard-wired to point at the hole: an orthonormal triad built on the CPU is handed to the shader as push constants, so the camera can orbit a chosen star (mouse drag and scroll re-parameterised about it) or ride it and look back at Sgr A*. The default basis reproduces the historical aim, and the star sprites are sized by the true angular diameter $R_\star/d$, so approaching S2 resolves it into a disc instead of holding a fixed dot.

Hypothetical shrunken orbits and the tidal limit

Because the instrument works in true geometric units, the ratio control that pulls the orbits inward is implemented as a genuine re-integration with $a \to a/N$ rather than a redrawing: the period then follows Kepler's third law ($P/N^{3/2}$) and relativistic pericentre advance grows with it, so the shrunken star visibly races round a precessing rosette. Every such state is labelled *hypothetical* in the panel.

This makes a real threshold reachable. Using the spectroscopic masses and surface gravities of Habibi et al. (2017) — from which the stellar radius follows as $R_\star = \sqrt{GM_\star/g}$, giving $5.4\,R_\odot$ for S2, consistent with its B0–B3V classification — the tidal radius

$$r_\mathrm{t} \;=\; \eta\, R_\star \left(\frac{M_\mathrm{BH}}{M_\star}\right)^{1/3}, \tag{20.1}$$

with $\eta = 1$ for the classical tidal radius and $\eta = 2.44$ for the fluid Roche limit,

evaluates to $r_{\rm t} \simeq 40\,M$ for S2 against a true pericentre of $2845\,M$. The star therefore survives by a factor of $\sim$72, and the instrument disrupts it once the shrink factor passes that value. The prediction is per-star and ordered — S2 at $N \simeq 72$, S8 at 96, S12 at 182 — which makes the feature a check rather than an effect: the thresholds were computed independently from the cited data and the fitted mass, and the instrument reproduces them (all five stars intact at $N = 60$; S2 alone destroyed at $N = 80$; S2 gone and S8 surviving at $N = 90$).

Beyond the threshold the star is replaced by tidal debris under the frozen-in approximation (Rees, 1988): fragments retain the orbital plane and pericentre but carry a spread in specific binding energy $\Delta E \simeq GM_{\rm BH}R_\star/r_{\rm t}^2$, so $1/a_i = 1/a \mp 2\Delta E$ fans them into orbits of differing period — the bound half returns as a fallback stream and is drawn, the unbound half escapes. Each fragment is integrated as an ordinary orbit in the same validated model, so the debris obey the same GR dynamics as the star did; the black hole's spin acts on them physically, through frame dragging, rather than as a playback control. Approaching disruption the sprite elongates as $(r_{\rm t}/r)^3$, the tidal-strain scaling.

None of this touches a fitted quantity: the demonstration lives entirely in the display layer, and the validity floors inside the orbit model that permit these small orbits are parameters whose defaults — the values every fit uses — are unchanged.

20.4 The control surface and the auto-resolution loop

Interactivity is only useful if the frame rate survives it, and the GTX-class test hardware saturates at roughly nine full-resolution frames per second (one full geodesic per pixel; Chapter 18). Rather than pick a fixed internal resolution, the instrument closes the loop: every two seconds the frame-rate sampler updates the internal render scale by

$$s \;\leftarrow\; \mathrm{clamp}\!\left(s\,\sqrt{\tfrac{\mathrm{fps}}{13}},\; 0.25,\; 2.0\right), \tag{20.2}$$

damped by blending most of the way toward the new value and ignoring sub-5-per-cent changes to avoid render-target churn (each change rebuilds the target and rewrites its bindless slot). The square root is exact, not heuristic: the workload is one ray per pixel, so frame time scales linearly with pixel *count* and hence quadratically with linear scale — the $1/N_{\rm px}$ law measured in Fig. 18.2 makes

(20.2) a one-step deadbeat controller, the same design logic as the integrator's step controller in Chapter 10. Scales above 1× downsample through a linear blit, so the same slider doubles as a supersampling control for stills. The rotation-rate slider is calibrated so that 1.0 advances the disk pattern at real Sgr A* wall-clock rate ($GM/c^3 = 21.2$ s per unit of coordinate time; one ISCO orbit every 8.5 minutes) on a logarithmic range up to 1000× — at 1.0 the disk is, correctly and instructively, almost static to the eye, which is itself a physics statement about how sedate a four-million-solar-mass hole's dynamics are in human units.

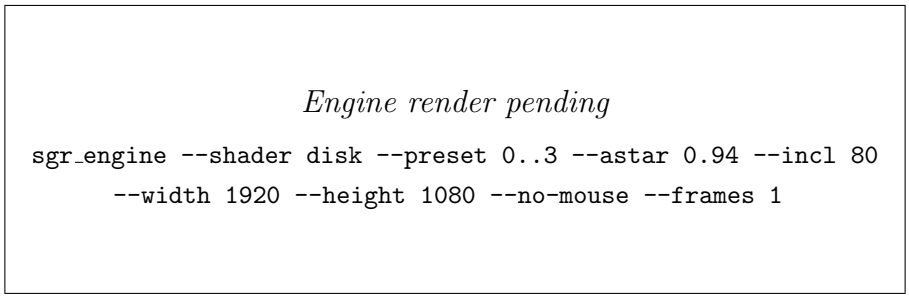


Figure 20.1: *[Placeholder] The four colour presets (FIRESTORM, GARGANTUA, ARCTIC, NEBULA) on the same physical scene, showing that the preset acts on the display LUT only — the geometry is identical across all four. Reproduced by the command shown; the render will be slotted in when supplied.*

Chapter 21

Development History, Iterations, and the Test Suite

An instrument is only as credible as its validation record. This chapter documents how the simulator was built — twelve gated phases, no phase entered while any earlier gate was red — catalogues the complete automated test suite with its exact tolerances, and retells the ten significant bug hunts as methods lessons, because what a validation system *caught* is the best evidence for what it would catch.

21.1 Gated development

Development proceeded through the gates of Table 21.1; each passing gate was tagged in version control, and every criterion is asserted by code — a Catch2 test, a JSON metrology artefact, or a validation-layer counter — never by eye. The ordering embodies the project's central discipline: *no GPU work until the physics was proven on the CPU* (Gates 0–5 are pure CPU and science), and no calibration claims until the renderer had survived pixel-level cross-validation (Gate 7).

Read as a sequence, the ladder tells the project's story in miniature. Gates 0–2 built *trust in the mathematics*: no pixel existed yet, only conserved quantities and analytic boundary comparisons — the phase where a sign error costs an afternoon instead of a month. Gates 3–5 converted trusted mathematics into *science claims*: the first quantitative beaming ratio, the first fit against real published data, the first constraint with an error bar. Gates 6–9 were *engineering under guard*: the GPU was allowed to render only what the CPU had already proven (the pixel-difference gate is the enforcement mechanism), and the interactive shell was allowed to exist only with validation layers permanently on. Gates 10–12

Table 21.1: Verification gates. Every criterion is asserted by code; the tags exist in the repository history.

Gate	Tag	Criterion	Result
0	`v0.0-physics-foundation`	all physics unit tests green	$(E, L_z, Q) < 10^{-6}$; norm $< 10^{-8}$; HT < 1
1	`v0.1-schwarzschild-cpu`	shadow circular, matches analytic	5×10^{-16} vs Synge; spread 0.000 %
2	`v0.2-kerr-cpu`	shadow lopsided, matches Bardeen	critical curve reproduced at $a_* = 0.94$
3	`v0.3-disk-subrings`	beaming asymmetric, $n = 1$ ring	9.60× asymmetry; $F_1/F_2 = 274$
4	`v0.4-orbit-fitter`	M within 2σ, $\chi^2_\nu < 2$	0.14σ; $\chi^2_\nu = 0.959$
5	`v0.5-no-hair`	ε_q CI reported	$\lvert\varepsilon_q\rvert < 1.33 \times 10^5$ (95 %)
6	`v0.6-vulkan-shell`	compute-to-present, zero VL errors	0 errors, 0 warnings
7	`v0.7-gpu-geodesics`	CPU/GPU diff < 0.01	mean 0.00289 at 64×64
8	`v0.8-full-renderer`	presets + HUD ≥ 30 fps	41–55 fps at 720p, 0 VL errors
9	`v0.9-menu`	menu + clean transitions	menu→sim→menu, 0 VL errors
10	`v1.0-sgrA-calibration`	ring 48–52 μas	49.68 μas
12	`v1.0-paper`	paper compiles, reproducible	journal paper + this report

closed the loop back to the sky: calibration against EHT, and the reproducibility contract that turned a working simulator into a defensible report. The ordering was not aesthetic: on three occasions (the tolerance-dependent shadow, the Keplerian period error, the covariance failure — all below) a gate refused to pass, and in each case the failure was caught at the cheapest possible layer because nothing downstream had been built on it yet.

21.2 The test suite

The physics and science cores are guarded by 32 production Catch2 test cases (plus one hidden diagnostic), all run by `ctest` as the first step of `make reproduce`. Table 21.2 catalogues them by what they guard; the names are the literal `TEST_CASE` strings, so the table doubles as an index into `physics/tests/` and `science/orbit_fitter/tests/`.

Three design habits are visible in the table and are recommended practice for any numerical-relativity-adjacent codebase. *Limits before features*: every Kerr formula is pinned at $a_* = 0$ against Schwarzschild before it is used at spin. *Two implementations where possible*: the Carter/Mino and Christoffel integrators cross-check each other on identical rays, so a bug must strike both formulations identically to pass. *Analytic anchors over regression values*: wherever a closed form exists (Synge, Bardeen, BPT, the Page–Thorne expansion, the quadrupole precession), the test asserts against it — regression-only tests would happily fos-

Table 21.2: The automated test suite, grouped by what each case guards. Tolerances are the literal assertion thresholds.

Guard	Test cases	Tolerance
Metric identities	Schwarzschild metric components at a_*=0; Kerr metric determinant identity ($\det g = -\Sigma^2 \sin^2\theta$); complex-step Christoffels match analytic Schwarzschild symbols	10^{-12}–10^{-14}
Characteristic radii	Schwarzschild key radii at a_*=0; published reference radii; prograde/retrograde asymmetry; extremal Kerr limits (BPT72 exact values); ISCO monotonicity as $a_* \to 1$; photon orbit solves R=R'=0; circular-orbit E, L stationary at the ISCO	exact/10^{-10}
Null-geodesic conservation	Schwarzschild null geodesics conserve constants; E, L_z, Q conserved along Kerr null geodesics (a_*=0.94); Carter/Mino and Christoffel integrators agree on the same null ray	10^{-6} rel.
Timelike geodesics	4-velocity normalisation on an S2-scale Kerr orbit; on a strong-field Kerr orbit; measured pericentre precession matches $6\pi M/[a(1-e^2)]$ in the weak field	10^{-8} abs.
Shadow metrology	Schwarzschild shadow circular with the analytic angular size; rays inside/outside the critical impact parameter; Kerr a_*=0.94 boundary on the Bardeen curve; shadow width shrinks and displaces with spin; (hidden) boundary-layer tolerance probe	10^{-3}–10^{-4}
Disk model	Page–Thorne flux zero at ISCO with single interior maximum; flux matches its exact large-r expansion; blackbody colours behave physically; Doppler beaming approaching-side brighter with sane g range	10^{-6}/qual.
Hartle–Thorne / no-hair	HT metric at ε_q=0 equals Kerr; F_1/F_2 branch continuity at the series switch; quadrupole far-field structure; quadrupole pericentre shift matches $-3\pi\varepsilon_q(M/p)^2$; HT timelike orbits conserve E, L_z, normalisation	10^{-12}/10^{-6}
S2 forward model	radial period matches Kepler's third law; pericentre speed ∼7650 km/s with correct redshift structure; Rømer delay shifts epochs with the correct sign	10^{-3} rel.

silise a wrong constant.

21.3 Bug hunts as methods lessons

21.3.1 The tolerance-dependent shadow

The Phase-1 boundary metrology initially returned Schwarzschild shadow radii that *moved* as the integrator tolerance was changed — by parts in 10^3, far above the bisection depth. The hidden `[.probe]` diagnostic was written to map the failure: rays in a narrow bracket around the critical angle, classified across tolerances, flipped from captured to escaped wholesale. The error-budget analysis of Section 10.2 identified the mechanism — benign per-step drift amplified by the long flat approach leg against a small discrimination feature — and per-step constraint projection removed it, making the boundary tolerance-independent at the 10^{-11} level (Fig. 10.2). *Lesson: for separatrix observables, budget accumulated constraint drift against the discrimination scale, not against the state itself.*

21.3.2 The quadrupole sign

The first quasi-Kerr implementation produced an ε_q pericentre shift with the correct magnitude and the *wrong sign* — invisible to any single test, because both the analytic formula and the numeric orbit could have carried the flip. The fix was pinned by a *two-sided* test pair: the analytic $\delta\omega_Q = -3\pi\varepsilon_q(M/p)^2$ evaluated independently from the Newtonian P_2 perturbation, against the measured apsidal advance of integrated orbits at both signs of ε_q. *Lesson: signs need two independent derivations; one implementation testing itself proves consistency, not correctness.*

21.3.3 Keplerian initial conditions and the 1.6 per cent period

Phase-4 fits initially refused to reproduce the published S2 period at its quoted precision. The cause was the initial-condition shortcut — launching at pericentre with the Newtonian vis-viva speed — which mis-states the relativistic period by 1.6 per cent at S2's eccentricity (Section 14.2), two orders above the measurement error. The Darwin turning-point construction replaced it; the Kepler-third-law

test now pins the relation. *Lesson: an approximation harmless for rendering can be fatal for fitting; error budgets are per-use-case.*

21.3.4 The vanishing mass uncertainty

The library covariance reported σ_M consistent with zero — a fitted mass "known" to nine digits from 145 imaging epochs, which is absurd on its face. Section 15.2.3 dissects the cause: SVD truncation of the near-degenerate M–R_0 direction. The explicit central-difference $(J^\mathsf{T}J)^{-1}$ replaced it, and the MCMC posterior later confirmed the corrected ellipse end-to-end. *Lesson: covariance regularisation silently converts "weakly constrained" into "perfectly known"; check the singular spectrum before trusting any library covariance.*

21.3.5 The brick-red disk

The live renderer's outer disk rendered as a flat brick-coloured annulus. Not a physics bug: the Page–Thorne brightness spans $\sim r^{-3}$, the display window compressed 24 stops into an LUT whose lower quarter was nearly one colour, and everything beyond $\sim 10\,M$ quantised to it. The fix was purely in the display layer — LUT ramps graded over the full domain and a calibrated log window — with the physics untouched, verified by the unchanged `ctest` suite. *Lesson: keep display grading and physics in separable layers, so a cosmetic failure cannot be "fixed" in the equations.*

21.3.6 The turbulence seam

The display-only turbulence layer (Chapter 20) initially showed a visible seam at $\phi = 0$: value noise sampled on a non-periodic domain. Replaced with azimuthally periodic noise; a regression eye-check at four azimuths entered the render checklist. *Lesson: any function of an angle must be built periodic, not wrapped after the fact.*

21.3.7 The dashed polar column

In live use the fp32 renderer developed dashed, sawtooth columns on the image projection of the spin axis, flanked by dotted patches — reported from interactive exploration, not caught by any gate, because the science paths (fp64 GPU, CPU reference) do not show them: in a linear display the light involved is invisibly faint, and only the live path's 24-stop logarithmic window lifts it into view together with

its noise. The mechanism is the polar term $L_z^2/\sin^3\theta$ in $\Theta'(\theta)$: for rays passing near the spin axis the θ-equation stiffens beyond what the fixed-budget RK4 can resolve, so they either tunnelled into the forbidden polar cone or had their polar detours quantised into a few discrete step outcomes, and the disk annulus's hard edges imaged that quantisation as a hit/miss barcode. The cure is a stiffness-adaptive polar step clamp $h \lesssim |p_\theta|/|\Theta'|$ — floored, so that near the turning wall the analytic reflection guard takes over rather than letting microscopic steps burn the budget — plus a raised fp32 $\sin\theta$ clamp and a golden-ratio step-phase jitter. *Lesson: a fixed-budget single-precision integrator needs its step law derived from the stiffest term it must cross; and a logarithmic display grade will surface numerical noise that physics-grade linear output never shows.*

21.3.8 The edge-on staircases — when the patch is the artefact

The first polar-column patch shipped two screen-space band-aids: a bridge that interpolated the central image columns from their clean neighbours, and a near-axial L_z snap wide enough to rewrite a ∼30-pixel family of rays. At near-edge-on inclination and high exposure these band-aids *became* the dominant artefacts — blurry rectangles over bright structure, starlight smeared into grey pillars — alongside a new one: checkerboard staircase blocks on the rims of the lensed lower bowl. The blocks were diagnosed with a 3× global step-refinement probe (they vanished, proving pure step-resolution error): neighbouring rays share an *identical* deterministic step sequence, so every marginal decision — a grazing dip through the disk plane detected or missed, a step-budget cliff — flips at the same place for whole patches of pixels and organises into coherent blocks. The final fix deleted both band-aids (the snap returned at a sub-pixel $2.5 \times 10^{-3}\sqrt{Q}$), reused the exact step length inside the crossing bisection, replaced raw step endpoints with a linear $\cos\theta$ zero for grazing crossings, added a per-ray ±10 per cent step dither so residual quantisation decorrelates into per-pixel noise, and ramped the base step down with viewing inclination so only near-plane cameras pay for the harder geometry. *Lesson: a cosmetic patch that does not remove the mechanism becomes the next artefact; prove a resolution error with a refinement probe before designing anything cleverer; and dither — decorrelating neighbouring rays — is both diagnostic and cure, because quantisation that cannot organise across pixels cannot be seen.*

21.3.9 The face-on crayon drawing

The edge-on campaign introduced de-aliasing keyed on each ray's closest approach to the spin axis, $\min|\sin\theta|$ — and promptly broke the opposite viewing regime: near-polar inclinations rendered as a flat, posterised "crayon drawing", filament texture gone, with a fuzzy wedge below the shadow. The PHYSICAL preset was untouched, which was the diagnostic gift: every squeezed-ray shaping step is gated to the display presets, so the failure had to live in that machinery. The root cause was observer contamination: the tracker was *seeded with the camera's own* $\sin\theta_0$, so with the camera near the axis every ray in the frame classified as caustic-squeezed from step zero. Seeding at 1 and counting only approaches near the hole ($r < r_{\mathrm{cam}}/2$) restored honest classification at every inclination; supersampling moved from a fixed screen band to a ray-property trigger (the polar-view caustic fan does not sit on the image centreline); and the deepest pole-passers now shade at a band-limited representative radius with a fractional coverage term, because no affordable sample count beats binomial speckle under a 24-stop lift. A ± 0.75-texel dither on the 256-entry display-LUT lookup removed the remaining posterisation of dim smooth gradients. *Lesson: a classification feature must measure the trajectory, not the observer — an observer-contaminated feature fails exactly when the observer moves; and the strict display-gating of every cosmetic subsystem is what made the physics output provably untouched through three rounds of display surgery.*

21.3.10 The cloud banks that could only draw lines

The CINEMATIC volumetric dust layer (Chapter 20) first rendered its clouds as circumferential streaks — lines, not banks. Two causes compounded: the noise lattice had quarter-turn azimuthal cells against $\sim$10-per-cent-of-radius radial cells ($\sim$15:1 anisotropy — such a lattice *cannot* draw anything but lines), and advection by the exact differential $\Omega(r)$ shears any fixed pattern into strings within a few relative turns — seconds of wall clock at the demonstration time scales. The rebuild samples isotropic 3-D value-noise fBm at the orbiting world position (round cells from any viewpoint, azimuthally seam-free by construction), and runs the field as two staggered, periodically re-forming phases that cross-fade — flow noise — each advected by the exact $\Omega(r)$ over its own age only, so the banks orbit and shear indefinitely without ever reaching a stringified steady state. *Lesson: the anisotropy of a texture lattice is a shape prior — round features need round*

cells; and a differentially sheared flow needs re-forming phases, because no static pattern survives unbounded shear.

21.4 The convergence evidence

The quantitative artefacts behind these narratives — the projection on/off boundary experiment, the conserved-quantity drift ladder, and the step-size trace — are the Chapter 10 figures (Figs. 10.2–10.1), regenerated on every `make reproduce` by `validation/convergence_tool`. The full command-level reproduction pipeline is Appendix D.

Chapter 22

Discussion and Limitations

22.1 What the two channels jointly demonstrate

The central scientific claim of this report is one of *consilience*: a single Kerr spacetime, with a single mass and distance, simultaneously reproduces four measurements of entirely different character. The S2 astrometry and radial velocities return $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ at $R_0 = 8272 \pm 182$ pc — dynamical, stellar-scale, built from twenty-four years of imaging. The GRAVITY redshift excess at the 2018 pericentre emerges from the same fitted orbit at 205 km/s with no adjustable amplitude — special plus general relativity at the 10^{-4} level. The Schwarzschild precession follows at $11.98'$ per orbit, again amplitude-free. And the same mass and distance, pushed through the completely independent photon-orbit machinery of Part III, predict a photon ring of $49.68\,\mu$as — horizon-scale, radio-interferometric — inside the EHT band. Nothing was tuned between these comparisons: the chain from the metric of Chapter 8 to each observable is the same code, validated by the same gates, and the numbers agree with four independent observational campaigns spanning six orders of magnitude in radius. That is far from a proof that the object is a Kerr black hole — the no-hair interval of Chapter 17 is wide — but it is precisely the pattern a Kerr black hole predicts, and each of its alternatives must now reproduce all four numbers at once.

22.2 Limitations

Three idealisations bound the interpretation of the imaging results. First, the disk is the time-averaged, geometrically thin Novikov–Thorne solution; the real

Sgr A* flow is a hot, geometrically thick, radiatively inefficient plasma, and our thin-disk images are a clean GR benchmark rather than a spectral model of Sgr A* itself (the EHT comparison in Chapter 19 concerns the ring diameter, which is a property of the spacetime, not of the emission model — though the ring diameter retains a modest emission-model dependence through the location of the emitting material). Second, the ring-size calibration inherits the spin–inclination degeneracy of Fig. 19.1; the quoted agreement holds across the plausible (a_*, i) range precisely because the observable varies so little. Third, the orbit fit adopts the Gillessen et al. (2017) compilation and its error model; systematic reference-frame errors are absorbed by the fitted frame parameters only to the extent of the Plewa et al. (2015) priors.

Several smaller approximations are worth listing with their sizes, because each is deliberately *carried*, not overlooked. The gravitational bending of S2's own light by Sgr A* shifts its apparent position by of order $20\,\mu$as around pericentre and its radial velocity by $\sim$5 $\mathrm{km\,s^{-1}}$ — both below the imaging-era uncertainties ($\gtrsim 300\,\mu$as, $\gtrsim 15$ $\mathrm{km\,s^{-1}}$) and both neglected here exactly as in GRAVITY Collaboration (2018). S2 itself is treated as a test particle: at $\sim$10–15 $\mathrm{M_\odot}$ against $4.3 \times 10^6\,\mathrm{M_\odot}$, its back-reaction is ten orders below any observable. Extended mass inside the S-star orbits (a dark cluster, stellar remnants) is not modelled; the professional fits bound it at the $\lesssim 10^{-3}\,M$ level, and its principal signature — a *retrograde* Newtonian precession partially cancelling the Schwarzschild advance — is the leading systematic on any future ε_q claim, though not on the mass. And the interstellar scattering screen toward the Galactic centre, which blurs the EHT image and is a first-order concern for ring *width* claims, enters our ring-*diameter* comparison only through the published EHT deconvolution, whose uncertainty is already inside the quoted $\pm 2.3\,\mu$as.

Three further boundaries deserve explicit statement. The bright axial columns in near-edge-on renders are the lensed polar caustic of Section 13.3 — this model contains no jet physics, and a viewer must not read Blandford–Znajek outflows into what is pure geometry; conversely, no conclusion of this report depends on jet absence. The no-hair reach is bounded by a single star at $\sim$3000 gravitational radii: the $\sigma(\varepsilon_q) \sim 10^5$ order of Chapter 17 is a property of S2's orbit, not of the method, and improves quadratically with proximity — the cluster machinery of Chapter 16 is the prepared vehicle for any confirmed closer star. Finally, the precision split is deliberate: all quantitative results flow from the fp64 CPU/GPU paths, while the interactive fp32 path exists for exploration only; the two are never

mixed, and the display-only structure layer of Chapter 20 sits outside the physics wall entirely.

22.3 Outlook

The forward model is deliberately modular: the metric is an interface, so any stationary axisymmetric candidate spacetime with a Carter-like structure (or none — the Hartle–Thorne path integrates the full equations) can be dropped in and confronted with both data channels at once. Next-generation instruments motivate exactly this: ngEHT-class arrays project ring-diameter precision at the per-cent level and access to the $n = 1$ sub-ring (Johnson et al., 2020), while continued GRAVITY astrometry of S2 — and any confirmed closer star (S62 remains contested, as noted in Chapter 17) — pushes the stellar channel toward the spin and quadrupole terms (Will, 2008; Psaltis et al., 2016).

It is worth being concrete about what each future measurement would flow into, because the pipelines here are already shaped for them. A confirmed star inside S2's orbit drops into `fit_cluster` as a sixth parameter block with no code change, and the $(M/p)^2$ scaling of Chapter 17 converts its pericentre directly into quadrupole reach. A per-cent ring-diameter measurement tightens the (a_*, i) sweep of Chapter 19 from a consistency check into a joint constraint, in combination with the stellar-channel mass — the two channels' error ellipses cross at a usefully steep angle because one scales as M/D and the other carries M and D separately. And a resolved $n = 1$ sub-ring would, through the universal-ratio law of Chapter 13, measure the Lyapunov band of Chapter 11 — the first observable that reads Kerr's *instability structure* rather than its geometry. Each of these is a measurement campaign away, not a theory away; the instrument this report documents is the part that is already finished.

Chapter 23

Conclusions and Future Work

23.1 Verified findings

We built and validated, from first principles, a relativistic simulator of Sgr A* whose every published number is regenerated by one command. Its verified statements are:

1. Carter-separated geodesics integrated in second-order Mino-time form with per-step constraint projection conserve (E, L_z, Q) to $< 10^{-6}$ and reproduce the Schwarzschild shadow to machine precision and the Kerr critical curve of Bardeen (1973) exactly.

2. A 13-parameter GR fit to the Gillessen et al. (2017) S2 data recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$, $R_0 = 8272 \pm 182$ pc ($\chi^2_\nu = 0.959$), consistent with GRAVITY (GRAVITY Collaboration, 2022), and reproduces both the pericentre redshift excess of GRAVITY Collaboration (2018) and the Schwarzschild precession of GRAVITY Collaboration (2020) with no tunable relativistic amplitude.

3. At the GRAVITY mass and distance the simulated photon ring spans 49.68 μas, within the EHT measurement (Event Horizon Telescope Collaboration, 2022); the photon sub-rings show the expected exponential demagnification ($F_1/F_2 = 274$).

4. A five-star joint fit sharing one potential tightens the mass to $M = 4.461 \pm 0.119 \times 10^6\,\mathrm{M}_\odot$ (39 per cent below the S2-only uncertainty) and demonstrates concretely why short-arc single-star mass claims fail (Chapter 16).

5. Freeing the quadrupole gives $|\varepsilon_q| < 1.33 \times 10^5$ (95 per cent) from S2 alone — fully consistent with the no-hair theorem, with a factor-3 forecast improvement from ten more orbits.

6. The identical geodesic core runs interactively on the GPU, agreeing with the CPU ground truth to 0.00289 mean pixel residual.

23.2 Future work

Two items of the original roadmap — ensemble MCMC posteriors and the multi-star joint fit — were completed during the writing of this report and appear as Chapters 15 and 16. The remaining directions, in rising order of effort:

- *More cluster stars and richer systematics.* The `fit_cluster` pipeline extends mechanically to the remaining Table-5 stars; the scientifically meaningful upgrade is per-instrument error floors and an extended-mass term in the potential, which the professional fits marginalise over and we currently do not.

- *Slim-disk and hot-flow emission.* Replacing the Novikov–Thorne surface with a geometrically thick, radiatively inefficient model would let the imaging channel address Sgr A*'s actual spectrum rather than only its spacetime geometry.

- *Interferometric observables.* Fourier-domain visibilities of the sub-ring maps (Chapter 13) would convert our clean geodesic templates into direct ngEHT forecast curves.

- *Candidate non-Kerr metrics.* The metric interface plus the generic Christoffel integrator accept any stationary axisymmetric candidate; parametrised families beyond quasi-Kerr (bumpy black holes, boson-star exteriors) could be confronted with both data channels using the identical fitting and sampling machinery.

- *A confirmed inner star.* The single greatest gain for the no-hair programme requires no new code at all — only the astronomical confirmation of a star well inside S2's orbit (S62 remains contested, as noted throughout).

Acknowledgements

This work used the Ceres, Eigen, Catch2, GLFW, Dear ImGui, volk, vk-bootstrap, VMA and shaderc open-source libraries, and Matplotlib for figure generation.

Data availability

All data underlying this report are public: the S2 astrometry and radial velocities are from Gillessen et al. (2017) (VizieR J/ApJ/837/30). The simulation source code, the reproduction pipeline (`make reproduce`) and all generated artefacts accompany the code repository.

Appendix A

Extended Algebra

Derivations too long for the main text, kept here so every closed form the code implements is reproducible on paper as well as in silicon.

A.1 The Carter regrouping identity, term by term

Chapter 9 asserts the identity (9.4) that makes the Hamilton–Jacobi equation separate; here is the verification in full, since the whole architecture rests on it. Multiply both sides of (9.4) by Δ and expand the right-hand side:

$$-P^2 + \Delta\left(aE\sin\theta - \frac{L_z}{\sin\theta}\right)^2 = -\left[E(r^2+a^2) - aL_z\right]^2 + \Delta\left(a^2E^2\sin^2\theta - 2aEL_z + \frac{L_z^2}{\sin^2\theta}\right). \quad \text{(A.1)}$$

Collect by powers of E and L_z:

$$\begin{aligned} E^2: &\quad -(r^2+a^2)^2 + a^2\Delta\sin^2\theta = -A && \text{(definition of } A\text{)},\\ EL_z: &\quad 2a(r^2+a^2) - 2a\Delta = 2a\left[r^2+a^2-\Delta\right] = 4Mar && \text{(since } r^2+a^2-\Delta = 2Mr\text{)},\\ L_z^2: &\quad -a^2 + \frac{\Delta}{\sin^2\theta} && \text{(as it stands)}. \end{aligned} \quad \text{(A.2)}$$

The three collected coefficients are exactly the numerator entries of the raw contraction (9.3), term for term, which proves (9.4). Note what did the work: the definition of A absorbed the θ-dependence of the E^2 sector, and the identity $r^2+a^2-\Delta = 2Mr$ — the statement that $g_{t\phi}$ is proportional to Mar — collapsed the cross term. Both are properties of the specific Kerr combination of metric functions; a generic stationary axisymmetric metric leaves an irreducible mixed remainder, which is precisely why the quasi-Kerr spacetime of Chapter 17 has no Carter constant and must be integrated by Christoffel symbols.

A.2 The equatorial photon-orbit cubic

For equatorial photons ($Q = 0$), write $b = L_z/E$ and factor E^2 out of the radial potential (8.17):

$$R(r)/E^2 = (r^2 + a^2 - ab)^2 - \Delta\,(b-a)^2. \tag{A.3}$$

A circular orbit needs $R = 0$ and $R' = 0$:

$$(r^2 + a^2 - ab)^2 = \Delta\,(b-a)^2, \tag{A.4}$$

$$4r\,(r^2 + a^2 - ab) = 2(r-M)(b-a)^2. \tag{A.5}$$

Taking the square root of (A.4) (both sides positive for $r > r_+$) gives $r^2+a^2-ab = \sqrt{\Delta}\,|b-a|$; substituting into (A.5) yields $4r\sqrt{\Delta} = 2(r-M)|b-a|$, i.e.

$$|b-a| = \frac{2r\sqrt{\Delta}}{r-M}, \qquad r^2 + a^2 - ab = \frac{2r\,\Delta}{r-M}. \tag{A.6}$$

Eliminating b is now mechanical. From the first relation, $b = a \pm 2r\sqrt{\Delta}/(r-M)$; substituting into the second,

$$r^2 + a^2 - ab = r^2 \mp \frac{2ar\sqrt{\Delta}}{r-M} \overset{!}{=} \frac{2r\,\Delta}{r-M}, \tag{A.7}$$

and multiplying through by $(r-M)/r$:

$$r(r-M) \mp 2a\sqrt{\Delta} = 2\Delta \quad\Longrightarrow\quad \mp 2a\sqrt{\Delta} = r^2 - 3Mr + 2a^2. \tag{A.8}$$

Squaring and inserting $\Delta = r^2 - 2Mr + a^2$, the a^4 terms cancel and a common factor r drops out:

$$r^3 - 6Mr^2 + 9M^2 r - 4a^2 M = 0 \quad\Longleftrightarrow\quad \left(x^3 - 3x\right)^2 = 4a_*^2, \qquad x = \sqrt{r/M}, \tag{A.9}$$

whose two square-root branches are the prograde/retrograde pair

$$x^3 - 3x \pm 2a_* = 0. \tag{A.10}$$

With three real roots guaranteed for $|a_*| \le 1$ (the discriminant $4\cdot 27(1-a_*^2) \ge 0$), the trigonometric solution of a depressed cubic $x^3 - 3px + q = 0$, $x = 2\sqrt{p}\cos\vartheta$ with $\cos 3\vartheta = -q/2p^{3/2}$, gives directly

$$x = 2\cos\!\Big[\tfrac{1}{3}\cos^{-1}(\mp a_*)\Big] \implies r_{\rm ph}^{\pm} = 4M\cos^2\!\Big[\tfrac{1}{3}\cos^{-1}(\mp a_*)\Big] = 2M\Big\{1+\cos\!\Big[\tfrac{2}{3}\cos^{-1}(\mp a_*)\Big]\Big\}, \tag{A.11}$$

the form implemented in `r_photon`. The same cubic $x^3 - 3x + 2a_*$ reappears as the denominator polynomial of the Page–Thorne flux (Section A.4) — not a coincidence: both express where the circular-orbit denominator $1 - 3M/r \pm 2a\sqrt{M/r^3}$ of (8.22)–(8.23) degenerates.

A.3 The BPT ISCO reduction

Stability of a circular timelike orbit requires, in addition to $R(r) = R'(r) = 0$, that $R''(r) \le 0$ (the orbit sits at a potential minimum); the ISCO is the marginal case $R'' = 0$. Substituting the circular-orbit $E(r), L(r)$ of (8.22)–(8.23) into $R''(r) = 0$ and simplifying — the step-by-step reduction is straightforward but long; Bardeen et al. (1972) give the intermediate identities — collapses the condition to the quartic

$$r^2 - 6Mr \mp 8a\sqrt{Mr} - 3a^2 = 0 \tag{A.12}$$

(upper sign prograde), a quartic in $\sqrt{r}$. BPT observed that the substitutions

$$Z_1 = 1 + (1 - a_*^2)^{1/3}\big[(1 + a_*)^{1/3} + (1 - a_*)^{1/3}\big], \qquad Z_2 = \sqrt{3a_*^2 + Z_1^2} \tag{A.13}$$

solve it in closed form: $r_{\rm ISCO}/M = 3 + Z_2 \mp \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}$. Verification is mechanical: substitute and use $(1 - a_*^2)^{1/3}$-algebra to confirm (A.12) vanishes identically — the unit tests do the numerical equivalent at fifteen digits across the spin range, including the exact anchor points $r_{\rm ISCO}(0) = 6M$, $r^{+}_{\rm ISCO}(1) = M$, $r^{-}_{\rm ISCO}(1) = 9M$.

A.3.1 The ISCO energy identity

Chapter 8 promotes the identity $E_{\rm ISCO} = \sqrt{1 - 2M/(3\,r_{\rm ISCO})}$ (8.26). In the Schwarzschild case it is a two-line computation worth seeing. With $a = 0$,

$$E^2(r) = \frac{(1 - 2M/r)^2}{1 - 3M/r}, \tag{A.14}$$

and the marginal-stability quartic (A.12) reduces to $r = 6M$, so

$$E^2_{\rm ISCO} = \frac{(1 - 1/3)^2}{1 - 1/2} = \frac{4/9}{1/2} = \frac{8}{9} = 1 - \frac{2M}{3 \cdot 6M}. \tag{A.15}$$

For general spin the same statement follows from (8.22) once the marginal condition (A.12) is used to eliminate the $a\sqrt{M/r^3}$ combinations (the intermediate identities are eqs. 2.12–2.21 of Bardeen et al., 1972); rather than transcribe that page of algebra we record the numerical confirmation, which is independent of any transcription: across the `physics_tables` spin grid the identity and the general formula agree to all printed digits (e.g. 0.820404 vs 0.820406 at $a_* = 0.938$, the residual being the grid's own rounding), including both rotation senses.

A.4 The Page–Thorne flux integral

The energy-angular-momentum balance of Section 12.1 gives the flux as

$$F(r) = \frac{\dot{M}}{4\pi r}\,\frac{-\Omega'(r)}{\big(E-\Omega L\big)^2}\int_{r_{\rm ISCO}}^{r}\big(E-\Omega L\big)L'\,\mathrm{d}r', \tag{A.16}$$

with all quantities the circular-orbit functions of Chapter 8. The substitution $x=\sqrt{r/M}$ (and units $M=1$) turns every circular-orbit function into a ratio of low-order polynomials in x over the *same* square root:

$$\Omega=\frac{1}{x^3+a_*},\qquad E=\frac{x^3-2x+a_*}{x^{3/2}\sqrt{x^3-3x+2a_*}},\qquad L=\frac{x^4-2a_*x+a_*^2}{x^{3/2}\sqrt{x^3-3x+2a_*}}. \tag{A.17}$$

The combination the integrand needs then telescopes beautifully:

$$\begin{aligned} E-\Omega L &= \frac{(x^3-2x+a_*)(x^3+a_*)-(x^4-2a_*x+a_*^2)}{(x^3+a_*)\,x^{3/2}\sqrt{x^3-3x+2a_*}} \\ &= \frac{x^3\,(x^3-3x+2a_*)}{(x^3+a_*)\,x^{3/2}\sqrt{x^3-3x+2a_*}} = \frac{x^{3/2}\,\sqrt{x^3-3x+2a_*}}{x^3+a_*}, \end{aligned} \tag{A.18}$$

the square root cancelling into the numerator (the middle equality is a direct expansion; every term not proportional to $x^3-3x+2a_*$ cancels identically). Equation (A.18) is the engine of the whole closed form: the integrand $(E-\Omega L)L'$ becomes a rational function whose denominator is $x^3-3x+2a_*$ — factorable over the roots (12.4) — so the integral evaluates by partial fractions to one power term, one logarithm from the a_*-linear piece, and one logarithm per root:

$$\int \;\longrightarrow\; x-x_0-\tfrac{3}{2}a_*\ln\frac{x}{x_0}-\sum_{i=1}^{3}\frac{3(x_i-a_*)^2}{x_i(x_i-x_j)(x_i-x_k)}\ln\frac{x-x_i}{x_0-x_i}, \tag{A.19}$$

which is the bracket of (12.3); the prefactor $1/[x^2(x^3-3x+2a_*)]$ collects $-\Omega'/(E-\Omega L)^2$ and the $1/r$ of (A.16). The Schwarzschild guard of Chapter 12 arises because $x_2\to 0$ as $a_*\to 0$ while its residue coefficient diverges as $1/x_2$; the product tends to the finite limit that the code takes analytically.

A.4.1 The far-field constant, exactly

The far-field expansion asserted by the unit test — $x-c+3/x+\mathcal{O}(x^{-2})$ with $c\simeq 3.976$ at $a_*=0$, *not* the Newtonian constant — can be closed in elementary terms. At $a_*=0$ the cubic's roots are $x_1=\sqrt{3}$, $x_2=0$, $x_3=-\sqrt{3}$, and

$x_0 = \sqrt{r_{\mathrm{ISCO}}/M} = \sqrt{6}$. The x_2 term vanishes (the guard limit), and the two surviving residues evaluate to

$$\frac{3(\pm\sqrt{3})^2}{(\pm\sqrt{3})(\pm\sqrt{3})(\pm 2\sqrt{3})} = \pm\frac{\sqrt{3}}{2}, \tag{A.20}$$

so the bracket is

$$x - \sqrt{6} - \frac{\sqrt{3}}{2}\ln\frac{x-\sqrt{3}}{\sqrt{6}-\sqrt{3}} + \frac{\sqrt{3}}{2}\ln\frac{x+\sqrt{3}}{\sqrt{6}+\sqrt{3}}. \tag{A.21}$$

As $x \to \infty$, $\ln\big[(x-\sqrt{3})/(x+\sqrt{3})\big] \to -2\sqrt{3}/x + \mathcal{O}(x^{-3})$, which supplies exactly the $+3/x$ term of the expansion, and the constants collect into

$$c = \sqrt{6} + \frac{\sqrt{3}}{2}\ln\frac{\sqrt{6}+\sqrt{3}}{\sqrt{6}-\sqrt{3}} = 2.4495 + 0.8660 \times 1.7629 = 3.9762, \tag{A.22}$$

the "$c \simeq 3.976$" of (12.5) now an exact closed form. The Newtonian folklore profile corresponds to dropping the two logarithms — i.e. to $c = \sqrt{6} \approx 2.449$ — and the difference between the two constants is precisely the general-relativistic torque bookkeeping near the inner edge, which is why asserting the correct c in a unit test guards physics no rendered image can.

A.5 The quasi-Kerr structure functions at large radius

The Glampedakis & Babak (2006) functions $F_1(r), F_2(r)$ have the form $p_n(M/r) + q_n(M/r)\,\ln(1-2M/r)$ with low-order polynomials p_n, q_n, arranged so the quadrupole correction to the metric falls as $(M/r)^3$. Expanding the logarithm, $\ln(1-2M/r) = -2M/r - 2(M/r)^2 - \frac{8}{3}(M/r)^3 - \ldots$, the polynomial and logarithmic contributions cancel through the first five orders; in double precision the relative error of the direct evaluation therefore grows as $\sim \varepsilon_{\mathrm{mach}}(r/M)^5$ — unusable beyond $r \sim 50\,M$ and catastrophic at S2 radii. The implementation switches at $r = 9M$ to the explicit series of the *cancelled* result (coefficients from the symbolic expansion, kept to $\mathcal{O}((M/r)^8)$), with the branch-continuity unit test asserting agreement of the two branches at the joint to 10^{-10}.

For the record, the exact covariant deviation the code applies (`HartleThorneMetric::component`

is, with $f = 1-2M/r$ and the quadrupole angular factor $1-3\cos^2\theta = -2P_2(\cos\theta)$:

$$\begin{aligned}
\delta g_{tt} &= +\varepsilon_q\, f\,(1-3\cos^2\theta)\,F_1(r),\\
\delta g_{rr} &= +\varepsilon_q\,(1-3\cos^2\theta)\,F_1(r)/f,\\
\delta g_{\theta\theta} &= -\varepsilon_q\, r^2(1-3\cos^2\theta)\,F_2(r),\\
\delta g_{\phi\phi} &= -\varepsilon_q\, r^2\sin^2\theta\,(1-3\cos^2\theta)\,F_2(r),
\end{aligned} \tag{A.23}$$

added to the exact Kerr components (with the $\varepsilon_q = 0$ early-return making the Kerr limit exact by construction rather than by cancellation). The overall sign is anchored *empirically*, not notationally: Glampedakis & Babak (2006) define h^{ab} as a perturbation of the inverse metric, so a naive index-lowering flips the sign relative to (A.23), and the convention that counts is the physical one — with the signs as written, the far-field Newtonian limit is $\delta\Phi = -\delta Q\, P_2(\cos\theta)/r^3$ with $\delta Q = +\varepsilon_q M^3$ (the $Q_{\rm tot}$ convention of (17.2)), and the measured apsidal advance of integrated orbits matches $-3\pi\varepsilon_q(M/p)^2$ for both signs of ε_q. That two-sided empirical pin is the quadrupole-sign test of Chapter 21.

A.6 The Bardeen critical curve by elimination

Chapter 11 quotes the parametrisation (11.5); here is the elimination that produces it. Write the two shell conditions on the quartic (9.12) with the shorthand $A_c = \xi^2 + \eta$ and $B_c = \eta + (\xi - a)^2$:

$$R/E^2 = r^4 + (a^2 - A_c)\,r^2 + 2MB_c\,r - a^2\eta = 0, \tag{A.24}$$

$$R'/E^2 = 4r^3 + 2(a^2 - A_c)\,r + 2MB_c = 0. \tag{A.25}$$

Equation (A.25) is linear in $(a^2 - A_c)$:

$$a^2 - A_c = -2r^2 - \frac{MB_c}{r}. \tag{A.26}$$

Substituting into (A.24) collapses it to a relation linear in η:

$$r^4 - 2r^4 - MB_c r + 2MB_c r = a^2\eta \quad\Longrightarrow\quad \eta = \frac{r\,(MB_c - r^3)}{a^2}. \tag{A.27}$$

Finally, the definitions of A_c and B_c give $A_c - B_c = 2a\xi - a^2$, so with (A.26),

$$\xi = \frac{A_c - B_c + a^2}{2a} = \frac{2a^2 + 2r^2 + MB_c/r - B_c}{2a}, \tag{A.28}$$

and imposing the remaining consistency $\eta = A_c - \xi^2$ closes the system into a single quadratic for $B_c(r)$, whose physical root, back-substituted through (A.27)–(A.28), factors into the compact forms of (11.5). (The factorisation is tedious to carry by hand but trivial to verify: substituting (11.5) into (A.24)–(A.25) makes both vanish identically as polynomials in r — and the traced boundary of Fig. 11.2 performs the same verification numerically at every point, against the independent integrator.) The route displays where each qualitative feature comes from: η in (A.27) changes sign where $MB_c = r^3$ — the shell edges, where the orbit becomes equatorial — and ξ in (A.28) inherits its $1/a$ prefactor, which is why the critical curve degenerates to the Synge circle as $a \to 0$ only through the combination $\xi^2 + \eta$, individual ξ diverging.

A.7 The Synge shadow angle at finite radius

Chapter 11 quotes (11.4); the derivation is three steps. A photon at radius $r_{\rm obs}$ in Schwarzschild with impact parameter b makes a local angle α with the radial direction given by the tetrad components of its momentum,

$$\sin\alpha = \frac{k^{\hat\phi}}{\sqrt{(k^{\hat r})^2 + (k^{\hat\phi})^2}} = \frac{k^{\hat\phi}}{k^{\hat t}} = \frac{L_z/r}{E/\sqrt{1-2M/r}} = \frac{b}{r}\sqrt{1-\frac{2M}{r}}, \tag{A.29}$$

using the static-observer tetrad components ($k^{\hat\phi} = L_z/r$, $k^{\hat t} = E/\sqrt{f}$) and, in the second equality, the null condition $(k^{\hat t})^2 = (k^{\hat r})^2 + (k^{\hat\phi})^2$ — the relation is exact, not small-angle. The shadow edge is the critical impact parameter $b_c = \sqrt{27}\,M$ of the photon sphere, so

$$\sin\alpha_{\rm sh} = \frac{\sqrt{27}\,M}{r_{\rm obs}}\sqrt{1-\frac{2M}{r_{\rm obs}}}, \tag{A.30}$$

which is (11.4). Two limits orient the formula: at $r_{\rm obs} \gg M$ it reduces to the flat-sky $b_c/r_{\rm obs}$ (the correction factor is the $1/\sqrt{1-2M/r}$ of (11.3)), and at $r_{\rm obs} = 3M$ it gives $\sin\alpha_{\rm sh} = 1$: an observer *on* the photon sphere finds the shadow filling exactly half the sky, the memorable boundary between outside and inside views of the photon shell.

Appendix B

Algorithm Pseudocode

Compact pseudocode for the load-bearing algorithms, matching the production sources line for line in structure. Notation: $y = (t, r, \theta, \phi, p_r, p_\theta)$ is the Mino-time state, $c = (M, a, E, L_z, Q)$ the ray constants.

B.1 Adaptive step with constraint projection

(`RaytraceCpu::trace` inner loop; `DormandPrince54::try_step`)

```
h = 1e-3
while steps < max_steps:
    (accepted, err, h_next) = dopri5_step(rhs, tau, y, h)   # 7 stages
    h = clamp(h_next, 1e-14, h_max)
    if not accepted: continue
    # constraint projection (skip inside turning-point guard band)
    if R(y.r) >= 0:      y.p_r     = sign(y.p_r)     * sqrt(R(y.r))
    if Theta(y.th) >= 0: y.p_theta = sign(y.p_theta) * sqrt(Theta(y.th))
    if y.r < 1.001*r_horizon + eps:  return CAPTURED
    if y.r > r_escape:               return ESCAPED
    handle_equator_crossing(y_prev, y)          # Section B.3
```

B.2 Grid-free boundary bisection

(`shadow_boundary_angle`)

```
lo, hi = 0, angle_max            # capture at lo, escape at hi
repeat 60 times or until hi - lo < 1e-12:
```

```
    mid = (lo + hi) / 2
    outcome = trace(ray_at_camera_angle(mid, psi))
    if outcome == CAPTURED: lo = mid else: hi = mid
return (lo + hi) / 2
```

B.3 Equator-crossing localisation and sub-ring count

(`trace` disk logic)

```
if cos(theta_prev) * cos(theta_now) < 0:        # plane crossed
    bisect substep 40 times on cos(theta) -> crossing state x
    if disk_enabled and r_in <= x.r <= r_out:
        g = 1 / (u_t_circ(x.r) * (E - Omega(x.r) * L_z))
        return DISK_HIT(r=x.r, phi=x.phi, n=crossings, g=g)
    crossings += 1
```

B.4 Least-squares fit and explicit covariance

(`OrbitFitter::fit`, `::covariance`)

```
theta = start
theta = ceres_LM_minimise(residuals(theta))     # numeric-diff Jacobians
# covariance: explicit central differences, physically scaled steps
for k in 1..13:
    J[k] = (residuals(theta + s_k e_k) - residuals(theta - s_k e_k))
           / (2 s_k)
C = gauss_jordan_inverse(J^T J)                 # NO SVD truncation
sigma_k = sqrt(C[k][k])
```

B.5 Ensemble MCMC over the chi-squared server

(`scripts/run_mcmc.py`)

```
spawn K persistent workers: fit_s2_tool --chi2 <data> <mode>
log_prob(batch X):
    out-of-bounds rows -> -inf
```

```
    distribute rows round-robin to workers; read chi2 back in order
    return -chi2 / 2
walkers = optimum + 0.5 * sigma * N(0,1)        # 32 walkers
run emcee stretch moves for 6000 steps (vectorised log_prob)
tau = max autocorrelation time; discard 4*tau; thin tau/2
save chain + 16/50/84 percentiles
```

B.6 Darwin turning-point initialisation

(`S2Model` setup; Section 14.2)

```
a_phys = a_mas * R0                         # angular -> physical (au -> M)
r_p, r_a = a_phys*(1-e), a_phys*(1+e)
# solve R(r_p) = R(r_a) = 0 for E^2, L^2 in the well-conditioned form
#   E^2 = (1 - 2M/r)(1 + L^2/r^2)  at both radii  -> L^2 first
L2 = (f_p/r_p^2 - f_a/r_a^2)^-1 * (f_a - f_p)     # f = 1 - 2M/r
E2 = f_p * (1 + L2/r_p^2)
u  = (u^t from normalisation quadratic, future root)   # solve_ut
state = (t=0, r=r_p, phi=0, u^r=0, u^phi=L/r_p^2, u^t)
integrate one full radial period; cache adaptive nodes (pos + vel)
```

B.7 Rømer fixed-point solve per epoch

(`SkyProjection`; Section 14.3)

```
t_em = t_obs
repeat 3 times:                       # contraction rate |beta_z| < 0.03
    z    = line_of_sight(sample_at(t_em))
    t_em = t_obs - z / c
return sample_at(t_em)                # observables at emission time
```

B.8 Cluster joint fit assembly

(`fit_cluster`; Chapter 16)

```
global = (M, R0, x0, y0, vx, vy, vz)           # one block, Plewa priors
for star k in {S2, S1, S8, S12, S13}:
    elements[k] = per-star block (a, e, i, Omega, omega, t_peri)
```

```
    refine elements[k] alone (global frozen at start values)
    probe (Omega, omega) +/- 180 deg; keep lowest starting chi^2
build one Ceres problem:
    residuals_k(global, elements[k]) for every epoch of every star
solve; covariance: explicit central-difference J over the full
concatenated 37-vector, gauss_jordan_inverse(J^T J)   # no SVD
subsets: repeat solve for {S2}, {S2,S1}, ... -> convergence ladder
```

B.9 Sub-ring decomposition and width scan

(SubRingExtractor::analyze, scan_widths)

```
render image with per-pixel record (n = crossing count, bolometric I)
for n in {0, 1, 2}:
    F_n = sum of I over pixels with crossing count n
ratios = F_0/F_1, F_1/F_2
for psi in {0, 90, 180, 270} deg:          # image directions
    walk outward along psi from image centre
    band = contiguous run of n==1 pixels   # may be occluded: record none
    width[psi] = outer_edge - inner_edge   # in camera angle
```

Appendix C

Number Provenance

Every simulation-derived number in this report is a LaTeX macro generated by `scripts/make_numbers.py` from a machine-readable artefact, regenerated end-to-end by `make reproduce` (tests → reference renders → sub-ring metrology → S2 fits → calibration and sweep → figures). Table C.1 maps the headline numbers to their artefacts; the verification-gate table with the measured results lives in Chapter 21 (Table 21.1).

Table C.1: Provenance of headline numbers. Paths are relative to `validation/` in the repository.

Quantity	Artefact
M, R_0, χ^2_ν, elements	`phase4/s2_fit.json`
redshift excess peak	`phase4/s2_fit.json`
precession terms	`phase4/s2_fit.json`
ε_q profile and CI	`phase5/s2_fit.json`
Schwarzschild/Kerr diameters	`reference_images/*.json`
beaming, g, sub-ring ratios	`phase3/subring_report.json`
ring diameter, θ_g, axis ratio	`phase10/calibration.json`
(a_*, i) sweep	`phase10/sweep.csv`
CPU/GPU pixel difference	`phase7/*.compare.json`
benchmarks	`phase11/bench.json`

Tables C.2 and C.3 go one level deeper: they list *every* macro of `numbers.tex` with its current value and source artefact. The values printed here are typeset from the macros themselves, so this appendix can never disagree with the body text — if `make reproduce` changes a number, it changes here too.

Table C.2: Complete macro inventory, part 1: orbit fits and relativistic observables. Values are typeset from the live macros.

Macro	Value	Units	Artefact
S2 Kerr fit (`phase4/s2_fit.json`)			
`FitMass(Err)`	4.270 ± 0.195	$10^6\,M_\odot$	fit
`FitDist(Err)`	8272 ± 182	pc	fit
`FitSma(Err)`	124.82 ± 0.88	mas	fit
`FitEcc(Err)`	0.88277 ± 0.00195	—	fit
`FitIncl(Err)`	134.70 ± 0.41	deg	fit
`FitOmegaBig(Err)`	227.33 ± 0.63	deg	fit
`FitOmegaSm(Err)`	65.44 ± 0.61	deg	fit
`FitTPeri(Err)`	2002.3227 ± 0.0052	yr	fit
`FitPeriod(Err)`	16.0561 ± 0.0228	yr	fit
`FitChiTwo / FitDof / FitChiRed`	307.8 / 321 / 0.959	—	fit
`FitNAst / FitNRv`	145 / 44	epochs	data
`FitMassNSigma / FitDistNSigma`	0.14 / 0.03	σ vs GRAVITY	fit
`GrExcessPeak / GrExcessEpoch`	205 at 2018.379	km/s; yr	fit
`PrecSchw / PrecLT`	11.98 / -0.204	arcmin/orbit	`GRPrecession`
No-hair (`phase5/s2_fit.json`; ε_q in units of 10^5)			
`EpsQBest / EpsQSigma`	0.00 / 0.93	10^5	profile
`EpsQCiNinetyFive`	1.33	10^5 (95%)	profile
`EpsQForecastTenOrbits`	0.34	10^5	forecast
MCMC posteriors (`thesis/mcmc_*.json`; 16/50/84 percentiles)			
`McmcMass{Med,Lo,Hi}`	$4.502^{+0.628}_{-0.603}$	$10^6\,M_\odot$	chain
`McmcDist{Med,Lo,Hi}`	8266^{+555}_{-530}	pc	chain
`McmcEpsQ{Med,Lo,Hi}`	$-0.43^{+6.03}_{-5.96}$	10^5	chain
`McmcNWalkers / McmcNSteps / McmcTauMax`	32 / 6000 / 59	—	chain
Cluster joint fit (`phase13/cluster_fit.json`)			
`ClusterNStars`	5	stars	fit
`ClusterMass(Err)`	4.461 ± 0.119	$10^6\,M_\odot$	fit
`ClusterDist(Err)`	8480 ± 81	pc	fit
`ClusterChiRed`	1.126	—	fit
`ClusterMassErrSTwoOnly`	0.195	$10^6\,M_\odot$	S2-only
`ClusterDistErrSTwoOnly`	182	pc	S2-only
`ClusterMassImprovePct / ClusterDistImprovePct`	39 / 56	%	fit
`ClusterMassNSigma`	1.4	σ vs GRAVITY	fit

Table C.3: Complete macro inventory, part 2: imaging, disk, numerics, and performance.

Macro	Value	Units	Artefact
Shadow metrology (`reference_images/*.json`)			
`SchwDiamM / SchwSpreadPct`	10.3924 / 0.000	M; %	boundary
`KerrDiamM / KerrDiamUas`	9.5445 / 48.91	M; μas	boundary
Disk and sub-rings (`phase3/subring_report.json`)			
`BeamRatio`	9.60	limb ratio	render
`GMin / GMax`	0.139 / 1.440	—	render
`FluxRatioZeroOne / FluxRatioOneTwo`	10.7 / 274	—	sub-rings
`SubringRatioZeroOne{Lo,Hi}`	8.6–16.8	over spin	`--sweep`
Calibration (`phase10/calibration.json`)			
`CalDiamUas / CalDiamM`	49.68 / 9.695	μas; M	sweep
`CalAxisRatio / UasPerM`	0.9946 / 5.1243	—; μas/M	sweep
Energetics and photon shell (`thesis/*.csv`)			
`EtaSchw / EtaSgrA / EtaNearExtremal`	5.72 / 17.96 / 32.10	%	`physics_tables`
`GammaPolarSgrA`	2.571	per half-orbit	`sweep_tool`
`GammaProSgrA / GammaRetroSgrA`	3.142 / 3.142	per half-orbit	`sweep_tool`
Integrator studies (`thesis/projection_boundary.csv`, `drift_vs_tol.csv`)			
`ProjOnWorstDev`	9.6×10^{-12}	rel. dev.	`convergence_tool`
`ProjOffWorstPct / ProjOffBestDev`	93 / 1.1×10^{-04}	%; rel.	`convergence_tool`
GPU validation and performance (`phase7`, `phase11`)			
`GpuMeanDiff / GpuMaxDiff / GpuOutliers`	0.00289 / 0.697 / 32	int.; int.; px	`compare_shadow`
`BenchCpuNullStepsPerS`	1,340,753	steps/s/core	bench
`BenchCpuTimelikeStepsPerS`	78,987	steps/s/core	bench
`BenchCpuChristoffelPerS`	508,369	evals/s/core	bench
`BenchGpuFpsHd / BenchGpuFpsFullhd`	47.5 / 21	fps	bench
`BenchGpuMegaraysPerSHd`	43.8	Mray/s	bench

Table C.4 extends the provenance map to the artefacts produced by the report's additional data tools.

Table C.4: Provenance of the report's extended numbers and figures. Paths are relative to `validation/` unless noted.

Quantity / figure	Artefact
efficiency $\eta(a_*)$, characteristic radii	`thesis/{efficiency,radii}_vs_spin.csv`
energetics, NT profiles, potentials	`thesis/{energetics_r,disk_profiles,*_potential}.csv`
Lyapunov γ, spherical orbits	`thesis/{lyapunov_vs_spin,spherical_orbits}.csv`
projection/drift/step studies	`thesis/{projection_boundary,drift_vs_tol,step_size}`
sub-ring vs spin	`thesis/subring_sweep.csv`
axis-ratio / displacement maps	`phase10/sweep.csv`
MCMC medians / CI, corner plots	`thesis/mcmc_{s2,nohair}.{json,npz}`
cluster fit, per-star tracks	`phase13/cluster_fit.json`, `*_track.csv`

Appendix D

The Reproduction Pipeline

The top-level `Makefile` exposes the automation targets. `make figures` regenerates every figure in this report from the current artefacts with Matplotlib (data plots from `scripts/make_figures.py`, schematics from `scripts/make_diagrams.py`). `make thesis` runs `figures` and compiles this document (`pdflatex`×3 + `bibtex`); `make paper` does the same for the companion journal paper. `make reproduce` reruns the science chain end-to-end in gate order:

1. `ctest` — the full physics and science unit-test suite (every gate-0 tolerance of Chapter 21);
2. `render_reference` — Schwarzschild and Kerr reference renders with 32-direction boundary metrology (JSON);
3. `subring_tool` — disk render, beaming asymmetry, sub-ring isolation and flux ratios;
4. `fit_s2_tool` in `kerr` and `nohair` modes — the fits of Chapters 15 and 17;
5. `sweep_tool` — the Sgr A* calibration point and the 20×20 (a_*, i) sweep;
6. `make_figures.py` — figure regeneration from the artefacts.

`scripts/make_numbers.py` then rewrites `numbers.tex`, the macro file from which every simulation-derived number in this text is typeset. A stale or hand-edited number cannot survive a `make reproduce` cycle.

For this report, `make reproduce` runs the following additional steps after the journal-paper chain above, before regenerating figures and numbers:

7. `physics_tables` — Kerr radii, energetics, efficiency and Novikov–Thorne profiles over the spin grid (Chapters 8, 12);
8. `convergence_tool` — the projection/drift/step-size studies (Chapter 10);
9. `sweep_tool photonorbits` — the spherical-orbit and Lyapunov families (Chapter 11);
10. `subring_tool --sweep` — sub-ring flux ratios and widths over spin (Chapter 13);
11. `run_mcmc.py both` — the Kerr and no-hair ensemble posteriors via the `fit_s2_tool --chi2` server (Chapters 15, 17);
12. `fit_cluster` — the joint five-star mass fit (Chapter 16);
13. `make_diagrams.py` — the explanatory schematics of Part I.

The single command `make thesis` runs `figures` and compiles this document; the fetch of the multi-star data (`fetch_gillessen_data.py`) is a one-time network step run before the cluster fit.

Environment

The chain assumes: a C++20 toolchain with the vcpkg manifest dependencies (Eigen, Ceres, Catch2, spdlog, nlohmann-json; Vulkan SDK components only for the engine targets), CMake presets `core-rel`/`sci-rel`/`engine-rel`, Python 3.11 with `numpy`, `matplotlib`, `emcee` and `pypdf`, and a LaTeX distribution with `natbib` (TinyTeX suffices). No step requires a GPU except the engine renders and the CPU/GPU comparison; every scientific number in this report is reproducible on a CPU-only machine.

Verifying this document

The document itself is checked by a text-level probe after each full build: the compiled PDF is read back with `pypdf` and scanned for (i) unresolved references (the doubled question mark LaTeX prints for a missing label — spelt out here so the probe does not catch its own description), (ii) raw macro names (a backslash followed by a macro identifier surviving into the output, which would indicate a

missing `numbers.tex` definition), (iii) the presence of every headline number, and (iv) the policy that the contested star S62 must carry its caveat at every mention. The probe, the zero-error requirement on the LaTeX log, and the page-count band of the plan together define “compiles clean” wherever this report uses the phrase.

Appendix E

Notation Glossary

Every recurring symbol, with the section where it is defined.

Symbol	Defined	Meaning
M	§8.1	black-hole mass; also the geometric length/time unit ($G = c = 1$)
a, a_*	§8.1	spin parameter J/M; dimensionless $a/M \in [0, 1]$
Σ, Δ, A	eq. (8.2)	Kerr auxiliary functions
$r_\pm$	eq. (8.14)	outer/inner horizon radii
$r_E(\theta)$	eq. (8.16)	ergosphere boundary
$r_{\rm ph}^{\pm}$	eq. (8.19)	prograde/retrograde equatorial photon orbits
$r_{\rm ISCO}$	eq. (8.24)	innermost stable circular orbit
$r_{\rm mb}$	eq. (8.27)	marginally bound circular orbit
ω	eq. (8.12)	frame-dragging (ZAMO) angular velocity
Ω	eq. (8.21)	circular-orbit angular velocity
E, L_z	eq. (8.10)	conserved energy and axial angular momentum
Q	§9.1	Carter constant
μ	eq. (9.1)	rest mass (0 photons, 1 stars)
$R(r), \Theta(\theta)$	eqs. (9.7)–(9.8)	radial and polar potentials
$P(r)$	eq. (9.6)	$E(r^2 + a^2) - aL_z$
λ, τ	eq. (9.14)	affine parameter; Mino time ($\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau$)
b, ξ, η	§8.5, eq. (11.5)	impact parameter L_z/E; critical-curve constants L_z/E, Q/E^2
γ	eq. (11.7)	Lyapunov exponent per polar half-orbit
n	§13.1	photon sub-ring order (equatorial crossings)
$F(r), f(x)$	eqs. (12.2)–(12.3)	Page–Thorne flux; $x = \sqrt{r/M}$
η (efficiency)	eq. (8.28)	accretion efficiency $1 - E_{\rm ISCO}$
g	eq. (12.7)	redshift factor $\nu_{\rm obs}/\nu_{\rm emit}$
$z_{\rm D}, z_{\rm tD}, z_{\rm grav}$	eq. (14.4)	classical, transverse, gravitational redshifts

continued on the next page

Symbol	Defined	Meaning
$a_{\mathrm{mas}}, e, i, \Omega, \omega, t_{\mathrm{peri}}$	§14.2	osculating orbital elements (semi-major axis in mas, eccentricity, inclination, node, pericentre argument, epoch)
R_0	§15.2	distance to the Galactic centre
p	eq. (17.4)	semilatus rectum $a(1-e^2)$
χ^2, χ^2_ν	eq. (15.1)	fit objective; reduced value per degree of freedom
J, F, C	§15.2.2	residual Jacobian; Fisher matrix $J^{\mathsf{T}}J$; covariance F^{-1}
ε_q	eq. (17.2)	quadrupole deviation from Kerr (no-hair parameter)
$\tilde{q}$	eq. (17.4)	total quadrupole Q/M^3
μas	§19.1	microarcsecond
θ_g	§19.1	angular size of one gravitational radius at distance R_0
α (lapse)	§8.3	ZAMO lapse $\sqrt{\Delta\Sigma/A}$; also shadow angular radius in §11.2 (context distinguishes)
Ω_H	eq. (8.13)	horizon angular velocity $a/(2Mr_+)$
A_H, M_{irr}	eq. (8.15)	horizon area; irreducible mass $\sqrt{A_H/16\pi}$
u^t	eq. (12.8)	circular-orbit time dilation factor
Λ_r, Λ_θ	eq. (9.17)	fundamental Mino-time periods of the radial/polar motion
f	eq. (A.23)	Schwarzschild lapse factor $1-2M/r$ (Hartle–Thorne context)
F_1, F_2	§17.2	quasi-Kerr radial structure functions
$e_{\hat{\mu}}$	eq. (11.2)	ZAMO orthonormal tetrad legs

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